

Remark. Let $\mathcal{H}_\gamma(\omega)$ be the set of times at which the path $\omega \in C$ is Hölder continuous of order γ . Theorem 7.1.5 shows that $P(\mathcal{H}_\gamma = [0, \infty)) = 1$ for $\gamma < 1/2$. Exercise 7.1.5 shows that $P(\mathcal{H}_\gamma = \emptyset) = 1$ for $\gamma > 1/2$. Exercise 7.2.4 shows $P(t \in \mathcal{H}_{1/2}) = 0$ for each t , but B. Davis (1983) has shown $P(\mathcal{H}_{1/2} \neq \emptyset) = 1$. Perkins (1983) has computed the Hausdorff dimension of

$$\left\{ t \in (0, 1) : \limsup_{h \downarrow 0} \frac{|B_{t+h} - B_t|}{h^{1/2}} \leq c \right\}$$

Multidimensional Brownian motion

All of the result in this section have been for one-dimensional Brownian motion. To define a d -dimensional Brownian motion starting at $x \in \mathbf{R}^d$ we let $B_t^1, \dots, B_t^d$ be independent Brownian motions with $B_0^i = x_i$. As in the case $d = 1$ these are realized as probability measures P_x on $(C, \mathcal{C})$ where $C = \{\text{continuous } \omega : [0, \infty) \rightarrow \mathbf{R}^d\}$ and $\mathcal{C}$ is the σ -field generated by the coordinate maps. Since the coordinates are independent, it is easy to see that the finite dimensional distributions satisfy (7.1.2) with transition probability

$$p_t(x, y) = (2\pi t)^{-d/2} \exp(-|y - x|^2/2t) \quad (7.1.4)$$

EXERCISES

7.1.1. Given $s < t$ find $P(B(s) > 0, B(t) > 0)$. Hint: write the probability in terms of $B(s)$ and $B(t) - B(s)$.

7.1.2. Find $E(B_1^2 B_2 B_3)$.

7.1.3. Let $W = \int_0^t B_s ds$. Find EW and EW^2 . What is the distribution of W ?

7.1.4. $A \in \mathcal{F}_o$ if and only if there is a sequence of times $t_1, t_2, \dots$ in $[0, \infty)$ and a $B \in \mathcal{R}^{\{1, 2, \dots\}}$ so that $A = \{\omega : (\omega(t_1), \omega(t_2), \dots) \in B\}$. In words, all events in $\mathcal{F}_o$ depend on only countably many coordinates.

7.1.5. Looking at the proof of Theorem 7.1.6 carefully shows that if $\gamma > 5/6$ then B_t is not Hölder continuous with exponent γ at any point in $[0, 1]$. Show, by considering k increments instead of 3, that the last conclusion is true for all $\gamma > 1/2 + 1/k$.

7.1.6. Fix t and let $\Delta_{m,n} = B(tm2^{-n}) - B(t(m-1)2^{-n})$. Compute

$$E\left(\sum_{m \leq 2^n} \Delta_{m,n}^2 - t\right)^2$$

and use Borel-Cantelli to conclude that $\sum_{m \leq 2^n} \Delta_{m,n}^2 \rightarrow t$ a.s. as $n \rightarrow \infty$.

Remark. The last result is true if we consider a sequence of partitions $\Pi_1 \subset \Pi_2 \subset \dots$ with mesh $\rightarrow 0$. See Freedman (1971a) p. 42–46. However, the true quadratic variation, defined as the sup over all partitions, is ∞ .

7.2 Markov Property, Blumenthal's 0-1 Law

Intuitively, the Markov property says “if $s \geq 0$ then $B(t+s) - B(s)$, $t \geq 0$ is a Brownian motion that is independent of what happened before time s .” The first step in making this into a precise statement is to explain what we mean by “what happened before time s .” The first thing that comes to mind is

$$\mathcal{F}_s^o = \sigma(B_r : r \leq s)$$

For reasons that will become clear as we go along, it is convenient to replace $\mathcal{F}_s^o$ by

$$\mathcal{F}_s^+ = \cap_{t>s} \mathcal{F}_t^o$$

The fields $\mathcal{F}_s^+$ are nicer because they are **right continuous**:

$$\cap_{t>s} \mathcal{F}_t^+ = \cap_{t>s} (\cap_{u>t} \mathcal{F}_u^o) = \cap_{u>s} \mathcal{F}_u^o = \mathcal{F}_s^+$$

In words, the $\mathcal{F}_s^+$ allow us an “infinitesimal peek at the future,” i.e., $A \in \mathcal{F}_s^+$ if it is in $\mathcal{F}_{s+\epsilon}^o$ for any $\epsilon > 0$. If $f(u) > 0$ for all $u > 0$, then in $d = 1$ the random variable

$$\limsup_{t \downarrow s} \frac{B_t - B_s}{f(t-s)}$$

is measurable with respect to $\mathcal{F}_s^+$ but not $\mathcal{F}_s^o$. We will see below that there are no interesting examples, i.e., $\mathcal{F}_s^+$ and $\mathcal{F}_s^o$ are the same (up to null sets).

To state the Markov property, we need some notation. Recall that we have a family of measures P_x , $x \in \mathbf{R}^d$, on $(C, \mathcal{C})$ so that under P_x , $B_t(\omega) = \omega(t)$ is a Brownian motion starting at x . For $s \geq 0$, we define the **shift transformation** $\theta_s : C \rightarrow C$ by

$$(\theta_s \omega)(t) = \omega(s+t) \quad \text{for } t \geq 0$$

In words, we cut off the part of the path before time s and then shift the path so that time s becomes time 0.

Theorem 7.2.1. Markov property. *If $s \geq 0$ and Y is bounded and $\mathcal{C}$ measurable, then for all $x \in \mathbf{R}^d$*

$$E_x(Y \circ \theta_s | \mathcal{F}_s^+) = E_{B_s} Y$$

where the right-hand side is the function $\varphi(x) = E_x Y$ evaluated at $x = B_s$.

Proof. By the definition of conditional expectation, what we need to show is that

$$E_x(Y \circ \theta_s; A) = E_x(E_{B_s} Y; A) \quad \text{for all } A \in \mathcal{F}_s^+ \quad (7.2.1)$$

We will begin by proving the result for a carefully chosen special case and then use the monotone class theorem (MCT) to get the general case. Suppose $Y(\omega) = \prod_{1 \leq m \leq n} f_m(\omega(t_m))$, where $0 < t_1 < \dots < t_n$ and the f_m are bounded and measurable. Let $0 < h < t_1$, let $0 < s_1 \dots < s_k \leq s + h$, and let $A = \{\omega : \omega(s_j) \in A_j, 1 \leq j \leq k\}$, where $A_j \in \mathcal{R}$ for $1 \leq j \leq k$. From the definition of Brownian motion, it follows that

$$E_x(Y \circ \theta_s; A) = \int_{A_1} dx_1 p_{s_1}(x, x_1) \int_{A_2} dx_2 p_{s_2-s_1}(x_1, x_2) \cdots \\ \int_{A_k} dx_k p_{s_k-s_{k-1}}(x_{k-1}, x_k) \int dy p_{s+h-s_k}(x_k, y) \varphi(y, h)$$

where

$$\varphi(y, h) = \int dy_1 p_{t_1-h}(y, y_1) f_1(y_1) \cdots \int dy_n p_{t_n-t_{n-1}}(y_{n-1}, y_n) f_n(y_n)$$

For more details, see the proof of (5.2.3), which applies without change here. Using that identity on the right-hand side, we have

$$E_x(Y \circ \theta_s; A) = E_x(\varphi(B_{s+h}, h); A) \quad (7.2.2)$$

The last equality holds for all finite dimensional sets A so the $\pi - \lambda$ theorem, Theorem 2.1.6, implies that it is valid for all $A \in \mathcal{F}_{s+h}^o \supset \mathcal{F}_s^+$.

It is easy to see by induction on n that

$$\psi(y_1) = f_1(y_1) \int dy_2 p_{t_2-t_1}(y_1, y_2) f_2(y_2) \\ \cdots \int dy_n p_{t_n-t_{n-1}}(y_{n-1}, y_n) f_n(y_n)$$

is bounded and measurable. Letting $h \downarrow 0$ and using the dominated convergence theorem shows that if $x_h \rightarrow x$, then

$$\phi(x_h, h) = \int dy_1 p_{t_1-h}(x_h, y_1) \psi(y_1) \rightarrow \phi(x, 0)$$

as $h \downarrow 0$. Using (7.2.2) and the bounded convergence theorem now gives

$$E_x(Y \circ \theta_s; A) = E_x(\varphi(B_s, 0); A)$$

for all $A \in \mathcal{F}_s^+$. This shows that (7.2.1) holds for $Y = \prod_{1 \leq m \leq n} f_m(\omega(t_m))$ and the f_m are bounded and measurable.

The desired conclusion now follows from the monotone class theorem, 5.2.2. Let $\mathcal{H}$ = the collection of bounded functions for which (7.2.1) holds. $\mathcal{H}$ clearly has properties (ii) and (iii). Let $\mathcal{A}$ be the collection of sets of the form $\{\omega : \omega(t_j) \in A_j\}$, where $A_j \in \mathcal{R}$. The special case treated above shows (i) holds and the desired conclusion follows. $\square$

The reader will see many applications of the Markov property below, so we turn our attention now to a “triviality” that has surprising consequences. Since

$$E_x(Y \circ \theta_s | \mathcal{F}_s^+) = E_{B(s)}Y \in \mathcal{F}_s^o$$

it follows from Theorem 4.1.12 that

$$E_x(Y \circ \theta_s | \mathcal{F}_s^+) = E_x(Y \circ \theta_s | \mathcal{F}_s^o)$$

From the last equation, it is a short step to:

Theorem 7.2.2. *If $Z \in \mathcal{C}$ is bounded then for all $s \geq 0$ and $x \in \mathbf{R}^d$,*

$$E_x(Z | \mathcal{F}_s^+) = E_x(Z | \mathcal{F}_s^o)$$

Proof. As in the proof of Theorem 7.2.1, it suffices to prove the result when

$$Z = \prod_{m=1}^n f_m(B(t_m))$$

and the f_m are bounded and measurable. In this case, Z can be written as $X(Y \circ \theta_s)$, where $X \in \mathcal{F}_s^o$ and Y is $\mathcal{C}$ measurable, so

$$E_x(Z | \mathcal{F}_s^+) = X E_x(Y \circ \theta_s | \mathcal{F}_s^+) = X E_{B_s}Y \in \mathcal{F}_s^o$$

and the proof is complete. $\square$

If we let $Z \in \mathcal{F}_s^+$, then Theorem 7.2.2 implies $Z = E_x(Z | \mathcal{F}_s^o) \in \mathcal{F}_s^o$, so the two σ -fields are the same up to null sets. At first glance, this conclusion is not exciting. The fun starts when we take $s = 0$ in Theorem 7.2.2 to get:

Theorem 7.2.3. Blumenthal’s 0-1 law. *If $A \in \mathcal{F}_0^+$ then for all $x \in \mathbf{R}^d$,*

$$P_x(A) \in \{0, 1\}.$$

Proof. Using $A \in \mathcal{F}_0^+$, Theorem 7.2.2, and $\mathcal{F}_0^o = \sigma(B_0)$ is trivial under P_x gives

$$1_A = E_x(1_A | \mathcal{F}_0^+) = E_x(1_A | \mathcal{F}_0^o) = P_x(A) \quad P_x \text{ a.s.}$$

This shows that the indicator function 1_A is a.s. equal to the number $P_x(A)$, and the result follows. $\square$

In words, the last result says that the **germ field**, $\mathcal{F}_0^+$, is trivial. This result is very useful in studying the local behavior of Brownian paths. For the rest of the section we restrict our attention to $d = 1$.

Theorem 7.2.4. *If $\tau = \inf\{t \geq 0 : B_t > 0\}$ then $P_0(\tau = 0) = 1$.*

Proof. $P_0(\tau \leq t) \geq P_0(B_t > 0) = 1/2$ since the normal distribution is symmetric about 0. Letting $t \downarrow 0$, we conclude

$$P_0(\tau = 0) = \lim_{t \downarrow 0} P_0(\tau \leq t) \geq 1/2$$

so it follows from Theorem 7.2.3 that $P_0(\tau = 0) = 1$. $\square$

Once Brownian motion must hit $(0, \infty)$ immediately starting from 0, it must also hit $(-\infty, 0)$ immediately. Since $t \rightarrow B_t$ is continuous, this forces:

Theorem 7.2.5. *If $T_0 = \inf\{t > 0 : B_t = 0\}$ then $P_0(T_0 = 0) = 1$.*

Theorem 7.2.3 concerns the behavior of B_t as $t \rightarrow 0$. By using a trick, we can use this result to get information about the behavior as $t \rightarrow \infty$.

Theorem 7.2.6. *If B_t is a Brownian motion starting at 0, then so is the process defined by $X_0 = 0$ and $X_t = tB(1/t)$ for $t > 0$.*

Proof. Here we will check the second definition of Brownian motion. To do this, we note: (i) If $0 < t_1 < \dots < t_n$, then $(X(t_1), \dots, X(t_n))$ has a multivariate normal distribution with mean 0. (ii) $EX_s = 0$ and if $s < t$ then

$$E(X_s X_t) = stE(B(1/s)B(1/t)) = s$$

For (iii) we note that X is clearly continuous at $t \neq 0$.

To handle $t = 0$, we begin by observing that the strong law of large numbers implies $B_n/n \rightarrow 0$ as $n \rightarrow \infty$ through the integers. To handle values in between integers, we note that Kolmogorov's inequality, Theorem 2.5.5, implies

$$P\left(\sup_{0 < k \leq 2^m} |B(n + k2^{-m}) - B_n| > n^{2/3}\right) \leq n^{-4/3}E(B_{n+1} - B_n)^2$$

Letting $m \rightarrow \infty$, we have

$$P\left(\sup_{u \in [n, n+1]} |B_u - B_n| > n^{2/3}\right) \leq n^{-4/3}$$

Since $\sum_n n^{-4/3} < \infty$, the Borel-Cantelli lemma implies $B_u/u \rightarrow 0$ as $u \rightarrow \infty$. Taking $u = 1/t$, we have $X_t \rightarrow 0$ as $t \rightarrow 0$. $\square$

Theorem 7.2.6 allows us to relate the behavior of B_t as $t \rightarrow \infty$ and as $t \rightarrow 0$. Combining this idea with Blumenthal's 0-1 law leads to a very useful result. Let

$$\begin{aligned}\mathcal{F}'_t &= \sigma(B_s : s \geq t) = \text{the future at time } t \\ \mathcal{T} &= \cap_{t \geq 0} \mathcal{F}'_t = \text{the tail } \sigma\text{-field.}\end{aligned}$$

Theorem 7.2.7. *If $A \in \mathcal{T}$ then either $P_x(A) \equiv 0$ or $P_x(A) \equiv 1$.*

Remark. Notice that this is stronger than the conclusion of Blumenthal's 0-1 law. The examples $A = \{\omega : \omega(0) \in D\}$ show that for A in the germ σ -field $\mathcal{F}_0^+$, the value of $P_x(A)$, $1_D(x)$ in this case, may depend on x .

Proof. Since the tail σ -field of B is the same as the germ σ -field for X , it follows that $P_0(A) \in \{0, 1\}$. To improve this to the conclusion given, observe that $A \in \mathcal{F}_1'$, so 1_A can be written as $1_D \circ \theta_1$. Applying the Markov property gives

$$\begin{aligned} P_x(A) &= E_x(1_D \circ \theta_1) = E_x(E_x(1_D \circ \theta_1 | \mathcal{F}_1)) = E_x(E_{B_1} 1_D) \\ &= \int (2\pi)^{-1/2} \exp(-(y-x)^2/2) P_y(D) dy \end{aligned}$$

Taking $x = 0$, we see that if $P_0(A) = 0$, then $P_y(D) = 0$ for a.e. y with respect to Lebesgue measure, and using the formula again shows $P_x(A) = 0$ for all x . To handle the case $P_0(A) = 1$, observe that $A^c \in \mathcal{T}$ and $P_0(A^c) = 0$, so the last result implies $P_x(A^c) = 0$ for all x . $\square$

The next result is a typical application of Theorem 7.2.7.

Theorem 7.2.8. *Let B_t be a one-dimensional Brownian motion starting at 0 then with probability 1,*

$$\limsup_{t \rightarrow \infty} B_t / \sqrt{t} = \infty \quad \liminf_{t \rightarrow \infty} B_t / \sqrt{t} = -\infty$$

Proof. Let $K < \infty$. By Exercise 2.3.1 and scaling

$$P_0(B_n / \sqrt{n} \geq K \text{ i.o.}) \geq \limsup_{n \rightarrow \infty} P_0(B_n \geq K\sqrt{n}) = P_0(B_1 \geq K) > 0$$

so the 0-1 law in Theorem 7.2.7 implies the probability is 1. Since K is arbitrary, this proves the first result. The second one follows from symmetry. $\square$

From Theorem 7.2.8, translation invariance, and the continuity of Brownian paths it follows that we have:

Theorem 7.2.9. *Let B_t be a one-dimensional Brownian motion and let $A = \cap_n \{B_t = 0 \text{ for some } t \geq n\}$. Then $P_x(A) = 1$ for all x .*

In words, one-dimensional Brownian motion is recurrent. For any starting point x , it will return to 0 “infinitely often,” i.e., there is a sequence of times $t_n \uparrow \infty$ so that $B_{t_n} = 0$. We have to be careful with the interpretation of the phrase in quotes since starting from 0, B_t will hit 0 infinitely many times by time $\epsilon > 0$.

Last rites. With our discussion of Blumenthal's 0-1 law complete, the distinction between $\mathcal{F}_s^+$ and $\mathcal{F}_s^o$ is no longer important, so we will make one final improvement in our σ -fields and remove the superscripts. Let

$$\begin{aligned}\mathcal{N}_x &= \{A : A \subset D \text{ with } P_x(D) = 0\} \\ \mathcal{F}_s^x &= \sigma(\mathcal{F}_s^+ \cup \mathcal{N}_x) \\ \mathcal{F}_s &= \cap_x \mathcal{F}_s^x\end{aligned}$$

$\mathcal{N}_x$ are the **null sets** and $\mathcal{F}_s^x$ are the completed σ -fields for P_x . Since we do not want the filtration to depend on the initial state, we take the intersection of all the σ -fields. The reader should note that it follows from the definition that the $\mathcal{F}_s$ are right-continuous.

EXERCISES

7.2.1. Let $T_0 = \inf\{s > 0 : B_s = 0\}$ and let $R = \inf\{t > 1 : B_t = 0\}$. R is for right or return. Use the Markov property at time 1 to get

$$P_x(R > 1 + t) = \int p_1(x, y) P_y(T_0 > t) dy \quad (7.2.3)$$

7.2.2. Let $T_0 = \inf\{s > 0 : B_s = 0\}$ and let $L = \sup\{t \leq 1 : B_t = 0\}$. L is for left or last. Use the Markov property at time $0 < t < 1$ to conclude

$$P_0(L \leq t) = \int p_t(0, y) P_y(T_0 > 1 - t) dy \quad (7.2.4)$$

7.2.3. If $a < b$, then with probability one a is the limit of local maximum of B_t in (a, b) . So the set of local maxima of B_t is almost surely a dense set. However, unlike the zero set it is countable.

7.2.4. (i) Suppose $f(t) > 0$ for all $t > 0$. Use Theorem 7.2.3 to conclude that $\limsup_{t \downarrow 0} B(t)/f(t) = c$, P_0 a.s., where $c \in [0, \infty]$ is a constant. (ii) Show that if $f(t) = \sqrt{t}$ then $c = \infty$, so with probability one Brownian paths are not Hölder continuous of order $1/2$ at 0.

7.3 Stopping Times, Strong Markov Property

Generalizing the definition from discrete time, we call a random variable S taking values in $[0, \infty]$ a **stopping time** if for all $t \geq 0$, $\{S < t\} \in \mathcal{F}_t$. In the last definition, we have obviously made a choice between $\{S < t\}$ and $\{S \leq t\}$. This makes a big difference in discrete time but none in continuous time (for a right continuous filtration $\mathcal{F}_t$):

If $\{S \leq t\} \in \mathcal{F}_t$ then $\{S < t\} = \cup_n \{S \leq t - 1/n\} \in \mathcal{F}_t$.

If $\{S < t\} \in \mathcal{F}_t$ then $\{S \leq t\} = \cap_n \{S < t + 1/n\} \in \mathcal{F}_t$.

The first conclusion requires only that $t \rightarrow \mathcal{F}_t$ is increasing. The second relies on the fact that $t \rightarrow \mathcal{F}_t$ is right continuous. Theorem 7.3.2 and 7.3.3 below show that when checking something is a stopping time, it is nice to know that the two definitions are equivalent.

Theorem 7.3.1. *If G is an open set and $T = \inf\{t \geq 0 : B_t \in G\}$ then T is a stopping time.*

Proof. Since G is open and $t \rightarrow B_t$ is continuous, $\{T < t\} = \cup_{q < t} \{B_q \in G\}$, where the union is over all rational q , so $\{T < t\} \in \mathcal{F}_t$. Here we need to use the rationals to get a countable union, and hence a measurable set. $\square$

Theorem 7.3.2. *If T_n is a sequence of stopping times and $T_n \downarrow T$ then T is a stopping time.*

Proof. $\{T < t\} = \cup_n \{T_n < t\}$. $\square$

Theorem 7.3.3. *If T_n is a sequence of stopping times and $T_n \uparrow T$ then T is a stopping time.*

Proof. $\{T \leq t\} = \cap_n \{T_n \leq t\}$. $\square$

Theorem 7.3.4. *If K is a closed set and $T = \inf\{t \geq 0 : B_t \in K\}$ then T is a stopping time.*

Proof. Let $B(x, r) = \{y : |y - x| < r\}$, let $G_n = \cup_{x \in K} B(x, 1/n)$ and let $T_n = \inf\{t \geq 0 : B_t \in G_n\}$. Since G_n is open, it follows from Theorem 7.3.1 that T_n is a stopping time. I claim that as $n \uparrow \infty$, $T_n \uparrow T$. To prove this, notice that $T \geq T_n$ for all n , so $\lim T_n \leq T$. To prove $T \leq \lim T_n$, we can suppose that $T_n \uparrow t < \infty$. Since $B(T_n) \in \bar{G}_n$ for all n and $B(T_n) \rightarrow B(t)$, it follows that $B(t) \in K$ and $T \leq t$. $\square$

Theorems 7.3.1 and 7.3.4 will take care of all the hitting times we will consider. Using 7.3.2 with the second result one can show that the hitting time of a union of closed sets is a stopping time. It is known that the hitting time of a Borel set is a stopping time but proving that is very difficult.

We will need the next result in the proof of the strong Markov property

Theorem 7.3.5. *Let S be a stopping time and let $S_n = ([2^n S] + 1)/2^n$ where $[x] =$ the largest integer $\leq x$. That is,*

$$S_n = (m + 1)2^{-n} \text{ if } m2^{-n} \leq S < (m + 1)2^{-n}$$

In words, we stop at the first time of the form $k2^{-n}$ after S (i.e., $> S$). From the verbal description, it should be clear that S_n is a stopping time.

Proof. If $m/2^{-n} < \leq (m + 1)/2^{-n}$ then $\{S_n < t\} = \{S \leq m/2^n\} \in \mathcal{F}_{m/2^n} \subset \mathcal{F}_t$. It is impossible to have $S_n = 0$. $\square$

Our next goal is to state and prove the strong Markov property. To do this, we need to generalize two definitions from Section 4.1. Given a nonnegative random variable $S(\omega)$ we define the random shift θ_S , which “cuts off the part of ω before $S(\omega)$ and then shifts the path so that time $S(\omega)$ becomes time 0.” In symbols, we set

$$(\theta_S \omega)(t) = \begin{cases} \omega(S(\omega) + t) & \text{on } \{S < \infty\} \\ \Delta & \text{on } \{S = \infty\} \end{cases}$$

where Δ is an extra point we add to C . As in Section 6.3, we will usually explicitly restrict our attention to $\{S < \infty\}$, so the reader does not have to worry about the second half of the definition.

The second quantity $\mathcal{F}_S$, “the information known at time S ,” is a little more subtle. Imitating the discrete time definition from Section 4.1, we let

$$\mathcal{F}_S = \{A : A \cap \{S \leq t\} \in \mathcal{F}_t \text{ for all } t \geq 0\}$$

In words, this makes the reasonable demand that the part of A that lies in $\{S \leq t\}$ should be measurable with respect to the information available at time t . Again we have made a choice between $\leq t$ and $< t$, but as in the case of stopping times, this makes no difference, and it is useful to know that the two definitions are equivalent.

Remark. When $\mathcal{F}_t$ is right continuous, the last definition is unchanged if we replace $\{S \leq t\}$ by $\{S < t\}$.

We will now derive some properties of $\mathcal{F}_S$. The first is intuitively obvious: at a later time we have more information.

Theorem 7.3.6. *If $S \leq T$ are stopping times then $\mathcal{F}_S \subset \mathcal{F}_T$.*

Proof. If $A \in \mathcal{F}_S$ then $A \cap \{T \leq t\} = (A \cap \{S \leq t\}) \cap \{T \leq t\} \in \mathcal{F}_t$. $\square$

Theorem 7.3.7. *If $T_n \downarrow T$ are stopping times then $\mathcal{F}_T = \cap \mathcal{F}(T_n)$.*

Proof. Theorem 7.3.6 implies $\mathcal{F}(T_n) \supset \mathcal{F}_T$ for all n . To prove the other inclusion, let $A \in \cap \mathcal{F}(T_n)$. Since $A \cap \{T_n < t\} \in \mathcal{F}_t$ and $T_n \downarrow T$, it follows that $A \cap \{T < t\} \in \mathcal{F}_t$. $\square$

The next result is obvious but takes a little work.

Theorem 7.3.8. $B_S \in \mathcal{F}_S$, i.e., the value of B_S is measurable with respect to the information known at time S !

Proof. To prove this, let $S_n = ([2^n S] + 1)/2^n$ be the stopping times defined in Theorem 7.3.5. If A is a Borel set.

$$\{B(S_n) \in A\} = \cup_{m=1}^{\infty} \{S_n = m/2^n, B(m/2^n) \in A\} \in \mathcal{F}_{S_n}$$

Now let $n \rightarrow \infty$ and use Theorem 7.3.7. $\square$

We are now ready to state the strong Markov property, which says that the Markov property holds at stopping times. It is interesting that the notion of Brownian motion dates to the very beginning of the 20th century, but the first proofs of the strong Markov property were given independently by Hunt (1956), and Dynkin and Yushkevich (1956). Hunt writes “Although mathematicians use this extended Markoff property, at least as a heuristic principle, I have nowhere found it discussed with rigor.”

Theorem 7.3.9. Strong Markov property. Let $(s, \omega) \rightarrow Y_s(\omega)$ be bounded and $\mathcal{R} \times \mathcal{C}$ measurable. If S is a stopping time, then for all $x \in \mathbf{R}^d$

$$E_x(Y_S \circ \theta_S | \mathcal{F}_S) = E_{B(S)} Y_S \text{ on } \{S < \infty\}$$

where the right-hand side is the function $\varphi(x, t) = E_x Y_t$ evaluated at $x = B(S)$, $t = S$.

Remark. The only facts about Brownian motion used here are that (i) it is a Markov process, and (ii) if f is bounded and continuous then $x \rightarrow E_x f(B_t)$ is continuous. In Markov process theory (ii) is called the Feller property. While Hunt’s proof only applies to Brownian motion, and Dynkin and Yushkevich proved the result in this generality.

Proof. We first prove the result under the assumption that there is a sequence of times $t_n \uparrow \infty$, so that $P_x(S < \infty) = \sum P_x(S = t_n)$. In this case, the proof is basically the same as the proof of Theorem 5.2.5. We break things down according to the value of S , apply the Markov property, and put the pieces back together. If we let $Z_n = Y_{t_n}(\omega)$ and $A \in \mathcal{F}_S$, then

$$E_x(Y_S \circ \theta_S; A \cap \{S < \infty\}) = \sum_{n=1}^{\infty} E_x(Z_n \circ \theta_{t_n}; A \cap \{S = t_n\})$$

Now if $A \in \mathcal{F}_S$, $A \cap \{S = t_n\} = (A \cap \{S \leq t_n\}) - (A \cap \{S \leq t_{n-1}\}) \in \mathcal{F}_{t_n}$, so it follows from the Markov property that the above sum is

$$= \sum_{n=1}^{\infty} E_x(E_{B(t_n)} Z_n; A \cap \{S = t_n\}) = E_x(E_{B(S)} Y_S; A \cap \{S < \infty\})$$

To prove the result in general, we let $S_n = ([2^n S] + 1)/2^n$ be the stopping time defined in Exercise 7.3.5. To be able to let $n \rightarrow \infty$, we restrict our attention to Y 's of the form

$$Y_s(\omega) = f_0(s) \prod_{m=1}^n f_m(\omega(t_m)) \quad (7.3.1)$$

where $0 < t_1 < \dots < t_n$ and $f_0, \dots, f_n$ are bounded and continuous. If f is bounded and continuous then the dominated convergence theorem implies that

$$x \rightarrow \int dy p_t(x, y) f(y)$$

is continuous. From this and induction, it follows that

$$\begin{aligned} \varphi(x, s) = E_x Y_s &= f_0(s) \int dy_1 p_{t_1}(x, y_1) f_1(y_1) \\ &\dots \int dy_n p_{t_n - t_{n-1}}(y_{n-1}, y_n) f_n(y_n) \end{aligned}$$

is bounded and continuous.

Having assembled the necessary ingredients, we can now complete the proof. Let $A \in \mathcal{F}_S$. Since $S \leq S_n$, Theorem 7.3.6 implies $A \in \mathcal{F}(S_n)$. Applying the special case proved above to S_n and observing that $\{S_n < \infty\} = \{S < \infty\}$ gives

$$E_x(Y_{S_n} \circ \theta_{S_n}; A \cap \{S < \infty\}) = E_x(\varphi(B(S_n), S_n); A \cap \{S < \infty\})$$

Now, as $n \rightarrow \infty$, $S_n \downarrow S$, $B(S_n) \rightarrow B(S)$, $\varphi(B(S_n), S_n) \rightarrow \varphi(B(S), S)$ and

$$Y_{S_n} \circ \theta_{S_n} \rightarrow Y_S \circ \theta_S$$

so the bounded convergence theorem implies that the result holds when Y has the form given in (7.3.1).

To complete the proof now, we will apply the monotone class theorem. As in the proof of Theorem 7.2.1, we let $\mathcal{H}$ be the collection of Y for which

$$E_x(Y_S \circ \theta_S; A) = E_x(E_{B(S)} Y_S; A) \quad \text{for all } A \in \mathcal{F}_S$$

and it is easy to see that (ii) and (iii) hold. This time, however, we take $\mathcal{A}$ to be the sets of the form $A = G_0 \times \{\omega : \omega(s_j) \in G_j, 1 \leq j \leq k\}$, where the G_j are open sets. To verify (i), we note that if $K_j = G_j^c$ and $f_j^n(x) = 1 \wedge n\rho(x, K_j)$, where $\rho(x, K) = \inf\{|x - y| : y \in K\}$ then f_j^n are continuous functions with $f_j^n \uparrow 1_{G_j}$ as $n \uparrow \infty$. The facts that

$$Y_s^n(\omega) = f_0^n(s) \prod_{j=1}^k f_j^n(\omega(s_j)) \in \mathcal{H}$$

and (iii) holds for $\mathcal{H}$ imply that $1_A \in \mathcal{H}$. This verifies (i) in the monotone class theorem and completes the proof. $\square$

EXERCISES

7.3.1. Let A be an F_σ , that is, a countable union of closed sets. Show that $T_A = \inf\{t : B_t \in A\}$ is a stopping time.

7.3.2. If S and T are stopping times, then $S \wedge T = \min\{S, T\}$, $S \vee T = \max\{S, T\}$, and $S + T$ are also stopping times. In particular, if $t \geq 0$, then $S \wedge t$, $S \vee t$, and $S + t$ are stopping times.

7.3.3. Let T_n be a sequence of stopping times. Show that

$$\sup_n T_n, \quad \inf_n T_n, \quad \limsup_n T_n, \quad \liminf_n T_n$$

are stopping times.

7.3.4. Let S be a stopping time, let $A \in \mathcal{F}_S$, and let $R = S$ on A and $R = \infty$ on A^c . Show that R is a stopping time.

7.3.5. Let S and T be stopping times. $\{S < T\}$, $\{S > T\}$, and $\{S = T\}$ are in $\mathcal{F}_S$ (and in $\mathcal{F}_T$).

7.3.6. Let S and T be stopping times. $\mathcal{F}_S \cap \mathcal{F}_T = \mathcal{F}_{S \wedge T}$.

7.4 Path Properties

In this section, we will use the strong Markov property to derive properties of the zero set $\{t : B_t = 0\}$, the hitting times $T_a = \inf\{t : B_t = a\}$, and $\max_{0 \leq s \leq t} B_s$ for one dimensional Brownian motion.

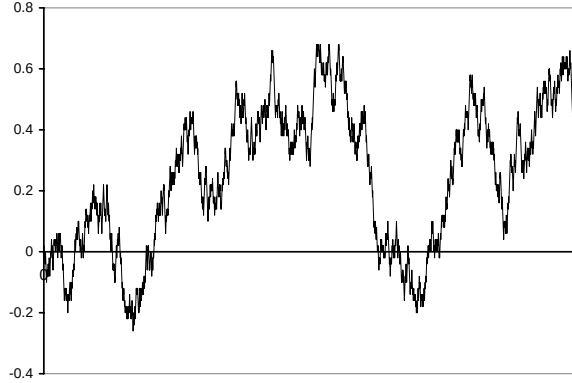


Figure 7.2: Simulation one-dimensional Brownian motion.

7.4.1 Zeros of Brownian Motion

Let $R_t = \inf\{u > t : B_u = 0\}$ and let $T_0 = \inf\{u > 0 : B_u = 0\}$. Now Theorem 7.2.9 implies $P_x(R_t < \infty) = 1$, so $B(R_t) = 0$ and the strong Markov property and Theorem 7.2.5 imply

$$P_x(T_0 \circ \theta_{R_t} > 0 | \mathcal{F}_{R_t}) = P_0(T_0 > 0) = 0$$

Taking expected value of the last equation, we see that

$$P_x(T_0 \circ \theta_{R_t} > 0 \text{ for some rational } t) = 0$$

From this, it follows that if a point $u \in \mathcal{Z}(\omega) \equiv \{t : B_t(\omega) = 0\}$ is isolated on the left (i.e., there is a rational $t < u$ so that $(t, u) \cap \mathcal{Z}(\omega) = \emptyset$), then it is, with probability one, a decreasing limit of points in $\mathcal{Z}(\omega)$. This shows that the closed set $\mathcal{Z}(\omega)$ has no isolated points and hence must be uncountable. For the last step, see Hewitt and Stromberg (1965), page 72.

If we let $|\mathcal{Z}(\omega)|$ denote the Lebesgue measure of $\mathcal{Z}(\omega)$ then Fubini's theorem implies

$$E_x(|\mathcal{Z}(\omega)| \cap [0, T]) = \int_0^T P_x(B_t = 0) dt = 0$$

So $\mathcal{Z}(\omega)$ is a set of measure zero.

The last four observations show that $\mathcal{Z}$ is like the Cantor set that is obtained by removing $(1/3, 2/3)$ from $[0, 1]$ and then repeatedly removing the middle third from the intervals that remain. The Cantor set is bigger however. Its Hausdorff dimension is $\log 2 / \log 3$, while $\mathcal{Z}$ has dimension $1/2$.

7.4.2 Hitting times

Theorem 7.4.1. *Under P_0 , $\{T_a, a \geq 0\}$ has stationary independent increments.*

Proof. The first step is to notice that if $0 < a < b$ then

$$T_b \circ \theta_{T_a} = T_b - T_a,$$

so if f is bounded and measurable, the strong Markov property, 7.3.9 and translation invariance imply

$$\begin{aligned} E_0(f(T_b - T_a) | \mathcal{F}_{T_a}) &= E_0(f(T_b) \circ \theta_{T_a} | \mathcal{F}_{T_a}) \\ &= E_a f(T_b) = E_0 f(T_{b-a}) \end{aligned}$$

To show that the increments are independent, let $a_0 < a_1 \dots < a_n$, let $f_i, 1 \leq i \leq n$ be bounded and measurable, and let $F_i = f_i(T_{a_i} - T_{a_{i-1}})$. Conditioning on $\mathcal{F}_{T_{a_{n-1}}}$ and using the preceding calculation we have

$$E_0 \left(\prod_{i=1}^n F_i \right) = E_0 \left(\prod_{i=1}^{n-1} F_i \cdot E_0(F_n | \mathcal{F}_{T_{a_{n-1}}}) \right) = E_0 \left(\prod_{i=1}^{n-1} F_i \right) E_0 F_n$$

By induction, it follows that $E_0 \prod_{i=1}^n F_i = \prod_{i=1}^n E_0 F_i$, which implies the desired conclusion. $\square$

The scaling relation (7.1.1) implies

$$T_a \stackrel{d}{=} a^2 T_1 \quad (7.4.1)$$

Combining Theorem 7.4.1 and (7.4.1), we see that $t_k = T_k - T_{k-1}$ are i.i.d. and

$$\frac{t_1 + \cdots + t_n}{n^2} \rightarrow T_1$$

so using Theorem 3.8.8, we see that T_a has a stable law. Since we are dividing by n^2 and $T_a \geq 0$, the index $\alpha = 1/2$ and the skewness parameter $\kappa = 1$, see (3.8.11).

Without knowing the theory mentioned in the previous paragraph, it is easy to determine the Laplace transform

$$\varphi_a(\lambda) = E_0 \exp(-\lambda T_a) \quad \text{for } a \geq 0$$

and reach the same conclusion. To do this, we start by observing that Theorem 7.4.1 implies

$$\varphi_x(\lambda) \varphi_y(\lambda) = \varphi_{x+y}(\lambda).$$

It follows easily from this that

$$\varphi_a(\lambda) = \exp(-ac(\lambda)) \quad (7.4.2)$$

Proof. Let $c(\lambda) = -\log \varphi_1(\lambda)$ so (7.4.2) holds when $a = 1$. Using the previous identity with $x = y = 2^{-m}$ and induction gives the result for $a = 2^{-m}$, $m \geq 1$. Then, letting $x = k2^{-m}$ and $y = 2^{-m}$ we get the result for $a = (k+1)2^{-m}$ with $k \geq 1$. Finally, to extend to $a \in [0, \infty)$, note that $a \rightarrow \phi_a(\lambda)$ is decreasing. $\square$

To identify $c(\lambda)$, we observe that (7.4.1) implies

$$E \exp(-T_a) = E \exp(-a^2 T_1)$$

so $ac(1) = c(a^2)$, i.e., $c(\lambda) = c(1)\sqrt{\lambda}$. Since all of our arguments also apply to σB_t we cannot hope to compute $c(1)$. Theorem 7.5.7 will show

$$E_0(\exp(-\lambda T_a)) = \exp(-a\sqrt{2\lambda}) \quad (7.4.3)$$

Our next goal is to compute the distribution of the hitting times T_a . This application of the strong Markov property shows why we want to allow the function Y that we apply to the shifted path to depend on the stopping time S .

Example 7.4.2. Reflection principle. Let $a > 0$ and let $T_a = \inf\{t : B_t = a\}$. Then

$$P_0(T_a < t) = 2P_0(B_t \geq a) \quad (7.4.4)$$

Intuitive proof. We observe that if B_s hits a at some time $s < t$, then the strong Markov property implies that $B_t - B(T_a)$ is independent of what happened before time T_a . The symmetry of the normal distribution and $P_a(B_u = a) = 0$ for $u > 0$ then imply

$$P_0(T_a < t, B_t > a) = \frac{1}{2}P_0(T_a < t) \quad (7.4.5)$$

Rearranging the last equation and using $\{B_t > a\} \subset \{T_a < t\}$ gives

$$P_0(T_a < t) = 2P_0(T_a < t, B_t > a) = 2P_0(B_t > a)$$

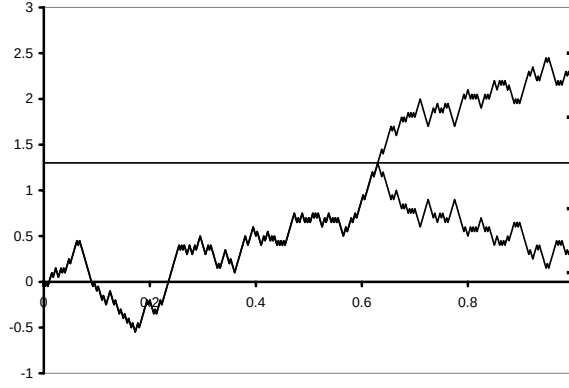


Figure 7.3: Proof by picture of the reflection principle.

Proof. To make the intuitive proof rigorous, we only have to prove (7.4.5). To extract this from the strong Markov property, Theorem 7.3.9, we let

$$Y_s(\omega) = \begin{cases} 1 & \text{if } s < t, \omega(t-s) > a \\ 0 & \text{otherwise} \end{cases}$$

We do this so that if we let $S = \inf\{s < t : B_s = a\}$ with $\inf \emptyset = \infty$, then

$$Y_S(\theta_S \omega) = \begin{cases} 1 & \text{if } S < t, B_t > a \\ 0 & \text{otherwise} \end{cases}$$

and the strong Markov property implies

$$E_0(Y_S \circ \theta_S | \mathcal{F}_S) = \varphi(B_S, S) \quad \text{on } \{S < \infty\} = \{T_a < t\}$$

where $\varphi(x, s) = E_x Y_s$. $B_S = a$ on $\{S < \infty\}$ and $\varphi(a, s) = 1/2$ if $s < t$, so taking expected values gives

$$\begin{aligned} P_0(T_a < t, B_t \geq a) &= E_0(Y_S \circ \theta_S; S < \infty) \\ &= E_0(E_0(Y_S \circ \theta_S | \mathcal{F}_S); S < \infty) = E_0(1/2; T_a < t) \end{aligned}$$

which proves (7.4.5). $\square$

Using (7.4.4), we can compute the probability density of T_a . We begin by noting that

$$P(T_a \leq t) = 2 P_0(B_t \geq a) = 2 \int_a^\infty (2\pi t)^{-1/2} \exp(-x^2/2t) dx$$

then change variables $x = (t^{1/2}a)/s^{1/2}$ to get

$$\begin{aligned} P_0(T_a \leq t) &= 2 \int_t^0 (2\pi t)^{-1/2} \exp(-a^2/2s) (-t^{1/2}a/2s^{3/2}) ds \\ &= \int_0^t (2\pi s^3)^{-1/2} a \exp(-a^2/2s) ds \end{aligned} \quad (7.4.6)$$

Using the last formula, we can compute:

Example 7.4.3. The distribution of $L = \sup\{t \leq 1 : B_t = 0\}$. By (7.2.4),

$$\begin{aligned} P_0(L \leq s) &= \int_{-\infty}^\infty p_s(0, x) P_x(T_0 > 1-s) dx \\ &= 2 \int_0^\infty (2\pi s)^{-1/2} \exp(-x^2/2s) \int_{1-s}^\infty (2\pi r^3)^{-1/2} x \exp(-x^2/2r) dr dx \\ &= \frac{1}{\pi} \int_{1-s}^\infty (sr^3)^{-1/2} \int_0^\infty x \exp(-x^2(r+s)/2rs) dx dr \\ &= \frac{1}{\pi} \int_{1-s}^\infty (sr^3)^{-1/2} rs/(r+s) dr \end{aligned}$$

Our next step is to let $t = s/(r+s)$ to convert the integral over $r \in [1-s, \infty)$ into one over $t \in [0, s]$. $dt = -s/(r+s)^2 dr$, so to make the calculations easier we first rewrite the integral as

$$= \frac{1}{\pi} \int_{1-s}^\infty \left(\frac{(r+s)^2}{rs} \right)^{1/2} \frac{s}{(r+s)^2} dr$$

and then change variables to get

$$P_0(L \leq s) = \frac{1}{\pi} \int_0^s (t(1-t))^{-1/2} dt = \frac{2}{\pi} \arcsin(\sqrt{s}) \quad (7.4.7)$$

Since L is the time of the last zero before time 1, it is remarkable that the density is symmetric about $1/2$ and blows up at 0. The arcsin may remind the reader of the limit theorem for $L_{2n} = \sup\{m \leq 2n : S_m = 0\}$ given in Theorem 4.9.5. We will see in Section 8.6 that our new result is a consequence of the old one.

EXERCISES

7.4.1. Let $B_t = (B_t^1, B_t^2)$ be a two-dimensional Brownian motion starting from 0. Let $T_a = \inf\{t : B_t^2 = a\}$. Show that (i) $B^1(T_a)$, $a \geq 0$ has independent increments and $B^1(T_a) =_d aB^1(T_1)$. Use this to conclude that $B^1(T_a)$ has a Cauchy distribution.

7.4.2. Use (7.2.3) to show that $R = \inf\{t > 1 : B_t = 0\}$ has probability density

$$P_0(R = 1 + t) = 1/(\pi t^{1/2}(1 + t))$$

7.4.3. (a) Generalize the proof of (7.4.5) to conclude that if $u < v \leq a$ then

$$P_0(T_a < t, u < B_t < v) = P_0(2a - v < B_t < 2a - u) \quad (7.4.8)$$

(b) Let (u, v) shrink down to x in (7.4.8) to conclude that for $a < x$

$$P_0(T_a < t, B_t = x) = p_t(0, 2a - x)$$

i.e., the (subprobability) density for B_t on $\{T_a < t\}$. (c) Use the fact that $\{T_a < t\} = \{M_t > a\}$ and differentiate with respect to a gives the joint density

$$f_{(M_t, B_t)}(a, x) = \frac{2(2a - x)}{\sqrt{2\pi t^3}} e^{-(2a-x)^2/2t}$$

7.4.4. Let $A_{s,t}$ be the event Brownian motion has at least one zero in $[s, t]$. Show that $P_0(A_{s,t}) = (2/\pi)\arccos(\sqrt{s/t})$. Hint: To get started note that

$$P_0(A_{s,t}) = 2 \int_0^\infty (2\pi s)^{-1/2} e^{-x^2/2} P_x(T_0 \leq t - s) dx$$

and use the formula for the density function of T_x . At the end you may need some trig to convert arctan into arccos.

Remark. Using the previous exercise it is easy to see that if $s \geq 1$ then $P(A_{s,s+h}) \leq Ch^{1/2}$. Thus if we divide the interval $[1, 2]$ into n equal pieces then expected number of intervals that contains zeros is $\leq Cn^{-1/2}$. It follows that the Hausdorff dimension of the zero set is at most $1/2$.

7.5 Martingales

At the end of Section 5.7 we used martingales to study the hitting times of random walks. The same methods can be used on Brownian motion once we prove:

Theorem 7.5.1. *Let X_t be a right continuous martingale adapted to a right continuous filtration. If T is a bounded stopping time, then $EX_T = EX_0$.*

Proof. Let n be an integer so that $P(T \leq n-1) = 1$. As in the proof of the strong Markov property, let $T_m = ([2^m T] + 1)/2^m$. $Y_k^m = X(k2^{-m})$ is a martingale with respect to $\mathcal{F}_k^m = \mathcal{F}(k2^{-m})$ and $S_m = 2^m T_m$ is a stopping time for $(Y_k^m, \mathcal{F}_k^m)$, so by Exercise 4.4.4

$$X(T_m) = Y_{S_m}^m = E(Y_{n2^m}^m | \mathcal{F}_{S_m}^m) = E(X_n | \mathcal{F}(T_m))$$

As $m \uparrow \infty$, $X(T_m) \rightarrow X(T)$ by right continuity and $\mathcal{F}(T_m) \downarrow \mathcal{F}(T)$ by Theorem 7.3.7, so it follows from Theorem 4.7.3 that

$$X(T) = E(X_n | \mathcal{F}(T))$$

Taking expected values gives $EX(T) = EX_n = EX_0$, since X_n is a martingale. $\square$

Theorem 7.5.2. B_t is a martingale w.r.t. the σ -fields $\mathcal{F}_t$ defined in Section 8.2.

Note: We will use these σ -fields in all of the martingale results but will not mention them explicitly in the statements.

Proof. The Markov property implies that

$$E_x(B_t | \mathcal{F}_s) = E_{B_s}(B_{t-s}) = B_s$$

since symmetry implies $E_y B_u = y$ for all $u \geq 0$. $\square$

From Theorem 7.5.2, it follows immediately that we have:

Theorem 7.5.3. If $a < x < b$ then $P_x(T_a < T_b) = (b-x)/(b-a)$.

Proof. Let $T = T_a \wedge T_b$. Theorem 7.2.8 implies that $T < \infty$ a.s. Using Theorems 7.5.1 and 7.5.2, it follows that $x = E_x B(T \wedge t)$. Letting $t \rightarrow \infty$ and using the bounded convergence theorem, it follows that

$$x = aP_x(T_a < T_b) + b(1 - P_x(T_a < T_b))$$

Solving for $P_x(T_a < T_b)$ now gives the desired result. $\square$

Theorem 7.5.4. $B_t^2 - t$ is a martingale.

Proof. Writing $B_t^2 = (B_s + B_t - B_s)^2$ we have

$$\begin{aligned} E_x(B_t^2 | \mathcal{F}_s) &= E_x(B_s^2 + 2B_s(B_t - B_s) + (B_t - B_s)^2 | \mathcal{F}_s) \\ &= B_s^2 + 2B_s E_x(B_t - B_s | \mathcal{F}_s) + E_x((B_t - B_s)^2 | \mathcal{F}_s) \\ &= B_s^2 + 0 + (t - s) \end{aligned}$$

since $B_t - B_s$ is independent of $\mathcal{F}_s$ and has mean 0 and variance $t - s$. $\square$

Theorem 7.5.5. Let $T = \inf\{t : B_t \notin (a, b)\}$, where $a < 0 < b$.

$$E_0 T = -ab$$

Proof Theorem 7.5.1 and 7.5.4 imply $E_0(B^2(T \wedge t)) = E_0(T \wedge t)$. Letting $t \rightarrow \infty$ and using the monotone convergence theorem gives $E_0(T \wedge t) \uparrow E_0T$. Using the bounded convergence theorem and Theorem 7.5.3, we have

$$E_0B^2(T \wedge t) \rightarrow E_0B_T^2 = a^2 \frac{b}{b-a} + b^2 \frac{-a}{b-a} = ab \frac{a-b}{b-a} = -ab \quad \square$$

Theorem 7.5.6. $\exp(\theta B_t - (\theta^2 t/2))$ is a martingale.

Proof. Bringing $\exp(\theta B_s)$ outside

$$\begin{aligned} E_x(\exp(\theta B_t) | \mathcal{F}_s) &= \exp(\theta B_s) E(\exp(\theta(B_t - B_s)) | \mathcal{F}_s) \\ &= \exp(\theta B_s) \exp(\theta^2(t-s)/2) \end{aligned}$$

since $B_t - B_s$ is independent of $\mathcal{F}_s$ and has a normal distribution with mean 0 and variance $t-s$. $\square$

Theorem 7.5.7. If $T_a = \inf\{t : B_t = a\}$ then $E_0 \exp(-\lambda T_a) = \exp(-a\sqrt{2\lambda})$.

Proof. Theorem 7.5.1 and 7.5.6 imply that $1 = E_0 \exp(\theta B(T \wedge t) - \theta^2(T \wedge t)/2)$. Taking $\theta = \sqrt{2\lambda}$, letting $t \rightarrow \infty$ and using the bounded convergence theorem gives $1 = E_0 \exp(a\sqrt{2\lambda} - \lambda T_a)$. $\square$

Theorem 7.5.8. If $u(t, x)$ is a polynomial in t and x with

$$\frac{\partial u}{\partial t} + \frac{1}{2} \frac{\partial^2 u}{\partial x^2} = 0 \quad (7.5.1)$$

then $u(t, B_t)$ is a martingale.

Proof. Let $p_t(x, y) = (2\pi)^{-1/2} t^{-1/2} \exp(-(y-x)^2/2t)$. The first step is to check that p_t satisfies the heat equation: $\partial p_t / \partial t = (1/2) \partial^2 p_t / \partial y^2$.

$$\begin{aligned} \frac{\partial p}{\partial t} &= -\frac{1}{2} t^{-1} p_t + \frac{(y-x)^2}{2t^2} p_t \\ \frac{\partial p}{\partial y} &= -\frac{y-x}{t} p_t \\ \frac{\partial^2 p}{\partial y^2} &= -\frac{1}{t} p_t + \frac{(y-x)^2}{t^2} p_t \end{aligned}$$

Interchanging $\partial/\partial t$ and $\int$, and using the heat equation

$$\begin{aligned} \frac{\partial}{\partial t} E_x u(t, B_t) &= \int \frac{\partial}{\partial t} (p_t(x, y) u(t, y)) dy \\ &= \int \frac{1}{2} \frac{\partial}{\partial y^2} p_t(x, y) u(t, y) + p_t(x, y) \frac{\partial}{\partial t} u(t, y) dy \end{aligned}$$

Integrating by parts twice the above

$$= \int p_t(x, y) \left(\frac{\partial}{\partial t} + \frac{1}{2} \frac{\partial}{\partial y^2} \right) u(t, y) dy = 0$$

Since $u(t, y)$ is a polynomial there is no question about the convergence of integrals and there is no contribution from the boundary terms when we integrate by parts.

At this point we have shown that $t \rightarrow E_x u(t, B_t)$ is constant. To check the martingale property we need to show that

$$E(u(t, B_t) | \mathcal{F}_s) = u(s, B_s)$$

To do this we note that $v(r, x) = u(s+r, x)$ satisfies $\partial v / \partial r = (1/2) \partial v^2 / \partial x^2$, so using the Markov property

$$\begin{aligned} E(u(t, B_t) | \mathcal{F}_s) &= E(v(t-s, B_{t-s}) \circ \theta_s | \mathcal{F}_s) \\ &= E_{B(s)} v(t-s, B_{t-s}) = v(0, B_s) = u(s, B_s) \end{aligned}$$

where in the next to last step we have used the previously proved constancy of the expected value. $\square$

Examples of polynomials that satisfy (7.5.1) are x , $x^2 - t$, $x^3 - 3tx$, $x^4 - 6x^2t + 3t^2 \dots$. The result can also be extended to $\exp(\theta x - \theta^2 t/2)$. The only place we used the fact that $u(t, x)$ was a polynomial was in checking the convergence of the integral to justify integration by parts.

Theorem 7.5.9. *If $T = \inf\{t : B_t \notin (-a, a)\}$ then $ET^2 = 5a^4/3$.*

Proof. Theorem 7.5.1 implies

$$E(B(T \wedge t)^4 - 6(T \wedge t)B(T \wedge t)^2) = -3E(T \wedge t)^2.$$

From Theorem 7.5.5, we know that $ET = a^2 < \infty$. Letting $t \rightarrow \infty$, using the dominated convergence theorem on the left-hand side, and the monotone convergence theorem on the right gives

$$a^4 - 6a^2 ET = -3E(T^2)$$

Plugging in $ET = a^2$ gives the desired result. $\square$

EXERCISES

7.5.1. Let $T = \inf\{B_t \notin (-a, a)\}$. Show that

$$E_0 \exp(-\lambda T) = 1 / \cosh(a\sqrt{2\lambda}).$$

7.5.2. The point of this exercise is to get information about the amount of time it takes Brownian motion with drift $-b$, $X_t \equiv B_t - bt$ to hit level a . Let $\tau = \inf\{t : B_t = a + bt\}$, where $a > 0$. (i) Use the martingale $\exp(\theta B_t - \theta^2 t/2)$ with $\theta = b + (b^2 + 2\lambda)^{1/2}$ to show

$$E_0 \exp(-\lambda \tau) = \exp(-a\{b + (b^2 + 2\lambda)^{1/2}\})$$

Letting $\lambda \rightarrow 0$ gives $P_0(\tau < \infty) = \exp(-2ab)$.

7.5.3. Let $\sigma = \inf\{t : B_t \notin (a, b)\}$ and let $\lambda > 0$. Use the strong Markov property to show

$$E_x \exp(-\lambda T_a) = E_x(e^{-\lambda \sigma}; T_a < T_b) + E_x(e^{-\lambda \sigma}; T_b < T_a) E_b \exp(-\lambda T_a)$$

(ii) Interchange the roles of a and b to get a second equation, use Theorem 7.5.7, and solve to get

$$\begin{aligned} E_x(e^{-\lambda \sigma}; T_a < T_b) &= \sinh(\sqrt{2\lambda}(b-x)) / \sinh(\sqrt{2\lambda}(b-a)) \\ E_x(e^{-\lambda \sigma}; T_b < T_a) &= \sinh(\sqrt{2\lambda}(x-a)) / \sinh(\sqrt{2\lambda}(b-a)) \end{aligned}$$

7.5.4. If $T = \inf\{t : B_t \notin (a, b)\}$, where $a < 0 < b$ and $a \neq -b$, then T and B_T^2 are not independent, so we cannot calculate ET^2 as we did in the proof of Theorem 7.5.9. Use the Cauchy-Schwarz inequality to estimate $E(TB_T^2)$ and conclude (i) $ET^2 \leq 4 E(B_T^4)$ and (ii) $EB_T^4 \leq 36ET^2$

7.5.5. Find a martingale of the form $B_t^6 - c_1 t B_t^4 + c_2 t^2 B_t^2 - c_3 t^3$ and use it to compute the third moment of $T = \inf\{t : B_t \notin (-a, a)\}$.

7.5.6. Show that $(1+t)^{-1/2} \exp(B_t^2/2(1+t))$ is a martingale and use this to conclude that $\limsup_{t \rightarrow \infty} B_t / ((1+t) \log(1+t))^{1/2} \leq 1$ a.s.

7.6 Itô's formula*

Our goal in this section is to prove three versions of Itô's formula. The first one, for one dimensional Brownian motion is

Theorem 7.6.1. *Suppose that $f \in C^2$, i.e., it has two continuous derivatives. Then with probability one, for all $t \geq 0$,*

$$f(B_t) - f(B_0) = \int_0^t f'(B_s) dB_s + \frac{1}{2} \int_0^t f''(B_s) ds \quad (7.6.1)$$

Of course, as part of the derivation we have to explain the meaning of the first term on the right-hand side. In contrast if A_s is continuous and has bounded variation on each finite interval, and $f \in C^1$:

$$f(A_t) - f(A_0) = \int_0^t f'(A_s) dA_s$$

where the Riemann-Stieltjes integral on the right-hand side is

$$\lim_{n \rightarrow \infty} \sum_{i=1}^{k(n)} f'(A_{t_{i-1}^n}) [A(t_i^n) - A(t_{i-1}^n)]$$

over a sequence of partitions $0 = t_0^n < t_1^n < \dots < t_{k(n)}^n = t$ with mesh $\max_i t_i^n - t_{i-1}^n \rightarrow 0$.

Proof. To derive (7.6.1) we let $t_i^n = ti/2^n$ for $0 \leq i \leq k(n) = 2^n$. From calculus we know that for any a and b , there is a $c(a, b)$ in between a and b such that

$$f(b) - f(a) = (b - a)f'(a) + \frac{1}{2}(b - a)^2 f''(c(a, b)) \quad (7.6.2)$$

Using the calculus fact, it follows that

$$\begin{aligned} f(B_t) - f(B_0) &= \sum_i f(B_{t_{i+1}^n}) - f(B_{t_i^n}) \\ &= \sum_i f'(B_{t_i^n})(B_{t_{i+1}^n} - B_{t_i^n}) \\ &\quad + \frac{1}{2} \sum_i g_i^n(\omega)(B_{t_{i+1}^n} - B_{t_i^n})^2 \end{aligned} \quad (7.6.3)$$

where $g_i^n(\omega) = f''(c(B_{t_i^n}, B_{t_{i+1}^n}))$. Comparing (7.6.3) with (7.6.1), it becomes clear that we want to show

$$\sum_i f'(B_{t_i^n})(B_{t_{i+1}^n} - B_{t_i^n}) \rightarrow \int_0^t f'(B_s) dB_s \quad (7.6.4)$$

$$\frac{1}{2} \sum_i g_i(\omega)(B_{t_{i+1}^n} - B_{t_i^n})^2 \rightarrow \frac{1}{2} \int_0^t f''(B_s) ds \quad (7.6.5)$$

in probability as $n \rightarrow \infty$.

We will prove the result first under the assumption that $|f'(x)|, |f''(x)| \leq K$. Let

$$I_m^1 \equiv \sum_i f'(B_{t_i^m})(B_{t_{i+1}^m} - B_{t_i^m})$$

In order to compare I_m^1 and I_n^1 with $n \geq m$, we want to further subdivide the intervals of length $t2^{-m}$ to be of length $t2^{-n}$. To leave the sum unchanged we have to redefine things. Let $i(n, m, j) = [2^m j / 2^n] / 2^m$ and $H_j^m = f'(B_{i(n, m, j)})$. If we let $K_j^n = f'(B_{t_j^n})$ then we have

$$I_m^1 - I_n^1 = \sum_j (H_j^m - K_j^n)(B_{t_{j+1}^n} - B_{t_j^n})$$

Squaring we have

$$\begin{aligned} (I_m^1 - I_n^1)^2 &= \sum_j (H_j^m - K_j^n)^2 (B_{t_{j+1}^n} - B_{t_j^n})^2 \\ &\quad + 2 \sum_{j < k} (H_j^m - K_j^n)(H_k^m - K_k^n)(B_{t_{j+1}^n} - B_{t_j^n})(B_{t_{k+1}^n} - B_{t_k^n}) \end{aligned}$$

The next step is to take expected value. The expected value of the second sum vanishes because if we take expected value of the j, k term with respect to $\mathcal{F}_{t_k^n}$ the first three terms can be taken outside the expected value

$$(H_j^m - K_j^n)(H_k^m - K_k^n)(B_{t_{j+1}^n} - B_{t_j^n})E(B_{t_{k+1}^n} - B_{t_k^n} | \mathcal{F}_{t_k^n}) = 0$$

Conditioning on $\mathcal{F}_{t_j^n}$ in the first sum we have

$$E(I_m^1 - I_n^1)^2 = \sum_j E(H_j^m - K_j^n)^2 \cdot 2^{-n}t \quad (7.6.6)$$

Since $|i(n, m, j)/2^m - j/2^n| \leq 1/2^m$ and $|f''(x)| \leq K$, we have

$$\sup_{j \leq 2^n} |H_j^m - K_j^n| \leq K \sup\{|B_s - B_r| : 0 \leq r \leq s \leq t, |s - r| \leq 2^{-m}\}$$

Since Brownian motion is uniformly continuous on $[0, t]$ and $E \sup_{s \leq t} |B_s|^2 < \infty$ it follows from the dominated convergence theorem that

$$\lim_{m, n \rightarrow \infty} E(I_m^1 - I_n^1)^2 = 0$$

This shows that I_m^1 is a Cauchy sequence in L^2 so there is an I_∞^1 so that $I_m^1 \rightarrow I_\infty^1$ in L^2 . The limit defines $\int_0^t f'(B_s) dB_s$ as a limit of approximating Riemann sums.

To prove (7.6.5), we let $G_s^n = g_i^n(\omega) = f''(c(B_{t_i^n}, B_{t_{i+1}^n}))$ when $s \in (t_i^n, t_{i+1}^n]$ and let

$$A_s^n = \sum_{t_{i+1}^n \leq s} (B_{t_{i+1}^n} - B_{t_i^n})^2$$

so that

$$\sum_i g_i^n(\omega)(B_{t_{i+1}^n} - B_{t_i^n})^2 = \int_0^t G_s^n dA_s^n$$

and what we want to show is

$$\int_0^t G_s^n dA_s^n \rightarrow \int_0^t f''(B_s) ds \quad (7.6.7)$$

To do this we begin by observing that continuity of f'' implies that if $s_n \rightarrow s$ as $n \rightarrow \infty$ then $G_{s_n}^n \rightarrow f''(B_s)$, while Exercise 7.1.6 implies that A_s^n converges almost surely to s . At this point, we can fix ω and deduce (7.6.7) from the following simple result:

Lemma 7.6.2. *If (i) measures μ_n on $[0, t]$ converge weakly to μ_∞ , a finite measure, and (ii) g_n is a sequence of functions with $|g_n| \leq K$ that have the property that whenever $s_n \in [0, t] \rightarrow s$ we have $g_n(s_n) \rightarrow g(s)$, then as $n \rightarrow \infty$*

$$\int g_n d\mu_n \rightarrow \int g d\mu_\infty$$

Proof. By letting $\mu'_n(A) = \mu_n(A)/\mu_n([0, t])$, we can assume that all the μ_n are probability measures. A standard construction (see Theorem 3.2.8) shows that there is a sequence of random variables X_n with distribution μ_n so that $X_n \rightarrow X_\infty$ a.s. as $n \rightarrow \infty$. The convergence of g_n to g implies $g_n(X_n) \rightarrow g(X_\infty)$, so the result follows from the bounded convergence theorem. $\square$

Lemma 7.6.2 is the last piece in the proof of Theorem under the additional assumptions that $|f'(x)|, |f''(x)| \leq M$. Tracing back through the proof, we see that Lemma 7.6.2 implies (7.6.7), which in turn completes the proof of (7.6.5). So adding (7.6.4) and using (7.6.3) gives that

$$f(B_t) - f(B_0) = \int_0^t f'(B_s) dB_s + \frac{1}{2} \int_0^t f''(B_s) ds \quad \text{a.s.}$$

To remove the assumed bounds on the derivatives, we let M be large and define f_M with $f_M = f$ on $[-M, M]$ and $|f'_M|, |f''_M| \leq M$. The formula holds for f_M but on $\{\sup_{s \leq t} |B_s| \leq M\}$ there is no difference in the formula between using f_M and f . $\square$

Example 7.6.3. Taking $f = x^2$ in (7.6.1) we have

$$B_t^2 - B_0^2 = \int_0^t 2B_s dB_s + t$$

so if $B_0 = 0$, we have

$$\int_0^t 2B_s dB_s = B_t^2 - t$$

In contrast to the calculus formula $\int_0^a 2x dx = a^2$.

Direct proof. Using the subdivision we introduced previous we can write

$$\begin{aligned} B_t^2 &= \left(\sum_{i=0}^{2^n-1} (B(t_{i+1}^n) - B(t_i^n)) \right)^2 \\ &\quad + \sum_{i=0}^{2^n-1} \sum_{j=0}^{2^n-1} (B(t_{i+1}^n) - B(t_i^n))(B(t_{j+1}^n) - B(t_j^n)) \end{aligned}$$

If we sum over $j < i$ this becomes

$$\sum_{i=0}^{2^n-1} B(t^n - i)(B(t_{i+1}^n) - B(t_i^n)) \rightarrow \int_0^t B_s dB_s$$

We get the same limit for the sum over $i < j$. When we sum over $i = j$

$$\sum_{i=0}^{2^n-1} (B(t_{i+1}^n) - B(t_i^n))^2 \rightarrow t$$

To give one reason why the answer cannot be $B_t^2/2$, we note that

Theorem 7.6.4. *If g is continuous and $E \int_0^t |g(B_s)|^2 ds < \infty$ then $\int_0^t g(B_s) dB_s$ is a continuous martingale.*

If g grows too fast at ∞ then the integrability condition might not hold. To deal with this we introduce a weaker concept: M_t is a local martingale if there is an increasing sequence of stopping times $T_n \uparrow \infty$ so that $M(t \wedge T_n)$ is a martingale. If g is continuous then the stochastic integral is always a local martingale.

Proof. We first prove the result assuming $|g| \leq K$. Let $t_i^n = ti/2^n$ for $0 \leq i \leq 2^n$, let $J = \max\{i : t_{i+1}^n \leq s\}$ and let

$$I_n^1(s) = \sum_{i=0}^J g(B_{t_i^n})(B_{t_{i+1}^n} - B_{t_i^n}) + g(B_{t_{J+1}^n})(B_s - B_{t_{J+1}^n})$$

It is easy to check that $I_n^1(s)$, $s \leq t$ is a continuous martingale with

$$EI_n^1(t)^2 = \sum_{0 \leq i < 2^n} E[g(B_{t_i^n})^2] \cdot t2^{-n}.$$

The L^2 maximal inequality implies that

$$E \left(\sup_{s \leq t} |I_m^1(s) - I_n^1(s)|^2 \right) \leq 4E(I_m^1 - I_n^1)^2$$

The right-hand side $\rightarrow 0$ by the previous proof, so $I_m^1(s) \rightarrow I_\infty^1(s)$ uniformly on $[0, t]$. Let $Y \in \mathcal{F}_r$ with $EY^2 < \infty$. Since $I_n^1(s)$ is a martingale, if $r < s < t$ we have

$$E[Y\{I_n^1(s) - I_n^1(r)\}] = 0$$

From this and the Cauchy-Schwarz inequality we conclude that

$$E[Y\{I_\infty^1(s) - I_\infty^1(r)\}] = 0$$

Since this holds for $Y = 1_A$ with $A \in \mathcal{F}_r$, we have proved the desired result under the assumption $|g| \leq K$.

Approximating g by $g_M = g$ on $[-M, M]$ and g_M bounded as before and stopping at time $T_M = \inf\{t : |B_t| \geq M\}$ we can conclude that if g is continuous then

$$\int_0^{s \wedge T_M} g(B_r) dB_r$$

is a continuous martingale with

$$E \left(\int_0^{s \wedge T_M} f'(B_r) dB_r \right)^2 = E \int_0^{s \wedge T_M} f'(B_r)^2 dr$$

Letting $M \rightarrow \infty$ it follows that if $E \int_0^t |f'(B_s)|^2 ds < \infty$ then $\int_0^t f'(B_s) dB_s$ is a continuous martingale. $\square$

Our second version of Itô's formula allows f to depend on t as well

Theorem 7.6.5. *Suppose $f \in C^2$, i.e., it has continuous partial derivatives up to order two. Then with probability 1, for all $t \geq 0$*

$$\begin{aligned} f(t, B_t) - f(0, B_0) &= \int_0^t \frac{\partial f}{\partial t}(s, B_s) ds + \int_0^t \frac{\partial f}{\partial x}(s, B_s) dB_s \\ &\quad + \frac{1}{2} \int_0^t \frac{\partial^2 f}{\partial x^2}(s, B_s) ds \end{aligned} \quad (7.6.8)$$

Proof. Perhaps the simplest way to give a completely rigorous proof of this result (and the next one) is to prove the result when f is a polynomial in t and x and then note that for any $f \in C^2$ and $M < \infty$, we can find polynomials ϕ_n so that ϕ_n and all of its partial derivatives up to order two converge to those of f uniformly on $[0, t] \times [-M, M]$. Here, we will content ourselves to explain why this result is true.

Expanding to second order and being a little less precise about the error term

$$\begin{aligned} f(t, b) - f(s, a) &\approx \frac{\partial f}{\partial t}(s, a)(t - s) + \frac{\partial f}{\partial x}(s, a)(b - a) \\ &\quad + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(s, a)(b - a)^2 + \frac{\partial^2 f}{\partial t \partial x}(s, a)(t - s)(b - a) + \frac{1}{2} \frac{\partial^2 f}{\partial t^2}(s, a)(t - s)^2 \end{aligned}$$

Let t be a fixed positive number and let $t_k^n = tk/2^n$. As before, we first prove the result under the assumption that all of the first and second order partial derivatives are bounded by K . The details of the convergence of the first three terms to their limits is similar to the previous proof. To see that the fifth term vanishes in the limit, use the fact that $t_{i+1}^n - t_i^n = 2^{-n}$ to conclude that

$$\sum_i \left| \frac{\partial^2 f}{\partial x^2}(t_i^n, B(t_i^n)) \right| (t_{i+1}^n - t_i^n)^2 \leq K t 2^{-n} \rightarrow 0$$

To treat the fourth term, we first consider

$$V_n^{1,1}(t) = \sum_i (t_{i+1}^n - t_i^n) |B(t_{i+1}^n) - B(t_i^n)|$$

$EV_n^{1,1}(t) = \sum_i C_1 (t_{i+1}^n - t_i^n)^{3/2}$ where $C_1 = E\chi^{1/2}$ and χ is a standard normal. Arguing as before,

$$E \sum_i \left| \frac{\partial^2 f}{\partial x^2}(t_i^n, B(t_i^n)) \right| (t_{i+1}^n - t_i^n) |B(t_{i+1}^n) - B(t_i^n)| \leq KC_1 t \cdot 2^{-n/2} \rightarrow 0$$

Using Markov's inequality now we see that if $\epsilon > 0$

$$P \left(\sum_i \left| \frac{\partial^2 f}{\partial x^2}(t_i^n, B(t_i^n)) \right| (t_{i+1}^n - t_i^n) |B(t_{i+1}^n) - B(t_i^n)| > \epsilon \right) \leq KC_1 t 2^{-n/2} / \epsilon \rightarrow 0$$

and hence the sum converges to 0 in probability.

To remove the assumption that the partial derivatives are bounded, let $T_M = \min\{t : |B_t| > M\}$ and use the previous calculation to conclude that the formula holds up to time $T_M \wedge t$. $\square$

Example 7.6.6. Using (7.6.8) we can prove Theorem 7.5.8: if $u(t, x)$ is a polynomial in t and x with $\partial u / \partial t + (1/2) \partial^2 u / \partial x^2 = 0$ then Itô's formula implies

$$u(t, B_t) - u(0, B_0) = \int_0^t \frac{\partial u}{\partial x}(s, B_s) dB_s$$

Since $\partial u / \partial x$ is a polynomial it satisfies the integrability condition in Theorem 7.6.4, $u(t, B_t)$ is a martingale. To get a new conclusion, let $u(t, x) = \exp(\mu t + \sigma x)$. Exponential Brownian motion $X_t = u(t, B_t)$ is often used as a model of the price of a stock. Since

$$\frac{\partial u}{\partial t} = \mu u \quad \frac{\partial u}{\partial x} = \sigma u \quad \frac{\partial^2 u}{\partial x^2} = \sigma^2 u$$

Itô's formula gives

$$X_t - X_0 = \int_0^t \left(\mu - \frac{\sigma^2}{2} \right) X_s ds + \int_0^t \sigma X_s dB_s$$

From this we see that if $\mu = \sigma^2/2$ then X_t is a martingale, but this formula gives more information about how X_t behaves.

Our third and final version of Itô's formula is for multi-dimensional Brownian motion. To simplify notation we write $D_i f = \partial f / \partial x_i$ and $D_{ij} = \partial^2 f / \partial x_i \partial x_j$

Theorem 7.6.7. Suppose $f \in C^2$. Then with probability 1, for all $t \geq 0$

$$\begin{aligned} f(t, B_t) - f(0, B_0) &= \int_0^t \frac{\partial f}{\partial t}(s, B_s) ds + \sum_{i=1}^d \int_0^t D_i f(B_s) dB_s^i \\ &\quad + \frac{1}{2} \sum_{i=1}^d \int_0^t D_{ii} f(B_s) ds \end{aligned} \quad (7.6.9)$$

Proof. Let $x_0 = t$ and write $D_0 f = \partial f / \partial x_0$. Expanding to second order and again not worrying about the error term

$$\begin{aligned} f(y) - f(x) &\approx \sum_i D_i f(x)(y_i - x_i) \\ &\quad + \frac{1}{2} \sum_i D_{ii} f(x)(y_i - x_i)^2 + \sum_{i < j} D_{ij} f(x)(y_i - x_i)(y_j - x_j) \end{aligned}$$

Let t be a fixed positive number and let $t_i^n = ti/2^n$. The details of the convergence of the first two terms to their limits are similar to the previous two proofs. To handle the third term, we let

$$F_{i,j}(t_k^n) = D_{ij} f(B(t_k^n))$$

and note that independence of coordinates implies

$$\begin{aligned} E(F_{i,j}(t_k^n)(B^i(t_{k+1}^n) - B^i(t_k^n)) \cdot (B^j(t_{k+1}^n) - B^j(t_k^n)) | \mathcal{F}_{t_k^n}) \\ = F_{i,j}(t_k^n) E((B^i(t_{k+1}^n) - B^i(t_k^n)) \cdot (B^j(t_{k+1}^n) - B^j(t_k^n)) | \mathcal{F}_{t_k^n}) = 0 \end{aligned}$$

Fix $i < j$ and let $G_k = F_{i,j}(t_k^n)(B^i(t_{k+1}^n) - B^i(t_k^n)) \cdot (B^j(t_{k+1}^n) - B^j(t_k^n))$. The last calculation shows that

$$E \left(\sum_k G_k \right) = 0$$

To compute the variance of the sum we note that

$$E \left(\sum_k G_k \right)^2 = E \sum_k G_k^2 + 2E \sum_{k < \ell} G_k G_\ell$$

Again we first complete the proof under the assumption that $|\partial^2 f / \partial x_i \partial x_j| \leq C$. In this case, conditioning on $\mathcal{F}_{t_k^n}$ in order to bring $F_{i,j}^2(t_k^n)$ outside the first term,

$$\begin{aligned} E(F_{i,j}^2(t_k^n)(B^i(t_{k+1}^n) - B^i(t_k^n))^2 \cdot (B^j(t_{k+1}^n) - B^j(t_k^n))^2 | \mathcal{F}_{t_k^n}) \\ = F_{i,j}^2(t_k^n) E((B^i(t_{k+1}^n) - B^i(t_k^n))^2 \cdot (B^j(t_{k+1}^n) - B^j(t_k^n))^2 | \mathcal{F}_{t_k^n}) = F_{i,j}^2(t_k^n)(t_{k+1}^n - t_k^n)^2 \end{aligned}$$

and it follows that

$$E \sum_k G_k^2 \leq K^2 \sum_k (t_{k+1}^n - t_k^n)^2 \leq K^2 t 2^{-n} \rightarrow 0$$

To handle the cross terms, we condition on $\mathcal{F}_{t_\ell^n}$ to get

$$E(G_k G_\ell | \mathcal{F}_{t_k^n}) = G_k F_{i,j}(t_\ell^n) E((B^i(t_{\ell+1}^n) - B^i(t_\ell^n))(B^j(t_{\ell+1}^n) - B^j(t_\ell^n)) | \mathcal{F}_{t_\ell^n}) = 0$$

At this point we have shown $E(\sum_k G_k)^2 \rightarrow 0$ and the desired result follows from Markov's inequality. The assumption that the second derivatives are bounded can be removed as before. $\square$

EXERCISES

7.6.1. Suppose $f \in C^2$. Show that $f(B_t)$ is a martingale if and only if $f(x) = a + bx$.

7.6.2. Show that $B_t^3 - 3tB_t$ and $B_t^3 - \int_0^t 3B_s ds$ are a martingale.

7.6.3. Let $\beta_{2k}(t) = E_0 B_t^{2k}$. Use Itô's formula to relate $\beta_{2k}(t)$ to $\beta_{2k-2}(t)$ and use this relationship to derive a formula for $\beta_{2k}(t)$

7.6.4. Apply Itô's formula to $|B_t|$.

7.6.5. Apply Itô's formula to $|B_t|^2$. Use this to conclude that $E_0 |B_t|^2 = td$.

Chapter 8

Applications to Random Walk

The key to the developments in the section is that we can embed sums of mean zero independent random variables (see Section 8.1) and martingales (see Section 8.2) into Brownian motion and derive central limit theorems. These results lead in Section 8.3 to CLTs for stationary sequences and in Section 8.4 to the convergence of rescaled empirical distributions to Brownian Bridge. Finally in Section 8.5 we use the embedding to prove a law of the iterated logarithm.

8.1 Donsker's Theorem

In this section we will prove Donsker's (1951) theorem. The key to its proof is:

Theorem 8.1.1. Skorokhod's representation theorem. *If $EX = 0$ and $EX^2 < \infty$ then there is a stopping time T for Brownian motion so that $B_T =_d X$ and $ET = EX^2$.*

Remark. The Brownian motion in the statement and all the Brownian motions in this section have $B_0 = 0$.

Proof. Suppose first that X is supported on $\{a, b\}$, where $a < 0 < b$. Since $EX = 0$, we must have

$$P(X = a) = \frac{b}{b - a} \quad P(X = b) = \frac{-a}{b - a}$$

If we let $T = T_{a,b} = \inf\{t : B_t \notin (a, b)\}$ then Theorem 7.5.3 implies $B_T =_d X$ and Theorem 7.5.5 tells us that

$$ET = -ab = EB_T^2$$

To treat the general case, we will write $F(x) = P(X \leq x)$ as a mixture of two point distributions with mean 0. Let

$$c = \int_{-\infty}^0 (-u) dF(u) = \int_0^{\infty} v dF(v)$$

If φ is bounded and $\varphi(0) = 0$, then using the two formulas for c

$$\begin{aligned} c \int \varphi(x) dF(x) &= \left(\int_0^\infty \varphi(v) dF(v) \right) \int_{-\infty}^0 (-u) dF(u) \\ &\quad + \left(\int_{-\infty}^0 \varphi(u) dF(u) \right) \int_0^\infty v dF(v) \\ &= \int_0^\infty dF(v) \int_{-\infty}^0 dF(u) (v\varphi(u) - u\varphi(v)) \end{aligned}$$

So we have

$$\int \varphi(x) dF(x) = c^{-1} \int_0^\infty dF(v) \int_{-\infty}^0 dF(u) (v-u) \left\{ \frac{v}{v-u} \varphi(u) + \frac{-u}{v-u} \varphi(v) \right\}$$

The last equation gives the desired mixture. If we let $(U, V) \in \mathbf{R}^2$ have

$$\begin{aligned} P\{(U, V) = (0, 0)\} &= F(\{0\}) \\ P((U, V) \in A) &= c^{-1} \iint_{(u,v) \in A} dF(u) dF(v) (v-u) \end{aligned} \quad (8.1.1)$$

for $A \subset (-\infty, 0) \times (0, \infty)$ and define probability measures by $\mu_{0,0}(\{0\}) = 1$ and

$$\mu_{u,v}(\{u\}) = \frac{v}{v-u} \quad \mu_{u,v}(\{v\}) = \frac{-u}{v-u} \quad \text{for } u < 0 < v$$

then

$$\int \varphi(x) dF(x) = E \left(\int \varphi(x) \mu_{U,V}(dx) \right)$$

We proved the last formula when $\varphi(0) = 0$, but it is easy to see that it is true in general. Letting $\varphi \equiv 1$ in the last equation shows that the measure defined in (8.1.1) has total mass 1.

From the calculations above it follows that if we have (U, V) with distribution given in (8.1.1) and an independent Brownian motion defined on the same space then $B(T_{U,V}) =_d X$. Sticklers for detail will notice that $T_{U,V}$ is not a stopping time for B_t since (U, V) is independent of the Brownian motion. This is not a serious problem since if we condition on $U = u$ and $V = v$, then $T_{u,v}$ is a stopping time, and this is good enough for all the calculations below. For instance, to compute $E(T_{U,V})$ we observe

$$E(T_{U,V}) = E\{E(T_{U,V}|(U, V))\} = E(-UV)$$

by Theorem 7.5.5. (8.1.1) implies

$$\begin{aligned} E(-UV) &= \int_{-\infty}^0 dF(u) (-u) \int_0^\infty dF(v) v (v-u) c^{-1} \\ &= \int_{-\infty}^0 dF(u) (-u) \left\{ -u + \int_0^\infty dF(v) c^{-1} v^2 \right\} \end{aligned}$$

since

$$c = \int_0^\infty v dF(v) = \int_{-\infty}^0 (-u) dF(u)$$

Using the second expression for c now gives

$$E(T_{U,V}) = E(-UV) = \int_{-\infty}^0 u^2 dF(u) + \int_0^\infty v^2 dF(v) = EX^2 \quad \square$$

From Theorem 8.1.1, it is only a small step to:

Theorem 8.1.2. *Let $X_1, X_2, \dots$ be i.i.d. with a distribution F , which has mean 0 and variance 1, and let $S_n = X_1 + \dots + X_n$. There is a sequence of stopping times $T_0 = 0, T_1, T_2, \dots$ such that $S_n =_d B(T_n)$ and $T_n - T_{n-1}$ are independent and identically distributed.*

Proof. Let $(U_1, V_1), (U_2, V_2), \dots$ be i.i.d. and have distribution given in (8.1.1) and let B_t be an independent Brownian motion. Let $T_0 = 0$, and for $n \geq 1$, let

$$T_n = \inf\{t \geq T_{n-1} : B_t - B(T_{n-1}) \notin (U_n, V_n)\} \quad \square$$

As a corollary of Theorem 8.1.2, we get:

Theorem 8.1.3. Central limit theorem. *Under the hypotheses of Theorem 8.1.2, $S_n/\sqrt{n} \Rightarrow \chi$, where χ has the standard normal distribution.*

Proof. If we let $W_n(t) = B(nt)/\sqrt{n} =_d B_t$ by Brownian scaling, then

$$S_n/\sqrt{n} \stackrel{d}{=} B(T_n)/\sqrt{n} = W_n(T_n/n)$$

The weak law of large numbers implies that $T_n/n \rightarrow 1$ in probability. It should be clear from this that $S_n/\sqrt{n} \Rightarrow B_1$. To fill in the details, let $\epsilon > 0$, pick δ so that

$$P(|B_t - B_1| > \epsilon \text{ for some } t \in (1 - \delta, 1 + \delta)) < \epsilon/2$$

then pick N large enough so that for $n \geq N$, $P(|T_n/n - 1| > \delta) < \epsilon/2$. The last two estimates imply that for $n \geq N$

$$P(|W_n(T_n/n) - W_n(1)| > \epsilon) < \epsilon$$

Since ϵ is arbitrary, it follows that $W_n(T_n/n) - W_n(1) \rightarrow 0$ in probability. Applying the converging together lemma, Lemma 3.2.13, with $X_n = W_n(1)$ and $Z_n = W_n(T_n/n)$, the desired result follows. $\square$

Let $\mathbf{N}$ be the nonnegative integers and let

$$S(u) = \begin{cases} S_k & \text{if } u = k \in \mathbf{N} \\ \text{linear on } [k, k+1] & \text{for } k \in \mathbf{N} \end{cases}$$

Later in this section, we will prove:

Theorem 8.1.4. Donsker's theorem. *Under the hypotheses of Theorem 8.1.2,*

$$S(n\cdot)/\sqrt{n} \Rightarrow B(\cdot),$$

i.e., the associated measures on $C[0, 1]$ converge weakly.

To motivate ourselves for the proof we will begin by extracting several corollaries. The key to each one is a consequence of the following result which follows from Theorem 3.10.1.

Theorem 8.1.5. *If $\psi : C[0, 1] \rightarrow \mathbf{R}$ has the property that it is continuous P_0 -a.s. then*

$$\psi(S(n\cdot)/\sqrt{n}) \Rightarrow \psi(B(\cdot))$$

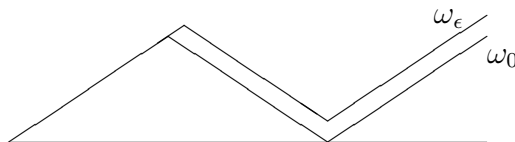
Example 8.1.6. Maxima. Let $\psi(\omega) = \max\{\omega(t) : 0 \leq t \leq 1\}$. Again, $\psi : C[0, 1] \rightarrow \mathbf{R}$ is continuous. This time Theorem 8.1.5 implies

$$\max_{0 \leq m \leq n} S_m/\sqrt{n} \Rightarrow M_1 \equiv \max_{0 \leq t \leq 1} B_t$$

To complete the picture, we observe that by (7.4.4) the distribution of the right-hand side is

$$P_0(M_1 \geq a) = P_0(T_a \leq 1) = 2P_0(B_1 \geq a)$$

Example 8.1.7. Last 0 before time n . Let $\psi(\omega) = \sup\{t \leq 1 : \omega(t) = 0\}$. This time, ψ is not continuous, for if ω_ϵ with $\omega_\epsilon(0) = 0$ is piecewise linear with slope 1 on $[0, 1/3 + \epsilon]$, -1 on $[1/3 + \epsilon, 2/3]$, and slope 1 on $[2/3, 1]$, then $\psi(\omega_0) = 2/3$ but $\psi(\omega_\epsilon) = 0$ for $\epsilon > 0$.



It is easy to see that if $\psi(\omega) < 1$ and $\omega(t)$ has positive and negative values in each interval $(\psi(\omega) - \delta, \psi(\omega) + \delta)$, then ψ is continuous at ω . By arguments in Subsection 8.4.1, the last set has P_0 measure 1. (If the zero at $\psi(\omega)$ was isolated on the left, it would not be isolated on the right.) It follows that

$$\sup\{m \leq n : S_{m-1} \cdot S_m \leq 0\}/n \Rightarrow L = \sup\{t \leq 1 : B_t = 0\}$$

The distribution of L is given in (7.4.7). The last result shows that the arcsine law, Theorem 4.9.5, proved for simple random walks holds when the mean is 0 and variance is finite.

Example 8.1.8. Occupation times of half-lines. Let

$$\psi(\omega) = |\{t \in [0, 1] : \omega(t) > a\}|.$$

The point $\omega \equiv a$ shows that ψ is not continuous, but it is easy to see that ψ is continuous at paths ω with $|\{t \in [0, 1] : \omega(t) = a\}| = 0$. Fubini's theorem implies that

$$E_0|\{t \in [0, 1] : B_t = a\}| = \int_0^1 P_0(B_t = a) dt = 0$$

so ψ is continuous P_0 -a.s. Using Theorem 8.1.5, we now get that

$$|\{u \leq n : S(u) > a\sqrt{n}\}|/n \Rightarrow |\{t \in [0, 1] : B_t > a\}|$$

As we will now show, with a little work, one can convert this into the more natural result

$$|\{m \leq n : S_m > a\sqrt{n}\}|/n \Rightarrow |\{t \in [0, 1] : B_t > a\}|$$

Proof. Application of Theorem 8.1.5 gives that for any a ,

$$|\{t \in [0, 1] : S(nt) > a\sqrt{n}\}| \Rightarrow |\{t \in [0, 1] : B_t > a\}|$$

To convert this into a result about $|\{m \leq n : S_m > a\sqrt{n}\}|$, we note that on $\{\max_{m \leq n} |X_m| \leq \epsilon\sqrt{n}\}$, which by Chebyshev's inequality has a probability $\rightarrow 1$, we have

$$\begin{aligned} |\{t \in [0, 1] : S(nt) > (a + \epsilon)\sqrt{n}\}| &\leq \frac{1}{n} |\{m \leq n : S_m > a\sqrt{n}\}| \\ &\leq |\{t \in [0, 1] : S(nt) > (a - \epsilon)\sqrt{n}\}| \end{aligned}$$

Combining this with the first conclusion of the proof and using the fact that $b \rightarrow |\{t \in [0, 1] : B_t > b\}|$ is continuous at $b = a$ with probability one, one arrives easily at the desired conclusion. $\square$

To compute the distribution of $|\{t \in [0, 1] : B_t > 0\}|$, observe that we proved in Theorem 4.9.6 that if S_n is symmetric simple random walk, then the left-hand side converges to the arcsine law, so the right-hand side has that distribution and is the limit for any random walk with mean 0 and finite variance. The last argument uses an idea called the “invariance principle” that originated with Erdős and Kac (1946, 1947): The asymptotic behavior of functionals of S_n should be the same as long as the central limit theorem applies.

Proof of Theorem 8.1.4. To simplify the proof and prepare for generalizations in the next section, let $X_{n,m}$, $1 \leq m \leq n$, be a triangular array of

random variables, $S_{n,m} = X_{n,1} + \cdots + X_{n,m}$ and suppose $S_{n,m} = B(\tau_m^n)$. Let

$$S_{n,(u)} = \begin{cases} S_{n,m} & \text{if } u = m \in \{0, 1, \dots, n\} \\ \text{linear for } u \in [m-1, m] & \text{when } m \in \{1, \dots, n\} \end{cases}$$

Lemma 8.1.9. *If $\tau_{[ns]}^n \rightarrow s$ in probability for each $s \in [0, 1]$ then*

$$\|S_{n,(n\cdot)} - B(\cdot)\| \rightarrow 0 \quad \text{in probability}$$

To make the connection with the original problem, let $X_{n,m} = X_m/\sqrt{n}$ and define $\tau_1^n, \dots, \tau_n^n$ so that $(S_{n,1}, \dots, S_{n,n}) =_d (B(\tau_1^n), \dots, B(\tau_n^n))$. If $T_1, T_2, \dots$ are the stopping times defined in the proof of Theorem 8.1.3, Brownian scaling implies $\tau_m^n =_d T_m/n$, so the hypothesis of Lemma 8.1.9 is satisfied.

Proof. The fact that B has continuous paths (and hence uniformly continuous on $[0, 1]$) implies that if $\epsilon > 0$ then there is a $\delta > 0$ so that $1/\delta$ is an integer and

$$(a) \quad P(|B_t - B_s| < \epsilon \text{ for all } 0 \leq s \leq 1, |t - s| < 2\delta) > 1 - \epsilon$$

The hypothesis of Lemma 8.1.9 implies that if $n \geq N_\delta$ then

$$P(|\tau_{[nk\delta]}^n - k\delta| < \delta \quad \text{for } k = 1, 2, \dots, 1/\delta) \geq 1 - \epsilon$$

Since $m \rightarrow \tau_m^n$ is increasing, it follows that if $s \in ((k-1)\delta, k\delta)$

$$\begin{aligned} \tau_{[ns]}^n - s &\geq \tau_{[n(k-1)\delta]}^n - k\delta \\ \tau_{[ns]}^n - s &\leq \tau_{[nk\delta]}^n - (k+1)\delta \end{aligned}$$

so if $n \geq N_\delta$,

$$(b) \quad P\left(\sup_{0 \leq s \leq 1} |\tau_{[ns]}^n - s| < 2\delta\right) \geq 1 - \epsilon$$

When the events in (a) and (b) occur

$$(c) \quad |S_{n,m} - B_{m/n}| < \epsilon \text{ for all } m \leq n$$

To deal with $t = (m + \theta)/n$ with $0 < \theta < 1$, we observe that

$$\begin{aligned} |S_{n,(nt)} - B_t| &\leq (1 - \theta)|S_{n,m} - B_{m/n}| + \theta|S_{n,m+1} - B_{(m+1)/n}| \\ &\quad + (1 - \theta)|B_{m/n} - B_t| + \theta|B_{(m+1)/n} - B_t| \end{aligned}$$

Using (c) on the first two terms and (a) on the last two, we see that if $n \geq N_\delta$ and $1/n < 2\delta$, then $\|S_{n,(n\cdot)} - B(\cdot)\| < 2\epsilon$ with probability $\geq 1 - 2\epsilon$. Since ϵ is arbitrary, the proof of Lemma 8.1.9 is complete. $\square$

To get Theorem 8.1.4 now, we have to show:

Lemma 8.1.10. *If φ is bounded and continuous then $E\varphi(S_{n,(n)}) \rightarrow E\varphi(B(\cdot))$.*

Proof. For fixed $\epsilon > 0$, let $G_\delta = \{\omega : \text{if } \|\omega - \omega'\| < \delta \text{ then } |\varphi(\omega) - \varphi(\omega')| < \epsilon\}$. Since φ is continuous, $G_\delta \uparrow C[0, 1]$ as $\delta \downarrow 0$. Let $\Delta = \|S_{n,(n)} - B(\cdot)\|$. The desired result now follows from Lemma 8.1.9 and the trivial inequality

$$|E\varphi(S_{n,(n)}) - E\varphi(B(\cdot))| \leq \epsilon + (2 \sup |\varphi(\omega)|) \{P(G_\delta^c) + P(\Delta \geq \delta)\} \quad \square$$

To accommodate our final example, we need a trivial generalization of Theorem 8.1.4. Let $C[0, \infty) = \{\text{continuous } \omega : [0, \infty) \rightarrow \mathbf{R}\}$ and let $\mathcal{C}[0, \infty)$ be the σ -field generated by the finite dimensional sets. Given a probability measure μ on $C[0, \infty)$, there is a corresponding measure $\pi_M \mu$ on $C[0, M] = \{\text{continuous } \omega : [0, M] \rightarrow \mathbf{R}\}$ (with $\mathcal{C}[0, M]$ the σ -field generated by the finite dimensional sets) obtained by “cutting off the paths at time M .” Let $(\psi_M \omega)(t) = \omega(t)$ for $t \in [0, M]$ and let $\pi_M \mu = \mu \circ \psi_M^{-1}$. We say that a sequence of probability measures μ_n on $C[0, \infty)$ converges weakly to μ if for all M , $\pi_M \mu_n$ converges weakly to $\pi_M \mu$ on $C[0, M]$, the last concept being defined by a trivial extension of the definitions for $M = 1$. With these definitions, it is easy to conclude:

Theorem 8.1.11. *$S(n\cdot)/\sqrt{n} \Rightarrow B(\cdot)$, i.e., the associated measures on $C[0, \infty)$ converge weakly.*

Proof. By definition, all we have to show is that weak convergence occurs on $C[0, M]$ for all $M < \infty$. The proof of Theorem 8.1.4 works in the same way when 1 is replaced by M . $\square$

Example 8.1.12. Let $N_n = \inf\{m : S_m \geq \sqrt{n}\}$ and $T_1 = \inf\{t : B_t \geq 1\}$. Since $\psi(\omega) = T_1(\omega) \wedge 1$ is continuous P_0 a.s. on $C[0, 1]$ and the distribution of T_1 is continuous, it follows from Theorem 8.1.5 that for $0 < t < 1$

$$P(N_n \leq nt) \rightarrow P(T_1 \leq t)$$

Repeating the last argument with 1 replaced by M and using Theorem 8.1.11 shows that the last conclusion holds for all t .

EXERCISES

8.1.1. Use Exercise 7.5.4 to conclude that $E(T_{U,V}^2) \leq 4EX^4$.

8.1.2. Suppose S_n is one-dimensional simple random walk and let

$$R_n = 1 + \max_{m \leq n} S_m - \min_{m \leq n} S_m$$

be the number of points visited by time n . Show that $R_n/\sqrt{n} \Rightarrow a$ limit.

8.2 CLT's for Martingales

In this section, we will prove central limit theorems for martingales. The key is an embedding theorem due to Strassen (1967). See his Theorem 4.3.

Theorem 8.2.1. *If S_n is a square integrable martingale with $S_0 = 0$, and B_t is a Brownian motion, then there is a sequence of stopping times $0 = T_0 \leq T_1 \leq T_2 \dots$ for the Brownian motion so that*

$$(S_0, S_1, \dots, S_k) \stackrel{d}{=} (B(T_0), B(T_1), \dots, B(T_k)) \quad \text{for all } k \geq 0$$

Proof. We include $S_0 = 0 = B(T_0)$ only for the sake of starting the induction argument. Suppose we have $(S_0, \dots, S_{k-1}) \stackrel{d}{=} (B(T_0), \dots, B(T_{k-1}))$ for some $k \geq 1$. The strong Markov property implies that $\{B(T_{k-1} + t) - B(T_{k-1}) : t \geq 0\}$ is a Brownian motion that is independent of $\mathcal{F}(T_{k-1})$. Let $\mu_k(s_0, \dots, s_{k-1}; \cdot)$ be a regular conditional distribution of $S_k - S_{k-1}$, given $S_j = s_j$, $0 \leq j \leq k-1$, that is, for each Borel set A

$$P(S_k - S_{k-1} \in A | S_0, \dots, S_{k-1}) = \mu_k(S_0, \dots, S_{k-1}; A)$$

By Exercises 4.1.18 and 4.1.16, this exists, and we have

$$0 = E(S_k - S_{k-1} | S_0, \dots, S_{k-1}) = \int x \mu_k(S_0, \dots, S_{k-1}; dx)$$

so the mean of the conditional distribution is 0 almost surely. Using Theorem 8.1.1 now we see that for almost every $\hat{S} \equiv (S_0, \dots, S_{k-1})$, there is a stopping time $\tau_{\hat{S}}$ (for $\{B(T_{k-1} + t) - B(T_{k-1}) : t \geq 0\}$) so that

$$B(T_{k-1} + \tau_{\hat{S}}) - B(T_{k-1}) \stackrel{d}{=} \mu_k(S_0, \dots, S_{k-1}; \cdot)$$

If we let $T_k = T_{k-1} + \tau_{\hat{S}}$ then $(S_0, \dots, S_k) \stackrel{d}{=} (B(T_0), \dots, B(T_k))$ and the result follows by induction. $\square$

Remark. While the details of the proof are fresh in the reader's mind, we would like to observe that if $E(S_k - S_{k-1})^2 < \infty$, then

$$E(\tau_{\hat{S}} | S_0, \dots, S_{k-1}) = \int x^2 \mu_k(S_0, \dots, S_{k-1}; dx)$$

since $B_t^2 - t$ is a martingale and $\tau_{\hat{S}}$ is the exit time from a randomly chosen interval $(S_{k-1} + U_k, S_{k-1} + V_k)$.

Before getting involved in the details of the central limit theorem, we show that the existence of the embedding allows us to give an easy proof of Theorem 4.5.2.

Theorem 8.2.2. *Let $\mathcal{F}_m = \sigma(S_0, S_1, \dots, S_m)$. $\lim_{n \rightarrow \infty} S_n$ exists and is finite on $\sum_{m=1}^{\infty} E((S_m - S_{m-1})^2 | \mathcal{F}_{m-1}) < \infty$.*

Proof. Let $\mathcal{B}_t$ be the filtration generated by Brownian motion, and let $t_m = T_m - T_{m-1}$. By construction we have

$$E((S_m - S_{m-1})^2 | \mathcal{F}_{m-1}) = E(t_m | \mathcal{B}(T_{m-1}))$$

Let $M = \inf\{n : \sum_{m=1}^{n+1} E(t_m | \mathcal{B}(T_{m-1})) > A\}$. M is a stopping time and by its definition has $\sum_{m=1}^M E(t_m | \mathcal{B}(T_{m-1})) \leq A$. Now $\{M \geq m\} \in \mathcal{F}_{m-1}$ so

$$\begin{aligned} E \sum_{m=1}^M t_m &= \sum_{m=1}^{\infty} E(t_m; M \geq m) \\ &= \sum_{m=1}^{\infty} E(E(t_m | \mathcal{B}(T_{m-1}); M \geq m)) = E \sum_{m=1}^M E(t_m | \mathcal{B}(T_{m-1})) \leq A \end{aligned}$$

From this we see that $\sum_{m=1}^M t_m < \infty$. As $A \rightarrow \infty$, $P(M < \infty) \downarrow 0$ so $T_{\infty} = \sum_{m=1}^{\infty} t_m < \infty$. Since $B(T_n) \rightarrow B(T_{\infty})$ as $n \rightarrow \infty$ the desired result follows. $\square$

Our first step toward the promised central limit theorem is to prove a result of Freedman (1971a). We say that $X_{n,m}$, $\mathcal{F}_{n,m}$, $1 \leq m \leq n$, is a **martingale difference array** if $X_{n,m} \in \mathcal{F}_{n,m}$ and $E(X_{n,m} | \mathcal{F}_{n,m-1}) = 0$ for $1 \leq m \leq n$, where $\mathcal{F}_{n,0} = \{\emptyset, \Omega\}$. Let $S_{n,m} = X_{n,1} + \dots + X_{n,m}$. Let $N = \{0, 1, 2, \dots\}$. Throughout the section, $S_{n,(n\cdot)}$ denotes the linear interpolation of $S_{n,m}$ defined by

$$S_{n,(u)} = \begin{cases} S_k & \text{if } u = k \in N \\ \text{linear for } u \in [k, k+1] & \text{when } k \in N \end{cases}$$

and $B(\cdot)$ is a Brownian motion with $B_0 = 0$. Let

$$V_{n,k} = \sum_{1 \leq m \leq k} E(X_{n,m}^2 | \mathcal{F}_{n,m-1})$$

Theorem 8.2.3. *Suppose $\{X_{n,m}, \mathcal{F}_{n,m}\}$ is a martingale difference array.*

If (i) for each t , $V_{n,[nt]} \rightarrow t$ in probability,

(i) $|X_{n,m}| \leq \epsilon_n$ for all m with $\epsilon_n \rightarrow 0$, and

then $S_{n,(n\cdot)} \Rightarrow B(\cdot)$.

Proof. (i) implies $V_{n,n} \rightarrow 1$ in probability. By stopping each sequence at the first time $V_{n,k} > 2$ and setting the later $X_{n,m} = 0$, we can suppose without loss of generality that $V_{n,n} \leq 2 + \epsilon_n^2$ for all n . By Theorem 8.2.1, we can find stopping times $T_{n,1}, \dots, T_{n,n}$ so that $(S_{n,1}, \dots, S_{n,n}) =_d (B(T_{n,1}), \dots, B(T_{n,n}))$. By Lemma 8.1.9, it suffices to show that $T_{n,[nt]} \rightarrow$

t in probability for each $t \in [0, 1]$. To do this we let $t_{n,m} = T_{n,m} - T_{n,m-1}$ (with $T_{n,0} = 0$) observe that by the remark after the proof of Theorem 8.2.1, $E(t_{n,m}|\mathcal{F}_{n,m-1}) = E(X_{n,m}^2|\mathcal{F}_{n,m-1})$. The last observation and hypothesis (i) imply

$$\sum_{m=1}^{[nt]} E(t_{n,m}|\mathcal{F}_{n,m-1}) \rightarrow t \quad \text{in probability}$$

To get from this to $T_{n,[nt]} \rightarrow t$ in probability, we observe

$$E \left(\sum_{m=1}^{[nt]} t_{n,m} - E(t_{n,m}|\mathcal{F}_{n,m-1}) \right)^2 = E \sum_{m=1}^{[nt]} \{t_{n,m} - E(t_{n,m}|\mathcal{F}_{n,m-1})\}^2$$

by the orthogonality of martingale increments, Theorem 4.4.7. Now

$$\begin{aligned} E(\{t_{n,m} - E(t_{n,m}|\mathcal{F}_{n,m-1})\}^2|\mathcal{F}_{n,m-1}) &\leq E(t_{n,m}^2|\mathcal{F}_{n,m-1}) \\ &\leq CE(X_{n,m}^4|\mathcal{F}_{n,m-1}) \leq C\epsilon_n^2 E(X_{n,m}^2|\mathcal{F}_{n,m-1}) \end{aligned}$$

by Exercise 7.5.4 and assumption (ii). Summing over n , taking expected values, and recalling we have assumed $V_{n,n} \leq 2 + \epsilon_n^2$, it follows that

$$E \left(\sum_{m=1}^{[nt]} t_{n,m} - E(t_{n,m}|\mathcal{F}_{n,m-1}) \right)^2 \rightarrow C\epsilon_n^2 EV_{n,n} \rightarrow 0$$

Unscrambling the definitions, we have shown $E(T_{n,[nt]} - V_{n,[nt]})^2 \rightarrow 0$, so Chebyshev's inequality implies $P(|T_{n,[nt]} - V_{n,[nt]}| > \epsilon) \rightarrow 0$, and using (ii) now completes the proof. $\square$

With Theorem 8.2.3 established, a truncation argument gives us:

Theorem 8.2.4. Lindeberg-Feller theorem for martingales. Suppose $X_{n,m}$, $\mathcal{F}_{n,m}$, $1 \leq m \leq n$ is a martingale difference array.

If (i) $V_{n,[nt]} \rightarrow t$ in probability for all $t \in [0, 1]$ and

(ii) for all $\epsilon > 0$, $\sum_{m \leq n} E(X_{n,m}^2 1_{(|X_{n,m}| > \epsilon)}|\mathcal{F}_{n,m-1}) \rightarrow 0$ in probability,

then $S_{n,(n)} \Rightarrow B(\cdot)$.

Remark 8.2.1. If $X_{n,m}$ on each row are independent this reduces to the martingale central limit theorem.

Proof. The first step is to truncate so that we can apply Theorem 8.2.3. Let

$$\hat{V}_n(\epsilon) = \sum_{m=1}^n E(X_{n,m}^2 1_{(|X_{n,m}| > \epsilon_n)}|\mathcal{F}_{n,m-1})$$

Lemma 8.2.5. *If $\epsilon_n \rightarrow 0$ slowly enough then $\epsilon_n^{-2} \hat{V}_n(\epsilon_n) \rightarrow 0$ in probability.*

Remark. The ϵ_n^{-2} in front is so that we can conclude

$$\sum_{m \leq n} P(|X_{n,m}| > \epsilon_n | \mathcal{F}_{n,m-1}) \rightarrow 0 \text{ in probability}$$

Proof. Let N_m be chosen so that $P(m^2 \hat{V}_n(1/m) > 1/m) \leq 1/m$ for $n \geq N_m$. Let $\epsilon_n = 1/m$ for $n \in [N_m, N_{m+1})$ and $\epsilon_n = 1$ if $n < N_1$. If $\delta > 0$ and $1/m < \delta$ then for $n \in [N_m, N_{m+1})$

$$P(\epsilon_n^{-2} \hat{V}_n(\epsilon_n) > \delta) \leq P(m^2 \hat{V}_n(1/m) > 1/m) \leq 1/m \quad \square$$

Let $\hat{X}_{n,m} = X_{n,m} 1_{(|X_{n,m}| > \epsilon_n)}$, $\bar{X}_{n,m} = X_{n,m} 1_{(|X_{n,m}| \leq \epsilon_n)}$ and

$$\tilde{X}_{n,m} = \bar{X}_{n,m} - E(\bar{X}_{n,m} | \mathcal{F}_{n,m-1})$$

Our next step is to show:

Lemma 8.2.6. *If we define $\tilde{S}_{n,(n \cdot)}$ in the obvious way then Theorem 8.2.3 implies $\tilde{S}_{n,(n \cdot)} \Rightarrow B(\cdot)$.*

Proof. Since $|\tilde{X}_{n,m}| \leq 2\epsilon_n$, we only have to check (ii) in Theorem 8.2.3. To do this, we observe that the conditional variance formula, Theorem 4.4.8, implies

$$E(\tilde{X}_{n,m}^2 | \mathcal{F}_{n,m-1}) = E(\bar{X}_{n,m}^2 | \mathcal{F}_{n,m-1}) - E(\bar{X}_{n,m} | \mathcal{F}_{n,m-1})^2$$

For the first term, we observe

$$E(\bar{X}_{n,m}^2 | \mathcal{F}_{n,m-1}) = E(X_{n,m}^2 | \mathcal{F}_{n,m-1}) - E(\hat{X}_{n,m}^2 | \mathcal{F}_{n,m-1})$$

For the second, we observe that $E(X_{n,m} | \mathcal{F}_{n,m-1}) = 0$ implies

$$E(\bar{X}_{n,m} | \mathcal{F}_{n,m-1})^2 = E(\hat{X}_{n,m} | \mathcal{F}_{n,m-1})^2 \leq E(\hat{X}_{n,m}^2 | \mathcal{F}_{n,m-1})$$

by Jensen's inequality, so it follows from (a) and (i) that

$$\sum_{m=1}^{[nt]} E(\tilde{X}_{n,m}^2 | \mathcal{F}_{n,m-1}) \rightarrow t \quad \text{for all } t \in [0, 1] \quad \square$$

Having proved Lemma 8.2.6, it remains to estimate the difference between $S_{n,(n \cdot)}$ and $\tilde{S}_{n,(n \cdot)}$. On $\{|X_{n,m}| \leq \epsilon_n \text{ for all } 1 \leq m \leq n\}$, we have

$$\|S_{n,(n \cdot)} - \tilde{S}_{n,(n \cdot)}\| \leq \sum_{m=1}^n |E(\bar{X}_{n,m} | \mathcal{F}_{n,m-1})| \quad (8.2.1)$$

To handle the right-hand side, we observe

$$\begin{aligned}
 \sum_{m=1}^n |E(\bar{X}_{n,m} | \mathcal{F}_{n,m-1})| &= \sum_{m=1}^n |E(\hat{X}_{n,m} | \mathcal{F}_{n,m-1})| \\
 &\leq \sum_{m=1}^n E(|\hat{X}_{n,m}| | \mathcal{F}_{n,m-1}) \\
 &\leq \epsilon_n^{-1} \sum_{m=1}^n E(\hat{X}_{n,m}^2 | \mathcal{F}_{n,m-1}) \rightarrow 0
 \end{aligned} \tag{8.2.2}$$

in probability by Lemma 8.2.5. To complete the proof now, it suffices to show

$$P(|X_{n,m}| > \epsilon_n \text{ for some } m \leq n) \rightarrow 0 \tag{8.2.3}$$

for with (8.2.2) and (8.2.1) this implies $\|S_{n,(n\cdot)} - \tilde{S}_{n,(n\cdot)}\| \rightarrow 0$ in probability. The proof of Theorem 8.2.3 constructs a Brownian motion with $\|\tilde{S}_{n,(n\cdot)} - B(\cdot)\| \rightarrow 0$, so the desired result follows from the triangle inequality and Lemma 8.1.10.

To prove (8.2.3), we will use Lemma 3.5 of Dvoretzky (1972).

Lemma 8.2.7. *If A_n is adapted to $\mathcal{G}_n$ then for any nonnegative $\delta \in \mathcal{G}_0$,*

$$P(\cup_{m=1}^n A_m | \mathcal{G}_0) \leq \delta + P\left(\sum_{m=1}^n P(A_m | \mathcal{G}_{m-1}) > \delta \mid \mathcal{G}_0\right)$$

Proof. We proceed by induction. When $n = 1$, the conclusion says

$$P(A_1 | \mathcal{G}_0) \leq \delta + P(P(A_1 | \mathcal{G}_0) > \delta | \mathcal{G}_0)$$

This is obviously true on $\Omega_- \equiv \{P(A_1 | \mathcal{G}_0) \leq \delta\}$ and also on $\Omega_+ \equiv \{P(A_1 | \mathcal{G}_0) > \delta\} \in \mathcal{G}_0$ since on Ω_+

$$P(P(A_1 | \mathcal{G}_0) > \delta | \mathcal{G}_0) = 1 \geq P(A_1 | \mathcal{G}_0)$$

To prove the result for n sets, observe that by the last argument the inequality is trivial on Ω_+ . Let $B_m = A_m \cap \Omega_-$. Since $\Omega_- \in \mathcal{G}_0 \subset \mathcal{G}_{m-1}$, $P(B_m | \mathcal{G}_{m-1}) = P(A_m | \mathcal{G}_{m-1})$ on Ω_- . (See Exercise 4.1.2.) Applying the result for $n - 1$ sets with $\gamma = \delta - P(B_1 | \mathcal{G}_0) \geq 0$,

$$P(\cup_{m=2}^n B_m | \mathcal{G}_1) \leq \gamma + P\left(\sum_{m=2}^n P(B_m | \mathcal{G}_{m-1}) > \gamma \mid \mathcal{G}_1\right)$$

Taking conditional expectation w.r.t. $\mathcal{G}_0$ and noting $\gamma \in \mathcal{G}_0$,

$$P(\cup_{m=2}^n B_m | \mathcal{G}_0) \leq \gamma + P\left(\sum_{m=2}^n P(B_m | \mathcal{G}_{m-1}) > \delta \mid \mathcal{G}_0\right)$$

$\cup_{2 \leq m \leq n} B_m = (\cup_{2 \leq m \leq n} A_m) \cap \Omega_-$ and another use of Exercise 4.1.2 shows

$$\sum_{1 \leq m \leq n} P(B_m | \mathcal{G}_{m-1}) = \sum_{1 \leq m \leq n} P(A_m | \mathcal{G}_{m-1})$$

on Ω_- . So, on Ω_- ,

$$P(\cup_{m=2}^n A_m | \mathcal{G}_0) \leq \delta - P(A_1 | \mathcal{G}_0) + P\left(\sum_{m=1}^n P(A_m | \mathcal{G}_{m-1}) > \delta | \mathcal{G}_0\right)$$

The result now follows from

$$P(\cup_{m=1}^n A_m | \mathcal{G}_0) \leq P(A_1 | \mathcal{G}_0) + P(\cup_{m=2}^n A_m | \mathcal{G}_0)$$

To see this, let $C = \cup_{2 \leq m \leq n} A_m$, observe that $1_{A_1 \cup C} \leq 1_{A_1} + 1_C$, and use the monotonicity of conditional expectations. $\square$

Proof of (8.2.3). Let $A_m = \{|X_{n,m}| > \epsilon_n\}$, $\mathcal{G}_m = \mathcal{F}_{n,m}$, and let δ be a positive number. Lemma 8.2.7 implies

$$P(|X_{n,m}| > \epsilon_n \text{ for some } m \leq n) \leq \delta + P\left(\sum_{m=1}^n P(|X_{n,m}| > \epsilon_n | \mathcal{F}_{n,m-1}) > \delta\right)$$

To estimate the right-hand side, we observe that “Chebyshev’s inequality” (Exercise 1.3 in Chapter 4) implies

$$\sum_{m=1}^n P(|X_{n,m}| > \epsilon_n | \mathcal{F}_{n,m-1}) \leq \epsilon_n^{-2} \sum_{m=1}^n E(\hat{X}_{n,m}^2 | \mathcal{F}_{n,m-1}) \rightarrow 0$$

so $\limsup_{n \rightarrow \infty} P(|X_{n,m}| > \epsilon_n \text{ for some } m \leq n) \leq \delta$. Since δ is arbitrary, the proof of (e) and hence of Theorem 8.2.4 is complete. $\square$

For applications, it is useful to have a result for a single sequence.

Theorem 8.2.8. Martingale central limit theorem. Suppose X_n , $\mathcal{F}_n$, $n \geq 1$, is a martingale difference sequence and let $V_k = \sum_{1 \leq n \leq k} E(X_n^2 | \mathcal{F}_{n-1})$.

If (i) $V_k/k \rightarrow \sigma^2 > 0$ in probability and

(ii) $n^{-1} \sum_{m \leq n} E(X_m^2 1_{(|X_m| > \epsilon \sqrt{n})}) \rightarrow 0$

then $S_{(n)}/\sqrt{n} \Rightarrow \sigma B(\cdot)$.

Proof. Let $X_{n,m} = X_m/\sigma\sqrt{n}$, $\mathcal{F}_{n,m} = \mathcal{F}_m$. Changing notation and letting $k = nt$, our first assumption becomes (i) of Theorem 8.2.4. To check (ii), observe that

$$E \sum_{m=1}^n E(X_{n,m}^2 1_{(|X_{n,m}| > \epsilon) | \mathcal{F}_{n,m-1}}) = \sigma^{-2} n^{-1} \sum_{m=1}^n E(X_m^2 1_{(|X_m| > \epsilon \sigma \sqrt{n})}) \rightarrow 0$$

$\square$

8.3 CLTs for Stationary Sequences

We begin by considering the martingale case.

Theorem 8.3.1. *Suppose X_n , $n \in \mathbf{Z}$, is an ergodic stationary sequence of square integrable martingale differences, i.e., $\sigma^2 = EX_n^2 < \infty$ and $E(X_n | \mathcal{F}_{n-1}) = 0$, where $\mathcal{F}_n = \sigma(X_m, m \leq n)$. Let $S_n = X_1 + \cdots + X_n$. Then*

$$S_{(n)}/n^{1/2} \Rightarrow \sigma B(\cdot)$$

Remark 8.3.1. This result was discovered independently by Billingsley (1961) and Ibragimov (1963).

Proof. $u_n \equiv E(X_n^2 | \mathcal{F}_{n-1})$ can be written as $\varphi(X_{n-1}, X_{n-2}, \dots)$, so (1.3) in Chapter 6 implies u_n is stationary and ergodic, and the ergodic theorem implies

$$n^{-1} \sum_{m=1}^n u_m \rightarrow Eu_0 = EX_0^2 \quad \text{a.s.}$$

The last conclusion shows that (i) of Theorem 8.2.8 holds. To verify (ii), we observe

$$n^{-1} \sum_{m=1}^n E(X_m^2 1_{(|X_m| > \epsilon \sqrt{n})}) = E(X_0^2 1_{(|X_0| > \epsilon \sqrt{n})}) \rightarrow 0$$

by the dominated convergence theorem. □

We prove our first central limit theorem for stationary sequences by reducing to the martingale case. Our proof is based on Scott (1973), but the key is an idea of Gordin (1969). In some applications of this result (see e.g., Exercise 8.3.1), it is convenient to use σ -fields richer than $\mathcal{F}_m = \sigma(X_n, n \leq m)$. So we adopt abstract conditions that satisfy the demands of martingale theory and ergodic theory:

- (i) $n \rightarrow \mathcal{F}_n$ is increasing, and $X_n \in \mathcal{F}_n$
- (ii) $\theta^{-1}\mathcal{F}_m = \mathcal{F}_{m+1}$

Theorem 8.3.2. *Suppose X_n , $n \in \mathbf{Z}$ is an ergodic stationary sequence with $EX_n = 0$. Let $\mathcal{F}_n$ satisfy (i), (ii), and*

$$\sum_{n \geq 1} \|E(X_0 | \mathcal{F}_{-n})\|_2 < \infty$$

where $\|Y\|_2 = (EY^2)^{1/2}$. Let $S_n = X_1 + \cdots + X_n$. Then $S_{(n)}/\sqrt{n} \Rightarrow \sigma B(\cdot)$ where

$$\sigma^2 = EX_0^2 + 2 \sum_{n=1}^{\infty} EX_0 X_n$$

and the series in the definition converges absolutely.

Proof. Suppose X_n , $n \in \mathbf{Z}$ is defined on sequence space $(\mathbf{R}^{\mathbf{Z}}, \mathcal{R}^{\mathbf{Z}}, P)$ with $X_n(\omega) = \omega_n$ and let $(\theta^n \omega)(m) = \omega(m+n)$. Let

$$\begin{aligned} H_n &= \{Y \in \mathcal{F}_n \text{ with } EY^2 < \infty\} \\ K_n &= \{Y \in H_n \text{ with } E(YZ) = 0 \text{ for all } Z \in H_{n-1}\} \end{aligned}$$

Geometrically, $H_0 \supset H_{-1} \supset H_{-2} \dots$ is a sequence of subspaces of L^2 and K_n is the orthogonal complement of H_{n-1} in H_n . If Y is a random variable, let $(\theta^n Y)(\omega) = Y(\theta^n \omega)$. Generalizing from the example $Y = f(X_{-j}, \dots, X_k)$, which has $\theta^n Y = f(X_{n-j}, \dots, X_{n+k})$, it is easy to see that if $Y \in H_k$, then $\theta^n Y \in H_{k+n}$, and hence if $Y \in K_j$ then $\theta^n Y \in K_{n+j}$.

If X_0 happened to be in K_0 then we would be happy since then $X_n = \theta^n X_0 \in K_n$ for all n , and taking $Z = 1_A \in H_{n-1}$ we would have $E(X_n 1_A) = 0$ for all $A \in \mathcal{F}_{n-1}$ and hence $E(X_n | \mathcal{F}_{n-1}) = 0$. The next best thing to having $X_0 \in K_0$ is to have

$$X_0 = Y_0 + Z_0 - \theta Z_0 \quad (8.3.1)$$

with $Y_0 \in K_0$ and $Z_0 \in L^2$, for then if we let

$$S_n = \sum_{m=1}^n X_m = \sum_{m=1}^n \theta^m X_0 \quad \text{and} \quad T_n = \sum_{m=1}^n \theta^m Y_0$$

then $S_n = T_n + \theta Z_0 - \theta^{n+1} Z_0$. The $\theta^m Y_0$ are a stationary ergodic martingale difference sequence (ergodicity follows from (1.3) in Chapter 6), so Theorem 8.3.1 implies

$$T_{(n)}/\sqrt{n} \Rightarrow \sigma B(\cdot) \quad \text{where } \sigma^2 = EY_0^2$$

To get rid of the other term, we observe $\theta Z_0/\sqrt{n} \rightarrow 0$ a.s. and

$$P\left(\sup_{1 \leq m \leq n} \theta^{m+1} Z_0 > \epsilon \sqrt{n}\right) \leq nP(Z_0 > \epsilon \sqrt{n}) \leq \epsilon^{-2} E(Z_0^2; Z_0 > \epsilon \sqrt{n}) \rightarrow 0$$

by dominated convergence. To solve (8.3.1) formally, we let

$$\begin{aligned} Z_0 &= \sum_{j=0}^{\infty} E(X_j | \mathcal{F}_{-1}) \\ \theta Z_0 &= \sum_{j=0}^{\infty} E(X_{j+1} | \mathcal{F}_0) \\ Y_0 &= \sum_{j=0}^{\infty} \{E(X_j | \mathcal{F}_0) - E(X_j | \mathcal{F}_{-1})\} \end{aligned}$$

and check that

$$Y_0 + Z_0 - \theta Z_0 = E(X_0 | \mathcal{F}_0) = X_0$$

To justify the last calculation, we need to know that the series in the definitions of Z_0 and Y_0 converge. Our assumption and shift invariance imply

$$\sum_{j=0}^{\infty} \|E(X_j|\mathcal{F}_{-1})\|_2 < \infty$$

so the triangle inequality implies that the series for Z_0 converges in L^2 . Since

$$\|E(X_j|\mathcal{F}_0) - E(X_j|\mathcal{F}_{-1})\|_2 \leq \|E(X_j|\mathcal{F}_0)\|_2$$

(the left-hand side is the projection onto $K_0 \subset H_0$), the series for Y_0 also converges in L^2 . Putting the pieces together, we have shown Theorem 8.3.2 with $\sigma^2 = EY_0^2$. To get the indicated formula for σ^2 , observe that conditioning and using Cauchy-Schwarz

$$|EX_0X_n| = |E(X_0E(X_n|\mathcal{F}_0))| \leq \|X_0\|_2 \|E(X_n|\mathcal{F}_0)\|_2$$

Shift invariance implies $\|E(X_n|\mathcal{F}_0)\|_2 = \|E(X_0|\mathcal{F}_{-n})\|_2$, so the series converges absolutely.

$$ES_n^2 = \sum_{j=1}^n \sum_{k=1}^n EX_jX_k = n EX_0^2 + 2 \sum_{m=1}^{n-1} (n-m) EX_0X_m$$

From this, it follows easily that

$$n^{-1}ES_n^2 \rightarrow EX_0^2 + 2 \sum_{m=1}^{\infty} EX_0X_m$$

To finish the proof, let $T_n = \sum_{m=1}^n \theta^m Y_0$, observe $\sigma^2 = EY_0^2$ and

$$n^{-1}E(S_n - T_n)^2 = n^{-1}E(\theta Z_0 - \theta^{n+1} Z_0)^2 \leq 4EZ_0^2/n \rightarrow 0$$

since $(a-b)^2 \leq (2a)^2 + (2b)^2$. $\square$

We turn now to examples. In the first one, it is trivial to check the hypothesis of Theorem 8.3.2.

Example 8.3.3. M-dependent sequences. Let X_n , $n \in \mathbf{Z}$, be a stationary sequence with $EX_n = 0$, $EX_n^2 < \infty$, and suppose $\{X_j, j \leq 0\}$ and $\{X_k, k > M\}$ are independent. In this case, $E(X_0|\mathcal{F}_{-n}) = 0$ for $n > M$, so Theorem 8.3.2 implies $S_{n,(n)}/\sqrt{n} \Rightarrow \sigma B(\cdot)$, where

$$\sigma^2 = EX_0^2 + 2 \sum_{m=1}^M EX_0X_m$$

Remark. For a bare-hands approach to this problem, see Theorem 7.3.1 in Chung (1974).

Exercise 8.3.1. Consider the special case of Example 8.3.3 in which ξ_i , $i \in \mathbf{Z}$, are i.i.d. and take values H and T with equal probability. Let $\eta_n = f(\xi_n, \xi_{n+1})$, where $f(H, T) = 1$ and $f(i, j) = 0$ otherwise. $S_n = \eta_1 + \dots + \eta_n$ counts the number of head runs in $\xi_1, \dots, \xi_{n+1}$. Apply Theorem 8.3.2 with $\mathcal{F}_n = \sigma(X_m : m \leq n+1)$ to show that there are constants μ and σ so that $(S_n - n\mu)/\sigma n^{1/2} \Rightarrow \chi$. What is the random variable Y_0 constructed in the proof of Theorem 8.3.2 in this case?

Example 8.3.4. Markov chains. Let ζ_n , $n \in \mathbf{Z}$ be an irreducible Markov chain on a countable state space S in which each ζ_n has the stationary distribution π . Let $X_n = f(\zeta_n)$, where $\sum f(x)\pi(x) = 0$ and $\sum f(x)^2\pi(x) < \infty$. Results in Chapter 6 imply that X_n is an ergodic stationary sequence. If we let $\mathcal{F}_{-n} = \sigma(\zeta_m, m \leq -n)$, then

$$E(X_0|\mathcal{F}_{-n}) = \sum_y p^n(\zeta_{-n}, y) f(y)$$

where $p^n(x, y)$ is the n step transition probability, so

$$\|E(X_0|\mathcal{F}_{-n})\|_2^2 = \sum_x \pi(x) \left(\sum_y p^n(x, y) f(y) \right)^2$$

When f is bounded, we can use $\sum f(x)\pi(x) = 0$ to get the following bound,

$$\|E(X_0|\mathcal{F}_{-n})\|_2^2 \leq \|f\|_\infty \sum_x \pi(x) \|p^n(x, \cdot) - \pi(\cdot)\|^2$$

where $\|f\|_\infty = \sup |f(x)|$ and $\|\cdot\|$ is the total variation norm.

When S is finite, all f are bounded. If the chain is aperiodic, Exercise 5.6 in Chapter 5 implies that

$$\sup_x \|p^n(x, \cdot) - \pi(\cdot)\| \leq Ce^{-\epsilon n}$$

and the hypothesis of Theorem 8.3.2 is satisfied. To see that the limiting variance σ^2 may be 0 in this case, consider the modification of Exercise 7.1 in which $X_n = (\xi_n, \xi_{n+1})$, $f(H, T) = 1$, $f(T, H) = -1$ and $f(H, H) = f(T, T) = 0$. In this case, $\sum_{m=1}^n f(X_m) \in \{-1, 0, 1\}$ so there is no central limit theorem.

Our last example is simple to treat directly, but will help us evaluate the strength of the conditions in Theorem 8.3.2 and in later theorems.

Example 8.3.5. Moving average process. Suppose

$$X_m = \sum_{k \geq 0} c_k \xi_{m-k} \quad \text{where} \quad \sum_{k \geq 0} c_k^2 < \infty$$

and the ξ_i , $i \in \mathbf{Z}$, are i.i.d. with $E\xi_i = 0$ and $E\xi_i^2 = 1$. If $\mathcal{F}_{-n} = \sigma(\xi_m ; m \leq -n)$ then

$$\|E(X_0|\mathcal{F}_{-n})\|_2 = \left\| \sum_{k \geq n} c_k \xi_{-k} \right\|_2 = \left(\sum_{k \geq n} c_k^2 \right)^{1/2}$$

If, for example, $c_k = (1+k)^{-p}$, $\|E(X_0|\mathcal{F}_{-n})\|_2 \sim n^{(1/2)-p}$, and Theorem 8.3.5 applies if $p > 3/2$.

Remark 8.3.2. Theorem 5.3 in Hall and Heyde (1980) shows that

$$\sum_{j \geq 0} \|E(X_j|\mathcal{F}_0) - E(X_j|\mathcal{F}_{-1})\|_2 < \infty$$

is sufficient for a central limit theorem. Using this result shows that $\sum |c_k| < \infty$ is sufficient for the central limit theorem in Example 7.3.

The condition in the improved result is close to the best possible. Suppose the ξ_i take values 1 and -1 with equal probability and let $c_k = (1+k)^{-p}$ where $1/2 < p < 1$. The Lindeberg-Feller theorem can be used to show that $S_n/n^{3/2-p} \Rightarrow \sigma\chi$. To check the normalization note that

$$\sum_{m=1}^n X_m = \sum_{j \leq n} a_{n,j} \xi_j$$

If $j \geq 0$ then $a_{n,j} = \sum_{i=0}^{n-j} c_k \approx (n-j)^{1-p}$, so using $0 \leq j \leq n/2$ the variance is at least n^{3-2p} . Further details are left to the reader.

8.3.1 Mixing Properties

The last three examples show that in many cases it is easy to verify the hypothesis of Theorem 8.3.2 directly. To connect Theorem 8.3.2 with other results in the literature, we will introduce sufficient conditions phrased in terms of quantitative measures of dependence between σ -fields. Our first is

$$\alpha(\mathcal{G}, \mathcal{H}) = \sup\{|P(A \cap B) - P(A)P(B)| : A \in \mathcal{G}, B \in \mathcal{H}\}$$

If $\alpha = 0$, $\mathcal{G}$ and $\mathcal{H}$ are independent so α measures the dependence between the σ -fields.

Lemma 8.3.6. *Let $p, q, r \in (1, \infty]$ with $1/p + 1/q + 1/r = 1$, and suppose $X \in \mathcal{G}$, $Y \in \mathcal{H}$ have $E|X|^p, E|Y|^q < \infty$. Then*

$$|EXY - EXEY| \leq 8\|X\|_p\|Y\|_q(\alpha(\mathcal{G}, \mathcal{H}))^{1/r}$$

Here, we interpret $x^0 = 1$ for $x > 0$ and $0^0 = 0$.

Proof. If $\alpha = 0$, X and Y are independent and the result is true, so we can suppose $\alpha > 0$. We build up to the result in three steps, starting with the case $r = \infty$.

$$|EXY - EXEY| \leq 2\|X\|_p\|Y\|_q \quad (8.3.2)$$

Proof. Hölder's inequality implies $|EXY| \leq \|X\|_p\|Y\|_q$, and Jensen's inequality implies

$$\|X\|_p\|Y\|_q \geq |E|X||E|Y|| \geq |EXEY|$$

so the result follows from the triangle inequality. $\square$

$$|EXY - EXEY| \leq 4\|X\|_\infty\|Y\|_\infty\alpha(\mathcal{G}, \mathcal{H}) \quad (8.3.3)$$

Proof. Let $\eta = \text{sgn}\{E(Y|\mathcal{G}) - EY\} \in \mathcal{G}$. $EXY = E(XE(Y|\mathcal{G}))$, so

$$\begin{aligned} |EXY - EXEY| &= |E(X\{E(Y|\mathcal{G}) - EY\})| \\ &\leq \|X\|_\infty E|E(Y|\mathcal{G}) - EY| \\ &= \|X\|_\infty E(\eta\{E(Y|\mathcal{G}) - EY\}) \\ &= \|X\|_\infty \{E(\eta Y) - E\eta EY\} \end{aligned}$$

Applying the last result with $X = Y$ and $Y = \eta$ gives

$$|E(Y\eta) - EY E\eta| \leq \|Y\|_\infty |E(\zeta\eta) - E\zeta E\eta|$$

where $\zeta = \text{sgn}\{E(\eta|\mathcal{H}) - E\eta\}$. Now $\eta = 1_A - 1_B$ and $\zeta = 1_C - 1_D$, so

$$\begin{aligned} |E(\zeta\eta) - E\zeta E\eta| &= |P(A \cap C) - P(B \cap C) - P(A \cap D) + P(B \cap D) \\ &\quad - P(A)P(C) + P(B)P(C) + P(A)P(D) - P(B)P(D)| \\ &\leq 4\alpha(\mathcal{G}, \mathcal{H}) \end{aligned}$$

Combining the last three displays gives the desired result. $\square$

$$|EXY - EXEY| \leq 6\|X\|_p\|Y\|_\infty\alpha(\mathcal{G}, \mathcal{H})^{1-1/p} \quad (8.3.4)$$

Proof. Let $C = \alpha^{-1/p}\|X\|_p$, $X_1 = X1_{(|X| \leq C)}$ and $X_2 = X - X_1$

$$\begin{aligned} |EXY - EXEY| &\leq |EX_1Y - EX_1EY| + |EX_2Y - EX_2EY| \\ &\leq 4\alpha C\|Y\|_\infty + 2\|Y\|_\infty E|X_2| \end{aligned}$$

by (8.3.3) and (8.3.2). Now

$$E|X_2| \leq C^{-(p-1)} E(|X|^p 1_{(|X| > C)}) \leq C^{-p+1} E|X|^p$$

Combining the last two inequalities and using the definition of C gives

$$|EXY - EXEY| \leq 4\alpha^{1-1/p}\|X\|_p\|Y\|_\infty + 2\|Y\|_\infty\alpha^{1-1/p}\|X\|_p^{-p+1+p}$$

which proves (8.3.4) $\square$

Finally, to prove Lemma 8.3.6, let $C = \alpha^{-1/q} \|Y\|_q$, $Y_1 = Y 1_{(|Y| \leq C)}$, and $Y_2 = Y - Y_1$.

$$\begin{aligned} |EXY - EXEY| &\leq |EXY_1 - EXEY_1| + |EXY_2 - EXEY_2| \\ &\leq 6C \|X\|_p \alpha^{1-1/p} + 2 \|X\|_p \|Y_2\|_\theta \end{aligned}$$

where $\theta = (1 - 1/p)^{-1}$ by (8.3.4) and (8.3.2). Now

$$E|Y_2|^\theta \leq C^{-q+\theta} E(|Y|^q 1_{(|Y| > C)}) \leq C^{-q+\theta} E|Y|^q$$

Taking the $1/\theta$ root of each side and recalling the definition of C

$$\|Y_2\|_\theta \leq C^{-(q-\theta)/\theta} \|Y\|_q^{q/\theta} \leq \alpha^{(q-\theta)/\theta q} \|Y\|_q$$

so we have

$$|EXY - EXEY| \leq 6\alpha^{-1/q} \|Y\|_q \|X\|_p \alpha^{1-1/p} + 2 \|X\|_p \alpha^{1/\theta-1/q} \|Y\|_q^{1/\theta+1/q}$$

proving the desired result. $\square$

Remark 8.3.3. The last proof is from Appendix III of Hall and Heyde (1980). They attribute (b) to Ibragimov (1962) and (c) and Lemma 8.3.6 to Davydov (1968).

Theorem 8.3.7. Suppose X_n , $n \in \mathbf{Z}$ is an ergodic stationary sequence with $EX_n = 0$, $E|X_0|^{2+\delta} < \infty$. Let $\alpha(n) = \alpha(\mathcal{F}_{-n}, \sigma(X_0))$, where $\mathcal{F}_{-n} = \sigma(X_m : m \leq -n)$, and suppose

$$\sum_{n=1}^{\infty} \alpha(n)^{\delta/2(2+\delta)} < \infty$$

If $S_n = X_1 + \cdots + X_n$, then $S_{(n)}/\sqrt{n} \Rightarrow \sigma B(\cdot)$, where

$$\sigma^2 = EX_0^2 + 2 \sum_{n=1}^{\infty} EX_0 X_n$$

Remark. If one strengthens the mixing condition one can weaken the summability condition. Let $\bar{\alpha}(n) = \alpha(\mathcal{F}_{-n}, \mathcal{F}'_0)$, where $\mathcal{F}'_0 = \sigma(X_k, k \geq 0)$. When $\bar{\alpha}(n) \downarrow 0$, the sequence is called **strong mixing**. Rosenblatt (1956) introduced the concept as a condition under which the central limit theorem for stationary sequences could be obtained. Ibragimov (1962) proved $S_n/\sqrt{n} \Rightarrow \sigma\chi$ where $\sigma^2 = \lim_{n \rightarrow \infty} ES_n^2/n$ under the assumption

$$\sum_{n=1}^{\infty} \bar{\alpha}(n)^{\delta/(2+\delta)} < \infty$$

See Ibragimov and Linnik (1971), Theorem 18.5.3, or Hall and Heyde (1980), Corollary 5.1, for a proof.

Proof. To use Lemma 8.3.6 to estimate the quantity in Theorem 8.3.2 we begin with

$$\|E(X|\mathcal{F})\|_2 = \sup\{E(XY) : Y \in \mathcal{F}, \|Y\|_2 = 1\} \quad (8.3.5)$$

Proof of (8.3.5). If $Y \in \mathcal{F}$ with $\|Y\|_2 = 1$, then using a by now familiar property of conditional expectation and the Cauchy-Schwarz inequality

$$EXY = E(E(XY|\mathcal{F})) = E(YE(X|\mathcal{F})) \leq \|E(X|\mathcal{F})\|_2 \|Y\|_2$$

Equality holds when $Y = E(X|\mathcal{F})/\|E(X|\mathcal{F})\|_2$. $\square$

Letting $p = 2 + \delta$ and $q = 2$ in Lemma 8.3.6, noticing

$$\frac{1}{r} = 1 - \frac{1}{p} - \frac{1}{q} = 1 - \frac{1}{2+\delta} - \frac{1}{2} = \frac{4 + 2\delta - 2 - (2 + \delta)}{(2 + \delta)2} = \frac{\delta}{(2 + \delta)2}$$

and recalling $EX_0 = 0$, shows that if $Y \in \mathcal{F}_{-n}$

$$|EX_0Y| \leq 8\|X_0\|_{2+\delta}\|Y\|_2 \alpha(n)^{\delta/2(2+\delta)}$$

Combining this with (8.3.5) gives

$$\|E(X_0|\mathcal{F}_{-n})\|_2 \leq 8\|X_0\|_{2+\delta} \alpha(n)^{\delta/2(2+\delta)}$$

and it follows that the hypotheses of Theorem 8.3.2 are satisfied. $\square$

In the M dependent case (Example 8.3.3), $\alpha(n) = 0$ for $n > M$, so Theorem 8.3.7 applies. As for Markov chains (Example 8.3.4), in this case

$$\begin{aligned} \alpha(n) &= \sup_{A,B} |P(X_{-n} \in A, X_0 \in B) - \pi(A)\pi(B)| \\ &= \sup_{A,B} \left| \sum_{x \in A, y \in B} \pi(x)p^n(x, y) - \pi(x)\pi(y) \right| \\ &\leq \sum_x \pi(x) 2\|p^n(x, \cdot) - \pi(\cdot)\| \end{aligned}$$

so the hypothesis of Theorem 8.3.7 can be checked if we know enough about the rate of convergence to equilibrium.

Finally, to see how good the conditions in Theorem 8.3.7 are, we consider the special case of the moving average process (Example 8.3.5) in which the ξ_i are i.i.d. standard normals. Let

$$\rho(\mathcal{G}, \mathcal{H}) = \sup\{\text{corr}(X, Y) : X \in \mathcal{G}, Y \in \mathcal{H}\}$$

where

$$\text{corr}(X, Y) = \frac{EXY - EXEY}{\|X - EX\|_2 \|Y - EY\|_2}$$

Clearly, $\alpha(\mathcal{G}, \mathcal{H}) \leq \rho(\mathcal{G}, \mathcal{H})$. Kolmogorov and Rozanov (1960) have shown that when $\mathcal{G}$ and $\mathcal{H}$ are generated by Gaussian random variables, then $\rho(\mathcal{G}, \mathcal{H}) \leq 2\pi\alpha(\mathcal{G}, \mathcal{H})$. They proved this by showing that $\rho(\mathcal{G}, \mathcal{H})$ is the angle between $L^2(\mathcal{G})$ and $L^2(\mathcal{H})$. Using the geometric interpretation, we see that if $|c_k|$ is decreasing, then $\alpha(n) \leq \bar{\alpha}(n) \leq |c_n|$, so Theorem 8.3.7 requires $\sum |c_n|^{(1/2)-\epsilon} < \infty$, but Ibragimov's result applies if $\sum |c_n|^{1-\epsilon} < \infty$. However as remarked earlier, the central limit theorem is valid if $\sum |c_n| < \infty$.

8.4 Empirical Distributions, Brownian Bridge

Let $X_1, X_2, \dots$ be i.i.d. with distribution F . Let

$$\hat{F}_n(x) = \frac{1}{n} |\{m \leq n : X_m \leq x\}|$$

be the empirical distribution. In Theorem 2.4.7 **add ref** we showed

Theorem 8.4.1. The Glivenko-Cantelli theorem. *As $n \rightarrow \infty$,*

$$\sup_x |F_n(x) - F(x)| \rightarrow 0 \quad a.s.$$

In this section, we will investigate the rate of convergence when F is continuous. We impose this restriction so we can reduce to the case of a uniform distribution on $(0,1)$ by setting $Y_n = F(X_n)$. Since $x \rightarrow F(x)$ is nondecreasing and continuous and no observations land in intervals of constancy of F , it is easy to see that if we let

$$\hat{G}_n(y) = \frac{1}{n} |\{m \leq n : Y_m \leq y\}|$$

then

$$\sup_x |\hat{F}_n(x) - F(x)| = \sup_{0 < y < 1} |\hat{G}_n(y) - y|$$

For the rest of the section then, we will assume $Y_1, Y_2, \dots$ is i.i.d. uniform on $(0,1)$. To be able to apply Donsker's theorem, we will transform the problem. Put the observations $Y_1, \dots, Y_n$ in increasing order: $U_1^n < U_2^n < \dots < U_n^n$. I claim that

$$\begin{aligned} \sup_{0 < y < 1} \hat{G}_n(y) - y &= \sup_{1 \leq m \leq n} \frac{m}{n} - U_m^n \\ \inf_{0 < y < 1} \hat{G}_n(y) - y &= \inf_{1 \leq m \leq n} \frac{m-1}{n} - U_m^n \end{aligned} \quad (8.4.1)$$

since the sup occurs at a jump of $\hat{G}_n$ and the inf right before a jump. For a picture, see Figure 8.1. We will show that

$$D_n \equiv n^{1/2} \sup_{0 < y < 1} |\hat{G}_n(y) - y|$$

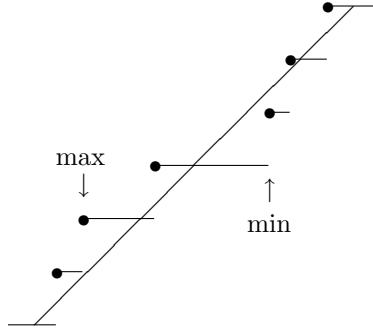


Figure 8.1: Picture proof of formulas in (8.4.1).

has a limit, so the extra $-1/n$ in the inf does not make any difference.

Our third and final maneuver is to give a special construction of the order statistics $U_1^n < U_2^n \dots < U_n^n$. Let $W_1, W_2, \dots$ be i.i.d. with $P(W_i > t) = e^{-t}$ and let $Z_n = W_1 + \dots + W_n$.

Lemma 8.4.2. $\{U_k^n : 1 \leq k \leq n\} \stackrel{d}{=} \{Z_k/Z_{n+1} : 1 \leq k \leq n\}$

Proof. We change variables $v = r(t)$, where $v_i = t_i/t_{n+1}$ for $i \leq n$, $v_{n+1} = t_{n+1}$. The inverse function is

$$s(v) = (v_1 v_{n+1}, \dots, v_n v_{n+1}, v_{n+1})$$

which has matrix of partial derivatives $\partial s_i / \partial v_j$ given by

$$\begin{pmatrix} v_{n+1} & 0 & \dots & 0 & v_1 \\ 0 & v_{n+1} & \dots & 0 & v_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & v_{n+1} & v_n \\ 0 & 0 & \dots & 0 & 1 \end{pmatrix}$$

The determinant of this matrix is v_{n+1}^n , so if we let $W = (V_1, \dots, V_{n+1}) = r(Z_1, \dots, Z_{n+1})$, the change of variables formula implies W has joint density

$$f_W(v_1, \dots, v_n, v_{n+1}) = \left(\prod_{m=1}^n \lambda e^{-\lambda v_{n+1}(v_m - v_{m-1})} \right) \lambda e^{-\lambda v_{n+1}(1 - v_n)} v_{n+1}^n$$

To find the joint density of $V = (V_1, \dots, V_n)$, we simplify the preceding formula and integrate out the last coordinate to get

$$f_V(v_1, \dots, v_n) = \int_0^\infty \lambda^{n+1} v_{n+1}^n e^{-\lambda v_{n+1}} dv_{n+1} = n!$$

for $0 < v_1 < v_2 \dots < v_n < 1$, which is the desired joint density. $\square$

We turn now to the limit law for D_n . As argued above, it suffices to consider

$$\begin{aligned} D'_n &= n^{1/2} \max_{1 \leq m \leq n} \left| \frac{Z_m}{Z_{n+1}} - \frac{m}{n} \right| \\ &= \frac{n}{Z_{n+1}} \max_{1 \leq m \leq n} \left| \frac{Z_m}{n^{1/2}} - \frac{m}{n} \cdot \frac{Z_{n+1}}{n^{1/2}} \right| \\ &= \frac{n}{Z_{n+1}} \max_{1 \leq m \leq n} \left| \frac{Z_m - m}{n^{1/2}} - \frac{m}{n} \cdot \frac{Z_{n+1} - n}{n^{1/2}} \right| \end{aligned} \quad (8.4.2)$$

If we let

$$B_n(t) = \begin{cases} (Z_m - m)/n^{1/2} & \text{if } t = m/n \text{ with } m \in \{0, 1, \dots, n\} \\ \text{linear} & \text{on } [(m-1)/n, m/n] \end{cases}$$

then

$$D'_n = \frac{n}{Z_{n+1}} \max_{0 \leq t \leq 1} \left| B_n(t) - t \left\{ B_n(1) + \frac{Z_{n+1} - Z_n}{n^{1/2}} \right\} \right|$$

The strong law of large numbers implies $Z_{n+1}/n \rightarrow 1$ a.s., so the first factor will disappear in the limit. To find the limit of the second, we observe that Donsker's theorem, Theorem 8.1.4, implies $B_n(\cdot) \Rightarrow B(\cdot)$, a Brownian motion, and computing second moments shows

$$(Z_{n+1} - Z_n)/n^{1/2} \rightarrow 0 \text{ in probability}$$

$\psi(\omega) = \max_{0 \leq t \leq 1} |\omega(t) - t\omega(1)|$ is a continuous function from $C[0, 1]$ to $\mathbf{R}$, so it follows from Donsker's theorem that:

Theorem 8.4.3. $D_n \Rightarrow \max_{0 \leq t \leq 1} |B_t - tB_1|$, where B_t is a Brownian motion starting at 0.

Remark. Doob (1949) suggested this approach to deriving results of Kolmogorov and Smirnov, which was later justified by Donsker (1952). Our proof follows Breiman (1968).

To identify the distribution of the limit in Theorem 8.4.3, we will first prove

$$\{B_t - tB_1, 0 \leq t \leq 1\} \stackrel{d}{=} \{B_t, 0 \leq t \leq 1 | B_1 = 0\} \quad (8.4.3)$$

a process we will denote by B_t^0 and call the **Brownian bridge**. The event $B_1 = 0$ has probability 0, but it is easy to see what the conditional probability should mean. If $0 = t_0 < t_1 < \dots < t_n < t_{n+1} = 1$, $x_0 = 0$, $x_{n+1} = 0$, and $x_1, \dots, x_n \in \mathbf{R}$, then

$$\begin{aligned} P(B(t_1) = x_1, \dots, B(t_n) = x_n | B(1) = 0) \\ = \frac{1}{p_1(0, 0)} \prod_{m=1}^{n+1} p_{t_m - t_{m-1}}(x_{m-1}, x_m) \end{aligned} \quad (8.4.4)$$

where $p_t(x, y) = (2\pi t)^{-1/2} \exp(-(y - x)^2/2t)$.

Proof of (8.4.3). Formula (8.4.4) shows that the f.d.d.'s of B_t^0 are multivariate normal and have mean 0. Since $B_t - tB_1$ also has this property, it suffices to show that the covariances are equal. We begin with the easier computation. If $s < t$ then

$$E((B_s - sB_1)(B_t - tB_1)) = s - st - st + st = s(1 - t) \quad (8.4.5)$$

For the other process, $P(B_s^0 = x, B_t^0 = y)$ is

$$\begin{aligned} & \frac{\exp(-x^2/2s)}{(2\pi s)^{1/2}} \cdot \frac{\exp(-(y-x)^2/2(t-s))}{(2\pi(t-s))^{1/2}} \cdot \frac{\exp(-y^2/2(1-t))}{(2\pi(1-t))^{1/2}} \cdot (2\pi)^{1/2} \\ &= (2\pi)^{-1} (s(t-s)(1-t))^{-1/2} \exp(-(ax^2 + 2bxy + cy^2)/2) \end{aligned} \quad (8.4.6)$$

where

$$\begin{aligned} a &= \frac{1}{s} + \frac{1}{t-s} = \frac{t}{s(t-s)} & b &= -\frac{1}{t-s} \\ c &= \frac{1}{t-s} + \frac{1}{1-t} = \frac{1-s}{(t-s)(1-t)} \end{aligned}$$

Recalling the discussion at the end of Section 3.9 and noticing

$$\begin{pmatrix} \frac{t}{s(t-s)} & \frac{-1}{(t-s)} \\ \frac{-1}{(t-s)} & \frac{1-s}{(t-s)(1-t)} \end{pmatrix}^{-1} = \begin{pmatrix} s(1-s) & s(1-t) \\ s(1-t) & t(1-t) \end{pmatrix}$$

(multiply the matrices!) shows (8.4.3) holds. $\square$

Remark. To do computations with the joint density it is useful to know that if (X, Y) have a bivariate normal distribution with mean 0 $EX^2 = \sigma_X^2$, $EY^2 = \sigma_Y^2$, and correlation ρ then the joint density is given by

$$\frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)}\left(\frac{x^2}{\sigma_X^2} - \frac{2xy}{\sigma_X\sigma_Y} + \frac{y^2}{\sigma_Y^2}\right)\right]$$

In (8.4.6) if $X = B_s^0$ and $Y = B_t^0$ then $\sigma_X = \sqrt{s(1-s)}$, $\sigma_Y = \sqrt{t(1-t)}$ and

$$\rho = \frac{s(1-t)}{\sqrt{s(1-s)t(1-t)}} = \sqrt{\frac{s(1-t)}{(1-s)t}}$$

so we have

$$1 - \rho^2 = \frac{(1-s)t - s(1-t)}{(1-s)t} = \frac{t-s}{(1-s)t}.$$

We leave it to the reader to check the well-known fact that the conditional distribution

$$(Y|X = x) = \text{normal}(x\rho\sigma_Y/\sigma_X, (1-\rho^2)\sigma_Y^2)$$

which in our special case implies

$$(B_t^0 | B_s^0 = x) = \text{normal}(x(1-t)/(1-s), (1-t)(1-s)/(1-s)) \quad (8.4.7)$$

Our final step in investigating the limit distribution of D_n is to compute the distribution of $\max_{0 \leq t \leq 1} |B_t^0|$. To do this, we first prove

Theorem 8.4.4. *The density function of B_t on $\{T_a \wedge T_b > t\}$ is*

$$P_x(T_a \wedge T_b > t, B_t = y) = \sum_{n=-\infty}^{\infty} P_x(B_t = y + 2n(b-a)) - P_x(B_t = 2b - y + 2n(b-a)) \quad (8.4.8)$$

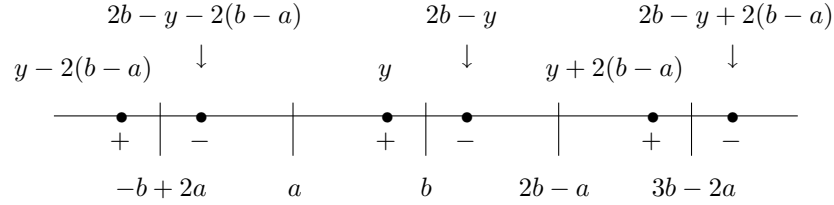


Figure 8.2: Picture of the infinite series in (8.4.8). Note that the array of + and - is anti-symmetric when seen from a or b .

Proof. We begin by observing that if $A \subset (a, b)$

$$P_x(T_a \wedge T_b > t, B_t \in A) = P_x(B_t \in A) - P_x(T_a < T_b, T_a < t, B_t \in A) - P_x(T_b < T_a, T_b < t, B_t \in A) \quad (8.4.9)$$

If we let $\rho_a(y) = 2a - y$ be reflection through a and observe that $\{T_a < T_b\}$ is $\mathcal{F}(T_a)$ measurable, then it follows from the proof of (7.4.5) that

$$P_x(T_a < T_b, T_a < t, B_t \in A) = P_x(T_a < T_b, B_t \in \rho_a A)$$

where $\rho_a A = \{\rho_a(y) : y \in A\}$. To get rid of the $T_a < T_b$, we observe that

$$P_x(T_a < T_b, B_t \in \rho_a A) = P_x(B_t \in \rho_a A) - P_x(T_b < T_a, B_t \in \rho_a A)$$

Noticing that $B_t \in \rho_a A$ and $T_b < T_a$ imply $T_b < t$ and using the reflection principle again gives

$$\begin{aligned} P_x(T_b < T_a, B_t \in \rho_a A) &= P_x(T_b < T_a, B_t \in \rho_b \rho_a A) \\ &= P_x(B_t \in \rho_b \rho_a A) - P_x(T_a < T_b, B_t \in \rho_b \rho_a A) \end{aligned}$$

Repeating the last two calculations n more times gives

$$\begin{aligned} P_x(T_a < T_b, B_t \in \rho_a A) &= \sum_{m=0}^n P_x(B_t \in \rho_a(\rho_b \rho_a)^m A) - P_x(B_t \in (\rho_b \rho_a)^{m+1} A) \\ &\quad + P_x(T_a < T_b, B_t \in (\rho_b \rho_a)^{n+1} A) \end{aligned}$$

Each pair of reflections pushes A further away from 0, so letting $n \rightarrow \infty$ shows

$$P_x(T_a < T_b, B_t \in \rho_a A) = \sum_{m=0}^{\infty} P_x(B_t \in \rho_a(\rho_b \rho_a)^m A) - P_x(B_t \in (\rho_b \rho_a)^{m+1} A)$$

Interchanging the roles of a and b gives

$$P_x(T_b < T_a, B_t \in \rho_b A) = \sum_{m=0}^{\infty} P_x(B_t \in \rho_b(\rho_a \rho_b)^m A) - P_x(B_t \in (\rho_a \rho_b)^{m+1} A)$$

Combining the last two expressions with (8.4.9) and using $\rho_c^{-1} = \rho_c$, $(\rho_a \rho_b)^{-1} = \rho_b^{-1} \rho_a^{-1}$ gives

$$P_x(T_a \wedge T_b > t, B_t \in A) = \sum_{m=-\infty}^{\infty} P_x(B_t \in (\rho_b \rho_a)^n A) - P_x(B_t \in \rho_a(\rho_b \rho_a)^n A)$$

To prepare for applications, let $A = (u, v)$ where $a < u < v < b$, notice that $\rho_b \rho_a(y) = y + 2(b - a)$, and change variables in the second sum to get

$$\begin{aligned} P_x(T_a \wedge T_b > t, u < B_t < v) &= \\ &\sum_{n=-\infty}^{\infty} \{P_x(u + 2n(b - a) < B_t < v + 2n(b - a)) \\ &\quad - P_x(2b - v + 2n(b - a) < B_t < 2b - u + 2n(b - a))\} \end{aligned} \quad (8.4.10)$$

Letting $u = y - \epsilon$, $v = y + \epsilon$, dividing both sides by 2ϵ , and letting $\epsilon \rightarrow 0$ (leaving it to the reader to check that the dominated convergence theorem applies) gives the desired result. $\square$

Setting $x = y = 0$, $t = 1$, and dividing by $(2\pi)^{-1/2} = P_0(B_1 = 0)$, we get a result for the Brownian bridge B_t^0 :

$$\begin{aligned} P_0 \left(a < \min_{0 \leq t \leq 1} B_t^0 < \max_{0 \leq t \leq 1} B_t^0 < b \right) & \\ &= \sum_{n=-\infty}^{\infty} e^{-(2n(b-a))^2/2} - e^{-(2b+2n(b-a))^2/2} \end{aligned} \quad (8.4.11)$$

Taking $a = -b$, we have

$$P_0 \left(\max_{0 \leq t \leq 1} |B_t^0| < b \right) = \sum_{m=-\infty}^{\infty} (-1)^m e^{-2m^2 b^2} \quad (8.4.12)$$

This formula gives the distribution of the Kolmogorov-Smirnov statistic, which can be used to test if an i.i.d. sequence $X_1, \dots, X_n$ has distribution F . To do this, we transform the data to $F(X_n)$ and look at the maximum discrepancy between the empirical distribution and the uniform. (8.4.12) tells us the distribution of the error when the X_i have distribution F .

EXERCISES

8.4.1. Use Exercise 7.4.8 and the reasoning that led to (8.4.11) to conclude

$$P \left(\max_{0 \leq t \leq 1} B_t^0 > b \right) = \exp(-2b^2)$$

8.4.2. Let B_t be a Brownian motion starting at 0. Show that $X_t = (1-t)B(t/1-t)$ is a Brownian bridge.

8.4.3. Show that Brownian bridge is a Markov process by checking that if $s_1 < \dots < s_n < s < t < 1$

$$P(B(t) = y | B(s) = x, B(s_n) = x_n, \dots, B(s_1) = x_1) = P(B(t) = y | B(s) = x)$$

8.4.4. Let B_t^0 be Brownian bridge. Show that $X_t = B_t^0/(1-t)$ is a martingale but is not L^2 bounded.

8.5 Laws of the Iterated Logarithm

Our first goal is to show:

Theorem 8.5.1. LIL for Brownian motion.

$$\limsup_{t \rightarrow \infty} B_t / (2t \log \log t)^{1/2} = 1 \quad a.s.$$

Here LIL is short for “law of the iterated logarithm,” a name that refers to the $\log \log t$ in the denominator. Once Theorem 8.5.1 is established, we can use the Skorokhod representation to prove the analogous result for random walks with mean 0 and finite variance.

Proof. The key to the proof is (7.4.4).

$$P_0 \left(\max_{0 \leq s \leq 1} B_s > a \right) = P_0(T_a \leq 1) = 2 P_0(B_1 \geq a) \quad (8.5.1)$$

To bound the right-hand side, we use the result in Lemma 1.2.6. For $x > 0$,

$$(x^{-1} - x^{-3}) \exp(-x^2/2) \leq \int_x^\infty \exp(-y^2/2) dy \leq x^{-1} \exp(-x^2/2) \quad (8.5.2)$$

and hence

$$\int_x^\infty \exp(-y^2/2) dy \sim \frac{1}{x} \exp(-x^2/2) \quad \text{as } x \rightarrow \infty \quad (8.5.3)$$

(8.5.3) and Brownian scaling imply that

$$P_0(B_t > (tf(t))^{1/2}) \sim \kappa f(t)^{-1/2} \exp(-f(t)/2)$$

where $\kappa = (2\pi)^{-1/2}$ is a constant that we will try to ignore below. The last result implies that if $\epsilon > 0$, then

$$\sum_{n=1}^{\infty} P_0(B_n > (nf(n))^{1/2}) \begin{cases} < \infty & \text{when } f(n) = (2 + \epsilon) \log n \\ = \infty & \text{when } f(n) = (2 - \epsilon) \log n \end{cases}$$

and hence by the Borel-Cantelli lemma that

$$\limsup_{n \rightarrow \infty} B_n / (2n \log n)^{1/2} \leq 1 \quad \text{a.s.}$$

To replace $\log n$ by $\log \log n$, we have to look along exponentially growing sequences. Let $t_n = \alpha^n$, where $\alpha > 1$.

$$\begin{aligned} P_0 \left(\max_{t_n \leq s \leq t_{n+1}} B_s > (t_n f(t_n))^{1/2} \right) &\leq P_0 \left(\max_{0 \leq s \leq t_{n+1}} B_s / t_{n+1}^{1/2} > \left(\frac{f(t_n)}{\alpha} \right)^{1/2} \right) \\ &\leq 2\kappa (f(t_n)/\alpha)^{-1/2} \exp(-f(t_n)/2\alpha) \end{aligned}$$

by (8.5.1) and (8.5.2). If $f(t) = 2\alpha^2 \log \log t$, then

$$\log \log t_n = \log(n \log \alpha) = \log n + \log \log \alpha$$

so $\exp(-f(t_n)/2\alpha) \leq C_\alpha n^{-\alpha}$, where C_α is a constant that depends only on α , and hence

$$\sum_{n=1}^{\infty} P_0 \left(\max_{t_n \leq s \leq t_{n+1}} B_s > (t_n f(t_n))^{1/2} \right) < \infty$$

Since $t \rightarrow (tf(t))^{1/2}$ is increasing and $\alpha > 1$ is arbitrary, it follows that

$$\limsup B_t / (2t \log \log t)^{1/2} \leq 1 \quad (8.5.4)$$

To prove the other half of Theorem 8.5.1, again let $t_n = \alpha^n$, but this time α will be large, since to get independent events, we will look at

$$P_0(B(t_{n+1}) - B(t_n) > (t_{n+1}f(t_{n+1}))^{1/2}) = P_0(B_1 > (\beta f(t_{n+1}))^{1/2})$$

where $\beta = t_{n+1}/(t_{n+1} - t_n) = \alpha/(\alpha - 1) > 1$. The last quantity is

$$\geq \frac{\kappa}{2}(\beta f(t_{n+1}))^{-1/2} \exp(-\beta f(t_{n+1})/2)$$

if n is large by (8.5.3). If $f(t) = (2/\beta^2) \log \log t$, then $\log \log t_n = \log n + \log \log \alpha$ so

$$\exp(-\beta f(t_{n+1})/2) \geq C_\alpha n^{-1/\beta}$$

where C_α is a constant that depends only on α , and hence

$$\sum_{n=1}^{\infty} P_0(B(t_{n+1}) - B(t_n) > (t_{n+1} f(t_{n+1}))^{1/2}) = \infty$$

Since the events in question are independent, it follows from the second Borel-Cantelli lemma that

$$B(t_{n+1}) - B(t_n) > ((2/\beta^2)t_{n+1} \log \log t_{n+1})^{1/2} \quad \text{i.o.} \quad (8.5.5)$$

From (8.5.4), we get

$$\limsup_{n \rightarrow \infty} B(t_n)/(2t_n \log \log t_n)^{1/2} \leq 1 \quad (8.5.6)$$

Since $t_n = t_{n+1}/\alpha$ and $t \rightarrow \log \log t$ is increasing, combining (8.5.5) and (8.5.6), and recalling $\beta = \alpha/(\alpha - 1)$ gives

$$\limsup_{n \rightarrow \infty} B(t_{n+1})/(2t_{n+1} \log \log t_{n+1})^{1/2} \geq \frac{\alpha - 1}{\alpha} - \frac{1}{\alpha^{1/2}}$$

Letting $\alpha \rightarrow \infty$ now gives the desired lower bound, and the proof of Theorem 8.5.1 is complete. $\square$

Turning to random walk, we will prove a result due to Hartman and Wintner (1941):

Theorem 8.5.2. *If $X_1, X_2, \dots$ are i.i.d. with $EX_i = 0$ and $EX_i^2 = 1$ then*

$$\limsup_{n \rightarrow \infty} S_n/(2n \log \log n)^{1/2} = 1$$

Proof. By Theorem 8.1.2, we can write $S_n = B(T_n)$ with $T_n/n \rightarrow 1$ a.s. As in the proof of Donsker's theorem, this is all we will use in the argument below. Theorem 8.5.2 will follow from Theorem 8.5.1 once we show

$$(S_{[t]} - B_t)/(t \log \log t)^{1/2} \rightarrow 0 \quad \text{a.s.} \quad (8.5.7)$$

To do this, we begin by observing that if $\epsilon > 0$ and $t \geq t_o(\omega)$

$$T_{[t]} \in [t/(1 + \epsilon), t(1 + \epsilon)] \quad (8.5.8)$$

To estimate $S_{[t]} - B_t$, we let $M(t) = \sup\{|B(s) - B(t)| : t/(1 + \epsilon) \leq s \leq t(1 + \epsilon)\}$. To control the last quantity, we let $t_k = (1 + \epsilon)^k$ and notice that if $t_k \leq t \leq t_{k+1}$

$$\begin{aligned} M(t) &\leq \sup\{|B(s) - B(t)| : t_{k-1} \leq s, t \leq t_{k+2}\} \\ &\leq 2 \sup\{|B(s) - B(t_{k-1})| : t_{k-1} \leq s \leq t_{k+2}\} \end{aligned}$$

Noticing $t_{k+2} - t_{k-1} = \delta t_{k-1}$, where $\delta = (1 + \epsilon)^3 - 1$, scaling implies

$$\begin{aligned} &P\left(\max_{t_{k-1} \leq s \leq t_{k+2}} |B(s) - B(t)| > (3\delta t_{k-1} \log \log t_{k-1})^{1/2}\right) \\ &= P\left(\max_{0 \leq r \leq 1} |B(r)| > (3 \log \log t_{k-1})^{1/2}\right) \\ &\leq 2\kappa(3 \log \log t_{k-1})^{-1/2} \exp(-3 \log \log t_{k-1}/2) \end{aligned}$$

by a now familiar application of (8.5.1) and (8.5.2). Summing over k and using (b) gives

$$\limsup_{t \rightarrow \infty} (S_{[t]} - B_t)/(t \log \log t)^{1/2} \leq (3\delta)^{1/2}$$

If we recall $\delta = (1 + \epsilon)^3 - 1$ and let $\epsilon \downarrow 0$, (a) follows and the proof is complete. $\square$

Strassen (1965) has shown an exact converse. If Theorem 8.5.2 holds then $EX_i = 0$ and $EX_i^2 = 1$. Another one of his contributions to this subject is

Theorem 8.5.3. Strassen's (1964) invariance principle. *Let $X_1, X_2, \dots$ be i.i.d. with $EX_i = 0$ and $EX_i^2 = 1$, let $S_n = X_1 + \dots + X_n$, and let $S_{(n\cdot)}$ be the usual linear interpolation. The limit set (i.e., the collection of limits of convergent subsequences) of*

$$Z_n(\cdot) = (2n \log \log n)^{-1/2} S(n\cdot) \quad \text{for } n \geq 3$$

is $\mathcal{K} = \{f : f(x) = \int_0^x g(y) dy \text{ with } \int_0^1 g(y)^2 dy \leq 1\}$.

Jensen's inequality implies $f(1)^2 \leq \int_0^1 g(y)^2 dy \leq 1$ with equality if and only if $f(t) = t$, so Theorem 8.5.3 contains Theorem 8.5.2 as a special case and provides some information about how the large value of S_n came about.

EXERCISES

8.5.1. Show that if $E|X_i|^\alpha = \infty$ for some $\alpha < 2$ then

$$\limsup_{n \rightarrow \infty} |X_n|/n^{1/\alpha} = \infty \quad \text{a.s.}$$

so the law of the iterated logarithm fails.

8.5.2. Give a direct proof that, under the hypotheses of Theorem 8.5.3, the limit set of $\{S_n/(2n \log \log n)^{1/2}\}$ is $[-1, 1]$.

Chapter 9

Multidimensional Brownian Motion

The point of this chapter is to establish the connection between Brownian motion and various parabolic and elliptic partial differential equations. The first section uses martingales to do computations for multidimensional Brownian motion.

9.1 Martingales

Let $\Delta f = \sum_{i=1}^d \partial^2 f / \partial x_i^2$ be the Laplacian of f . Combining Ito's formula (7.6.9) with Theorem 7.6.4

Theorem 9.1.1. *Suppose $v \in C^2$, i.e., all first and second order partial derivatives exist and are continuous, and $E \int_0^t |\nabla v(B_s)|^2 ds < \infty$ for all t . Then*

$$v(B_t) - \int_0^t \frac{1}{2} \Delta v(B_s) ds \quad \text{is a continuous martingale.}$$

Let $S_r = \inf\{t : |B_t| = r\}$, $r < R$, and $\tau = S_r \wedge S_R$. The first detail is to note that if $|x| < R$ then $P_x(S_R < \infty) = 1$. Once we know this we can conclude

Theorem 9.1.2. *If $|x| < R$ then $E_x S_R = (R^2 - |x|^2)/d$.*

Proof. It follows from Theorem 7.5.4 that $|B_t|^2 - dt = \sum_{i=1}^d (B_t^i)^2 - t$ is a martingale. Theorem 7.5.1 implies $|x|^2 = E|B_{S_R \wedge t}|^2 - dE(S_R \wedge t)$. Letting $t \rightarrow \infty$ gives the desired result. $\square$

To further investigate Brownian motion we introduce two new functions:

$$\varphi(x) = \begin{cases} \log |x| & d = 2 \\ |x|^{2-d} & d \geq 3 \end{cases}$$

We leave it to the reader to check that in each case $\Delta\varphi = 0$.

Lemma 9.1.3. $\varphi(x) = E_x\varphi(B_\tau)$

Proof. Define $\psi(x) = g(|x|)$ to be C^2 and have compact support, and have $\psi(x) = \phi(x)$ when $r < |x| < R$. Theorem 9.1.1 implies that $\psi(x) = E_x\psi(B_{t\wedge\tau})$. Letting $t \rightarrow \infty$ now gives the desired result. $\square$

Lemma 9.1.3 implies that

$$\varphi(x) = \varphi(r)P_x(S_r < S_R) + \varphi(R)(1 - P_x(S_r < S_R))$$

where $\varphi(r)$ is short for the value of $\varphi(x)$ on $\{x : |x| = r\}$. Solving now gives

$$P_x(S_r < S_R) = \frac{\varphi(R) - \varphi(x)}{\varphi(R) - \varphi(r)} \quad (9.1.1)$$

In $d = 2$, the last formula says

$$P_x(S_r < S_R) = \frac{\log R - \log |x|}{\log R - \log r} \quad (9.1.2)$$

If we fix r and let $R \rightarrow \infty$ in (9.1.2), the right-hand side goes to 1. So

$$P_x(S_r < \infty) = 1 \quad \text{for any } x \text{ and any } r > 0$$

It follows that two-dimensional Brownian motion is **recurrent** in the sense that if G is any open set, then $P_x(B_t \in G \text{ i.o.}) \equiv 1$.

If we fix R , let $r \rightarrow 0$ in (9.1.2), and let $S_0 = \inf\{t > 0 : B_t = 0\}$, then for $x \neq 0$

$$P_x(S_0 < S_R) \leq \lim_{r \rightarrow 0} P_x(S_r < S_R) = 0$$

Since this holds for all R and since the continuity of Brownian paths implies $S_R \uparrow \infty$ as $R \uparrow \infty$, we have $P_x(S_0 < \infty) = 0$ for all $x \neq 0$. To extend the last result to $x = 0$ we note that the Markov property implies

$$P_0(B_t = 0 \text{ for some } t \geq \epsilon) = E_0[P_{B_\epsilon}(T_0 < \infty)] = 0$$

for all $\epsilon > 0$, so $P_0(B_t = 0 \text{ for some } t > 0) = 0$, and thanks to our definition of $S_0 = \inf\{t > 0 : B_t = 0\}$, we have

$$P_x(S_0 < \infty) = 0 \quad \text{for all } x \quad (9.1.3)$$

Thus, in $d \geq 2$ Brownian motion will not hit 0 at a positive time even if it starts there. As a corollary we see that Brownian motion in $d = 3$ will not hit a straight line.

For $d \geq 3$, formula (9.1.1) says

$$P_x(S_r < S_R) = \frac{R^{2-d} - |x|^{2-d}}{R^{2-d} - r^{2-d}} \quad (9.1.4)$$

There is no point in fixing R and letting $r \rightarrow 0$, here. The fact that two dimensional Brownian motion does not hit 0 implies that three dimensional Brownian motion does not hit 0 and indeed will not hit the line $\{x : x_1 = x_2 = 0\}$. If we fix r and let $R \rightarrow \infty$ in (9.1.4) we get

$$P_x(S_r < \infty) = (r/|x|)^{d-2} < 1 \quad \text{if } |x| > r \quad (9.1.5)$$

From the last result it follows easily that for $d \geq 3$, Brownian motion is **transient**, i.e., it does not return infinitely often to any bounded set.

Theorem 9.1.4. *As $t \rightarrow \infty$, $|B_t| \rightarrow \infty$ a.s.*

Proof. Let $A_n = \{|B_t| > n^{1-\epsilon} \text{ for all } t \geq S_n\}$. The strong Markov property implies

$$P_x(A_n^c) = E_x(P_{B(S_n)}(S_{n^{1-\epsilon}} < \infty)) = (n^{1-\epsilon}/n)^{d-2} \rightarrow 0$$

as $n \rightarrow \infty$. Now $\limsup A_n = \bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} A_n$ has

$$P(\limsup A_n) \geq \limsup P(A_n) = 1$$

So infinitely often the Brownian path never returns to $\{x : |x| \leq n^{1-\epsilon}\}$ after time S_n and this implies the desired result. $\square$

The scaling relation (7.1.1) implies that $S_n =_d n^2 S_1$, so the proof of Theorem 9.1.4 suggests that

$$|B_t|/t^{(1-\epsilon)/2} \rightarrow \infty$$

Dvoretzky and Erdős (1951) have proved the following result about how fast Brownian motion goes to ∞ in $d \geq 3$.

Theorem 9.1.5. *Suppose $g(t)$ is positive and decreasing. Then*

$$P_0(|B_t| \leq g(t)\sqrt{t} \text{ i.o. as } t \uparrow \infty) = 1 \text{ or } 0$$

according as $\int_0^\infty g(t)^{d-2}/t \, dt = \infty$ or $< \infty$.

Here the absence of the lower limit implies that we are only concerned with the behavior of the integral “near ∞ .” A little calculus shows that

$$\int_0^\infty t^{-1} \log^{-\alpha} t \, dt = \infty \text{ or } < \infty$$

according as $\alpha \leq 1$ or $\alpha > 1$, so B_t goes to ∞ faster than

$$t^{1/2}(\log t)^{-\alpha/(d-2)} \quad \text{for any } \alpha > 1.$$

EXERCISES

9.1.1. Combine scaling with the 0-1 law to show that

$$\liminf_{t \rightarrow \infty} |B_t|/t^{1/2} = 0 \quad \text{a.s.}$$

9.1.2. Suppose $f \in C^2$ is nonnegative and superharmonic. Show that in dimensions 1 and 2, f must be constant. Give an example of the form $f(x) = g(|x|)$ in $d \geq 3$.

9.1.3. Suppose $f \in C^2$ is bounded and harmonic. Then f must be constant (in any dimension).

9.2 Heat Equation

In this section, we will consider the following equation:

- (a) $u \in C^{1,2}$ and $u_t = \frac{1}{2}\Delta u$ in $(0, \infty) \times \mathbf{R}^d$.
- (b) u is continuous at each point of $\{0\} \times \mathbf{R}^d$ and $u(0, x) = f(x)$, where f is bounded and continuous.

This equation derives its name from the fact that if the units of measurement are chosen suitably then the solution $u(t, x)$ gives the temperature at the point $x \in \mathbf{R}^d$ at time t when the temperature profile at time 0 is given by $f(x)$.

Obviously (b) cannot hold unless f is continuous. The results below can be generalized to unbounded f . We assume f is bounded to keep things simple. In this and the next five sections the assumptions stated at the beginning are in force throughout the section.

The first step in solving this equation, as it will be six times below, is to find a local martingale M_s , $s \in [0, t]$, i.e., a process so that there is an increasing sequence of stopping times so that $M(s \wedge T_n)$ is a martingale and $P(T_n < t) \rightarrow 0$.

Theorem 9.2.1. *If u satisfies (a), $M_s = u(t-s, B_s)$ is a local martingale on $[0, t]$.*

Proof. Applying Itô's formula, (7.6.9) gives

$$\begin{aligned} u(t-s, B_s) - u(t, B_0) &= \int_0^s -u_t(t-r, B_r) dr \\ &+ \int_0^s \nabla u(t-r, B_r) \cdot dB_r + \frac{1}{2} \int_0^s \Delta u(t-r, B_r) dr \end{aligned}$$

The result follows since $-u_t + \frac{1}{2}\Delta u = 0$ and the second term on the right-hand side is a local martingale by Theorem 7.6.4. Here we need the stopping times to make sure the integrability condition holds. $\square$

Our next step is to prove a uniqueness theorem.

Theorem 9.2.2. *If there is a solution that is bounded then it must be*

$$v(t, x) \equiv E_x f(B_t)$$

Here $\equiv$ means that the last equation defines v . We will always use u for a generic solution of the equation and v for our special solution.

Proof. If we now assume that u is bounded, then $M_s, 0 \leq s < t$, is a bounded martingale. The martingale convergence theorem implies that

$$M_t \equiv \lim_{s \uparrow t} M_s \quad \text{exists a.s.}$$

If u satisfies (b), this limit must be $f(B_t)$. Since M_s is uniformly integrable, it follows that $u(t, x) = E_x M_0 = E_x M_t = v(t, x)$. $\square$

Now that Theorem 9.2.2 has told us what the solution must be, the next logical step is to find conditions under which v is a solution. Here and in the other five examples it is easy to show that if v is smooth enough then it is a solution.

Theorem 9.2.3. *If $v \in C^{1,2}$ then it satisfies (a).*

Proof. If we think of running space time Brownian motion $(t - s, B_s)$ to construct the solution then we see

$$v(t + h, x) = E_x v(t, B_h)$$

Subtracting $v(t, x)$ from each side and using Taylor series

$$\begin{aligned} v(t + h, x) - v(t, x) = & E_x \left(\sum_{i=1}^d \frac{\partial v}{\partial x_i}(t, x) (B_i(h) - x_i) \right. \\ & \left. + \frac{1}{2} \sum_{1 \leq i, j \leq d} \frac{\partial^2 v}{\partial x_i \partial x_j}(t, x) (B_i(h) - x_i) (B_j(h) - x_j) \right) + o(h) \end{aligned}$$

Since for $i \neq j$, $B_i(t)$, $B_i^2(t) - t$, and $B_i(t)B_j(t)$ are martingales

$$v(t + h, x) - v(t, x) = \frac{1}{2} \sum_{i=1}^d \frac{\partial^2 v}{\partial x_i^2}(t, x) h + o(h)$$

Dividing by h and letting $h \rightarrow 0$ gives the desired result. $\square$

Theorem 9.2.4. *v satisfies (b).*

Proof. $(B_t - B_0) \stackrel{d}{=} t^{1/2} \chi$, where χ has a normal distribution with mean 0 and variance 1, so if $t_n \rightarrow 0$ and $x_n \rightarrow x$, the bounded convergence theorem implies that

$$v(t_n, x_n) = E f(x_n + t_n^{1/2} \chi) \rightarrow f(x)$$

which proves the desired result. $\square$

The final step in showing that v is a solution is to find conditions that guarantee that it is smooth.

Theorem 9.2.5. $v \in C^{1,2}$ and hence satisfies (a).

Proof. Our solution is

$$v(t, x) = E_x f(B_t) = \int (2\pi t)^{-d/2} e^{-|x-y|^2/2t} f(y) dy$$

Using calculus and some basic facts from analysis one can prove the desired result. The details are not exciting and are left to the reader. $\square$

9.3 Inhomogeneous Heat Equation

In this section, we will consider what happens when we add a bounded continuous function $g(x)$ to the heat equation. That is, we will study

- (a) $u \in C^{1,2}$ and $u_t = \frac{1}{2}\Delta u + g$ in $(0, \infty) \times \mathbf{R}^d$
- (b) u is continuous at each point of $\{0\} \times \mathbf{R}^d$ and $u(0, x) = f(x)$ is bounded and continuous.

We observed in the previous section that (b) cannot hold unless f is continuous. Here $g = u_t - (1/2)\Delta u$ so the equation in (a) cannot hold with $u \in C^{1,2}$ unless g is continuous.

Our first step in treating the new equation is to observe that if u_1 is a solution of the equation with $f = f_0$ and $g = 0$ which we studied in the last section, and u_2 is a solution of the equation with $f = 0$ and $g = g_0$ then $u_1 + u_2$ is a solution of the equation with $f = f_0$ and $g = g_0$ so we can restrict our attention to the case $f \equiv 0$.

Having made this simplification, we will now study the equation above by following the procedure used in the last section.

Theorem 9.3.1. *If u satisfies (a), then*

$$M_s = u(t-s, B_s) + \int_0^s g(B_r) dr$$

is a local martingale on $[0, t]$.

Proof. Applying Itô's formula as in the proof of Theorem 9.2.1 gives

$$\begin{aligned} u(t-s, B_s) - u(t, B_0) &= \int_0^s (-u_t + \frac{1}{2}\Delta u)(t-r, B_r) dr \\ &\quad + \int_0^s \nabla u(t-r, B_r) \cdot dB_r \end{aligned}$$

which proves the result, since $-u_t + \frac{1}{2}\Delta u = -g$ and the second term on the right-hand side is a local martingale by Theorem 7.6.4. $\square$

Again the next step is a uniqueness result.

Theorem 9.3.2. *If there is a solution that is bounded on $[0, T] \times \mathbf{R}^d$ for any $T < \infty$, it must be*

$$v(t, x) \equiv E_x \left(\int_0^t g(B_s) ds \right)$$

Proof. Under the assumptions on g and u , $M_s, 0 \leq s < t$, defined in Theorem 9.3.1 is a bounded martingale and $u(0, x) \equiv 0$ so

$$M_t \equiv \lim_{s \uparrow t} M_s = \int_0^t g(B_s) ds$$

Since M_s is uniformly integrable, $u(t, x) = E_x M_0 = E_x M_t = v(t, x)$. $\square$

Again, it is easy to show that if v is smooth enough it is a solution.

Theorem 9.3.3. *If $v \in C^{1,2}$, then it satisfies (a) in $(0, \infty) \times \mathbf{R}^d$.*

Proof. If we think of running space time Brownian motion $(t-s, B_s)$ to construct the solution then we see

$$v(t+h, x) = E_x \left(v(t, B_h) + \int_0^h g(B_s) ds \right)$$

Subtracting $v(t, x)$ from each side and using Taylor series

$$\begin{aligned} v(t+h, x) - v(t, x) &= E_x \left(\sum_{i=1}^d \frac{\partial v}{\partial x_i}(t, x) (B_i(h) - x_i) \right. \\ &\quad + \frac{1}{2} \sum_{1 \leq i, j \leq d} \frac{\partial^2 v}{\partial x_i \partial x_j}(t, x) (B_i(h) - x_i) (B_j(h) - x_j) \\ &\quad \left. + \int_0^h g(B_s) ds \right) + o(h) \end{aligned}$$

Since for $i \neq j$, $B_i(t)$, $B_i^2(t) - t$, and $B_i(t)B_j(t)$ are martingales

$$v(t+h, x) - v(t, x) = \frac{1}{2} \sum_{i=1}^d \frac{\partial^2 v}{\partial x_i^2}(t, x) h + E_x \int_0^h g(B_s) ds + o(h)$$

Dividing by h and letting $h \rightarrow 0$ gives $v_t = \frac{1}{2} \Delta V + g(x)$ which is the desired result. $\square$

Theorem 9.3.4. *v satisfies (b).*

Proof. If $|g| \leq M$, then as $t \rightarrow 0$

$$|v(t, x)| \leq E_x \int_0^t |g(B_s)| ds \leq Mt \rightarrow 0$$

which proves the desired result. $\square$

The final step in showing that v is a solution is to check that $v \in C^{1,2}$. The calculations are messy so we just state the result. Here and in what follows when we say a function is Hölder continuous, that means $|g(x) - g(y)| \leq C|x - y|^\alpha$ for some $\alpha > 0$.

Theorem 9.3.5. *If g is Hölder continuous then $v \in C^{1,2}$.*

The problem is that $v(t, x) = \int G_t(x, y)g(y) dy$ where

$$G_t(x, y) = \int_0^t (2\pi s)^{-d/2} e^{-|y-x|^2/2s} ds$$

Looking at the y with $|y - x|^2 \leq s$ leads to a lower bound

$$G_t(x, y) \geq C \int_{|y-x|^2}^t s^{-d/2} ds = O(|y - x|^{2-d})$$

From this it follows that first derivative blows up like δ^{1-d} and the second like δ^{-d} , which is not integrable near 0. However if g is Hölder continuity of g its smoothness can save the day. For details, see pages 12-13 of Friedman (1964).

9.4 Feynman-Kac Formula

In this section, we will consider what happens when we add cu to the right-hand side of the heat equation where $c(x)$ is bounded and continuous. That is, we will study

- (a) $u \in C^{1,2}$ and $u_t = \frac{1}{2}\Delta u + cu$ in $(0, \infty) \times \mathbf{R}^d$.
- (b) u is continuous at each point of $\{0\} \times \mathbf{R}^d$ and $u(0, x) = f(x)$, where f is bounded and continuous.

If $c(x) \leq 0$, then this equation describes heat flow with cooling. That is, $u(t, x)$ gives the temperature at the point $x \in \mathbf{R}^d$ at time t , when the heat at x at time t dissipates at the rate $-c(x)$.

The first step, as usual, is to find a local martingale.

Theorem 9.4.1. *Let $c_s = \int_0^s c(B_r) dr$. If u satisfies (a), then*

$$M_s = u(t - s, B_s) \exp(c_s)$$

is a local martingale on $[0, t]$

Proof. The first step in doing this is to generalize our third version of Itô's formula

$$\begin{aligned} & u(t - s, B_s) \exp(c_s) - u(t, B_0) \\ &= \int_0^s -u_t(t - r, B_r) \exp(c_r) dr + \int_0^s \exp(c_r) \nabla u(t - r, B_r) \cdot dB_r \\ &+ \int_0^s u(t - r, B_r) c(B_r) \exp(c_r) dr + \frac{1}{2} \int_0^s \Delta u(t - r, B_r) \exp(c_r) dr \end{aligned} \tag{9.4.1}$$

Proof of (9.4.1). As in the three previous proofs of Itô's formula, we subdivide the interval $[0, s]$ with $s_i^n = si/2^n$. We write the difference as

$$\begin{aligned} & [u(t - s_{i+1}^n, B(s_{i+1}^n)) - u(t - s_i^n, B(s_i^n))] \exp(c(s_{i+1}^n)) \\ & + u(t - s_i^n, B(s_i^n)) [\exp(c(s_{i+1}^n)) - \exp(c(s_i^n))] \end{aligned}$$

The limit of the first line gives the first, second, and fourth terms. Using

$$\exp(c(s_{i+1}^n)) - \exp(c(s_i^n)) = \exp(c(s_i^n)) \left\{ \exp \left(\int_{s_i^n}^{s_{i+1}^n} c(B_r) dr \right) - 1 \right\}$$

we see that the limit of the second line gives the third term. $\square$

Rearranging, the right-hand side of (9.4.1) is

$$\begin{aligned} & = \int_0^s \left(-u_t + cu + \frac{1}{2} \Delta u \right) (t - r, B_r) \exp(c_r) dr \\ & \quad + \int_0^s \exp(c_r) \nabla u(t - r, B_r) \cdot dB_r \end{aligned}$$

which proves the desired result, since $-u_t + cu + \frac{1}{2} \Delta u = 0$, so using Theorem 7.6.4 the second term is a local martingale. $\square$

The next step, again, is a uniqueness result.

Theorem 9.4.2. *If there is a solution that is bounded on $[0, T] \times \mathbf{R}^d$ for any $T < \infty$, it must be*

$$v(t, x) \equiv E_x \{ f(B_t) \exp(c_t^t) \} \quad (9.4.2)$$

Proof. Under our assumptions on c and u , M_s , $0 \leq s < t$, is a bounded martingale and $M_t \equiv \lim_{s \rightarrow t} M_s = f(B_t) \exp(c_t)$. Since M_s is uniformly integrable it follows that

$$u(t, x) = E_x M_0 = E_x M_t = v(t, x)$$

which proves the desired result. $\square$

As before, it is easy to show that if v is smooth enough it is a solution.

Theorem 9.4.3. *If $v \in C^{1,2}$, then it satisfies (a).*

Proof. When we write the solution in (9.4.2), we think about running space-time Brownian motion $(t - s, B_s)$ from time 0 to t , so we have

$$v(t + h, x) = E_x [v(t, B_h) \exp(c_h)]$$

Subtracting $v(t, x)$ from each side

$$v(t + h, x) - v(t, x) = E_x [v(t, B_h) \exp(c_h) - v(t, x)]$$

Adding and subtracting $v(t, B_h)$ and the right hand side and using Taylor's theorem

$$\begin{aligned} v(t, B_h) \exp(c_h) - v(t, x) &\approx \sum_{i=1}^d \frac{\partial v}{\partial x_i}(t, x)(B_h^i - x) \\ &+ \frac{1}{2} \sum_{1 \leq i, j \leq d} \frac{\partial^2 v}{\partial x_i \partial x_j}(t, x)(B_h^i - x)(B_h^j - x) + v(t, B_h)(\exp(c_h) - 1) + o(h) \end{aligned}$$

Since for $i \neq j$, $W_i(t)$, $W_j^2(t) = t$ and $W_i(t)W_j(t)$ are martingales.

$$E_x[v(t, B_h) \exp(c_h) - v(t, x)] \approx \sum_{i=1}^d \frac{\partial^2 v}{\partial x_i^2} h + E_x[v(t, B_h)(\exp(c_h) - 1)] + o(h)$$

Dividing each side by h and letting $h \rightarrow 0$ we get

$$\frac{\partial v}{\partial t} = \frac{1}{2} \Delta v(t, x) + c(x)v(t, x)$$

which proves the desired result. $\square$

Theorem 9.4.4. *v satisfies (b).*

Proof. If $|c| \leq M$, then $e^{-Mt} \leq \exp(c_t^t) \leq e^{Mt}$, so $\exp(c_t^t) \rightarrow 1$ as $t \rightarrow 0$. Letting $\|f\|_\infty = \sup_x |f(x)|$ this result implies

$$|E_x \exp(c_t^t) f(B_t) - E_x f(B_t)| \leq \|f\|_\infty E_x |\exp(c_t^t) - 1| \rightarrow 0$$

Theorem 9.2.4 implies that $(t, x) \rightarrow E_x f(B_t)$ is continuous at each point of $\{0\} \times \mathbf{R}^d$ and the desired result follows. $\square$

This brings us to the problem of determining when v is smooth enough to be a solution.

Theorem 9.4.5. *If f and c Hölder continuous, then $v \in C^{1,2}$ and, hence, satisfies (a).*

To solve the problem in this case, we use a trick to reduce our result to the previous case. We begin by observing that

$$c_s = \int_0^s c(B_r) dr$$

is continuous and locally of bounded variation. So using the ordinary change of variables formula if $h \in C^1$ then

$$h(c_t) - h(c_0) = \int_0^t h'(c_s) c(B_s) ds$$

Taking $h(x) = e^{-x}$ we have

$$\exp(-c_t) - 1 = - \int_0^t \exp(-c_s) c(B_s) ds$$

Multiplying by $-\exp(c_t)$ gives

$$\exp(c_t) - 1 = \int_0^t c(B_s) \exp(c_t - c_s) ds$$

Plugging in the definitions of c_t^t and c_s^t we have

$$\exp\left(\int_0^t c(B_r) dr\right) = 1 + \int_0^t c(B_s) \exp\left(\int_s^t c(B_r) dr\right) ds$$

Multiplying by $f(B_t)$, taking expected values, and using Fubini's theorem, which is justified since everything is bounded, gives

$$v(t, x) = E_x f(B_t) + \int_0^t E_x \left\{ c(B_s) \exp\left(\int_s^t c(B_r) dr\right) f(B_t) \right\} ds$$

Conditioning on $\mathcal{F}_s$, noticing $c(B_s) \in \mathcal{F}_s$, and using the Markov property

$$E_x \left(c(B_s) \exp\left(\int_s^t c(B_r) dr\right) f(B_t) \middle| \mathcal{F}_s \right) = c(B_s) v(t-s, B_s)$$

Taking the expected value of the last equation and plugging into the previous one, we have

$$\begin{aligned} v(t, x) &= E_x f(B_t) + \int_0^t E_x \{ c(B_s) v(t-s, B_s) \} ds \\ &\equiv v_1(t, x) + v_2(t, x) \end{aligned} \tag{9.4.3}$$

The first term on the right, $v_1(t, x)$, is $C^{1,2}$ by Theorem 9.2.3. The second term, $v_2(t, x)$, is of the form considered in the last section with $g(r, x) = c(x)v(r, x)$. If we start with the trivial observation that if c and f are bounded, then v is bounded on $[0, N] \times \mathbf{R}^d$, and apply (2.6b), we see that

$$|v_2(t, x) - v_2(t, y)| \leq C_N |x - y| \quad \text{whenever } t \leq N$$

To get a similar estimate for v_1 let $\bar{B}_t$ be a Brownian motion starting at 0 and observe that since f is Hölder continuous

$$\begin{aligned} |v_1(t, x) - v_1(t, y)| &= |E\{f(x + \bar{B}_t) - f(y + \bar{B}_t)\}| \\ &\leq E|f(x + \bar{B}_t) - f(y + \bar{B}_t)| \leq C|x - y|^\alpha \end{aligned}$$

Combining the last two estimates we see that $v(r, x)$ is Hölder continuous locally in t . Since c and v are bounded and we have supposed that c, x is Hölder continuous the triangle inequality implies $g(r, x)$ is Hölder continuous locally in t . The result then follows from smoothness results for the inhomogeneous heat equation.

9.5 Dirichlet problem

The developments here and in Section 9.7 follow port and Stone (1978). In this section, we will consider the Dirichlet problem. That is, given an open set G we will seek a function that satisfies

- (a) $u \in C^2$ and $\Delta u = 0$ in G .
- (b) At each point of ∂G , u is continuous and $u = f$, a bounded function.

Physically u is an equilibrium temperature distribution when ∂G is held at a fixed temperature profile f .

Let $\tau = \inf\{t > 0 : B_t \notin G\}$. We assume throughout this section that

$$P_x(\tau < \infty) = 1 \quad \text{for all } x \in G. \quad (9.5.1)$$

If u satisfies (a), then $M_t = u(B_t)$ is a local martingale on $[0, \tau)$. That is, there is an increasing sequence of stopping times T_n so that $u(B(t \wedge T_n))$ is a martingale. To turn a local martingale into an honest martingale we introduce a function $\gamma : [0, \infty) \rightarrow [0, \tau)$ stays constant for one unit of time when its value reaches T_n . That is, $\gamma(t) = t$ for $0 \leq t \leq T_1$, $\gamma(t) = T_1$ for $T_1 \leq t \leq T_1 + 1$, and in general

$$\begin{aligned} \gamma(t) &= t - k \text{ for } T_k + k \leq t \leq T_{k+1} + k \\ \gamma(t) &= T_{k+1} \text{ for } T_{k+1} + k \leq t \leq T_{k+1} + k + 1 \end{aligned}$$

Since $\gamma(t)$ is a stopping time and $\gamma(n) \leq T_n$.

Lemma 9.5.1. $M_t = u(B(\gamma(t)))$ is a martingale.

Theorem 9.5.2. If there is a solution of the Dirichlet problem that is bounded, it must be

$$v(x) \equiv E_x f(B_\tau)$$

Proof. Since u is bounded M_t is a bounded martingale, and hence $\lim_{t \rightarrow \infty} M_t$ exists. The boundary condition implies $M_\infty = f(B_\tau)$. Since the martingale is uniformly integrable, $u(x) = E_x M_\infty = E_x f(B_\tau)$. $\square$

Theorem 9.5.3. If $v \in C^2$, then it satisfies (a).

Proof. We learned this argument, which avoids the use of stochastic calculus, from Liggett (2010). Let $x \in D$ and $B = B(x, r) \subset D$. If τ_B is the exit time from B then the strong Markov property implies

$$v(x) = E_x v(B(\tau_B)).$$

Using Taylor's theorem

$$\begin{aligned} E_x v(B(\tau_B)) - v(x) &= E_x \sum_{i=1}^d \frac{\partial v}{\partial x_i}(x) [B_i(\tau_B) - x_i] \\ &\quad + \frac{1}{2} E_x \sum_{1 \leq i, j \leq d} \frac{\partial^2 v}{\partial x_i \partial x_j}(x) [B_i(\tau_B) - x_i] [B_j(\tau_B) - x_j] + o(r^2) \end{aligned}$$

Since for $i \neq j$, $B_i(t)$, $B_i^2(t) - t$ and $B_i(t)B_j(t)$ are martingales, we have

$$\begin{aligned} E_x[B_i(\tau_B) - x_i] &= 0 & E_x[B_i(\tau_B) - x_i][B_j(\tau_B) - x_j] &= 0 \quad \text{when } i \neq j \\ E_x[B_i(\tau_B) - x_i]^2 &= E_0[B_i^2(\tau_B)] = E_x\tau_B \end{aligned}$$

Using these results in the previous formula

$$0 = E_x v(B(\tau_B)) - v(x) = \frac{1}{2} \Delta v(x) E_x \tau_B + o(r^2)$$

which proves the desired result. $\square$

Theorem 9.5.4. $v \in C^\infty$.

Proof. Let $x \in D$ and $B = B(x, r) \subset D$. If τ_B is the exit time from B then the strong Markov property implies

$$v(x) = E_x v(B(\tau_B)) = \int_{\partial D(x, r)} \bar{v}(y) \pi(dy).$$

where π is surface measure on the boundary $\partial D(x, r)$ normalized to be a probability measure.

The last result is the **averaging property** of harmonic functions. Now a simple analytical argument takes over to show $v \in C^\infty$. To prove this we let ψ be a nonnegative infinitely differentiable function that vanishes on $[\delta^2, \infty)$ but is not $\equiv 0$. It is routine to show that

$$h(x) = \int_{D(x, \delta)} \psi(|y - x|^2) v(y) dy \in C^\infty$$

Changing to spherical coordinates and using the averaging property it is easy to see that $h(x)$ is a constant multiple of $v(x)$ so $v \in C^\infty$. $\square$

As the next example shows, v does not always satisfy (b).

Example 9.5.5. Punctured Disc. Let $d = 3$ and let $G = D - K$, where $D = \{x : |x| < 1\}$, and $K = \{x : x_1 = x_2 = 0, x_3 \geq 0\}$. Our uniqueness result and the fact that three dimensional Brownian motion never hits a line implies that the solution must be $v(x) = E_x f(B_{\tau(D)})$ where $\tau(D)$ is the exit time from D . If we let f be 0 on the boundary of D and $1 - x_3$ on K then the solution $v \equiv 0$ in G and boundary condition will not be satisfied

Example 9.5.6. Lebesgue's Thorn. Let $d \geq 3$ and let

$$G = (-1, 1)^d - \cup_{1 \leq n \leq \infty} \{[2^{-n}, 2^{-n+1}] \times [-a_n, a_n]^{d-1}\}$$

where the $n = \infty$ term is the single point $\{0\}$. We will now show that if $a_n \downarrow 0$ sufficiently fast, then $P_0(\tau = 0) = 0$. From this it will follow that if we define f to be 0 on the boundary of $(-1, 1)^d$ and $1 - x_1$ on the rest of the boundary, $E_0 f(B_\tau) < 1$ and the boundary condition is not satisfied.

Proof. $P_0((B_t^2, B_t^3) = (0, 0) \text{ for some } t > 0) = 0$, so with probability 1, a Brownian motion B_t starting at 0 will not hit

$$I_n = \{x : x_1 \in [2^{-n}, 2^{-n+1}], x_2 = x_3 = \dots = x_d = 0\}$$

Since B_t is transient in $d \geq 3$ it follows that for a.e. ω the distance between $\{B_s : 0 \leq s < \infty\}$ and I_n is positive. From the last observation, it follows immediately that if we let $T_n = \inf\{t : B_t \in [2^{-n}, 2^{-n+1}] \times [a_n, a_n]^{d-1}\}$ and pick a_n small enough, then $P_0(T_n < \infty) \leq 3^{-n}$. Now $\sum_{n=1}^{\infty} 3^{-n} = 3^{-1}(3/2) = 1/2$, so if we let $\tau = \inf\{t > 0 : B_t \notin G\}$ and $\sigma = \inf\{t > 0 : B_t \notin (-1, 1)^d\}$, then we have

$$P_0(\tau < \sigma) \leq \sum_{n=1}^{\infty} P_0(T_n < \infty) \leq \frac{1}{2}$$

Thus $P_0(\tau > 0) \geq P_0(\tau = \sigma) \geq 1/2$. Since $\tau = 0$ is measurable with respect to $\mathcal{F}_0^+$ we have $P_0(\tau = 0) = 1$. $\square$

A point $y \in \partial G$ is said to be a **regular point** if $P_y(\tau = 0) = 1$.

Theorem 9.5.7. *Let G be any open set. Suppose f is bounded and continuous and y is a regular point of ∂G . If $x_n \in G$ and $x_n \rightarrow y$, then $v(x_n) \rightarrow f(y)$.*

The first step is to show

Lemma 9.5.8. *If $t > 0$, then $x \rightarrow P_x(\tau \leq t)$ is lower semicontinuous. That is, if $x_n \rightarrow x$, then*

$$\liminf_{n \rightarrow \infty} P_{x_n}(\tau \leq t) \geq P_x(\tau \leq t)$$

Proof. By the Markov property

$$P_x(B_s \in G^c \text{ for some } s \in (\epsilon, t]) = \int p_\epsilon(x, y) P_y(\tau \leq t - \epsilon) dy$$

Since $y \rightarrow P_y(\tau \leq t - \epsilon)$ is bounded and measurable and

$$p_\epsilon(x, y) = (2\pi\epsilon)^{-d/2} e^{-|x-y|^2/2\epsilon}$$

it follows from the dominated convergence theorem that

$$x \rightarrow P_x(B_s \in G^c \text{ for some } s \in (\epsilon, t])$$

is continuous for each $\epsilon > 0$. Letting $\epsilon \downarrow 0$ shows that $x \rightarrow P_x(\tau \leq t)$ is an increasing limit of continuous functions and, hence, lower semicontinuous. $\square$

If y is regular for G and $t > 0$, then $P_y(\tau \leq t) = 1$, so it follows from Lemma 9.5.8 that if $x_n \rightarrow y$, then

$$\liminf_{n \rightarrow \infty} P_{x_n}(\tau \leq t) \geq 1 \quad \text{for all } t > 0$$

With this established, it is easy to complete the proof. Since f is bounded and continuous, it suffices to show that if $D(y, \delta) = \{x : |x - y| < \delta\}$, then

Lemma 9.5.9. *If y is regular for G and $x_n \rightarrow y$, then for all $\delta > 0$*

$$P_{x_n}(\tau < \infty, B_\tau \in D(y, \delta)) \rightarrow 1$$

Proof. Let $\epsilon > 0$ and pick t so small that

$$P_0\left(\sup_{0 \leq s \leq t} |B_s| > \frac{\delta}{2}\right) < \epsilon$$

Since $P_{x_n}(\tau \leq t) \rightarrow 1$ as $x_n \rightarrow y$, it follows from the choices above that

$$\begin{aligned} \liminf_{n \rightarrow \infty} P_{x_n}(\tau < \infty, B_\tau \in D(y, \delta)) &\geq \liminf_{n \rightarrow \infty} P_{x_n}\left(\tau \leq t, \sup_{0 \leq s \leq t} |B_s - x_n| \leq \frac{\delta}{2}\right) \\ &\geq \liminf_{n \rightarrow \infty} P_{x_n}(\tau \leq t) - P_0\left(\sup_{0 \leq s \leq t} |B_t| > \frac{\delta}{2}\right) > 1 - \epsilon \end{aligned}$$

Since ϵ was arbitrary, this proves the lemma and hence Theorem 9.5.7. $\square$

The next result shows that if G^c is not too small near y , then y is regular.

Theorem 9.5.10. Cone Condition. *If there is a cone V having vertex y and an $r > 0$ such that $V \cap D(y, r) \subset G^c$, then y is a regular point.*

Proof. The first thing to do is to define a **cone with vertex y , pointing in direction v , with opening a** as follows:

$$V(y, v, a) = \{x : x = y + \theta(v + z) \text{ where } \theta \in (0, \infty), z \perp v, \text{ and } |z| < a\}$$

Now that we have defined a cone, the rest is easy. Since the normal distribution is spherically symmetric,

$$P_y(B_t \in V(y, v, a)) = \epsilon_a > 0$$

where ϵ_a is a constant that depends only on the opening a . Let $r > 0$ be such that $V(y, v, a) \cap D(y, r) \subset G^c$. The continuity of Brownian paths implies

$$\lim_{t \rightarrow 0} P_y\left(\sup_{s \leq t} |B_s - y| > r\right) = 0$$

Combining the last two results with a trivial inequality we have

$$\epsilon_a \leq \liminf_{t \downarrow 0} P_y(B_t \in G^c) \leq \lim_{t \downarrow 0} P_y(\tau \leq t) \leq P_y(\tau = 0)$$

and it follows from Blumenthal's zero-one law that $P_y(\tau = 0) = 1$. $\square$

9.5.1 Exit distributions

Half-space. Let $\tau = \inf\{t : B_t \notin H\}$ where $H = \{(x, y) : x \in \mathbf{R}^{d-1}, y > 0\}$. For $\theta \in \mathbf{R}^{d-1}$ let

$$h_\theta(x, y) = \frac{C_d y}{(|x - \theta|^2 + y^2)^{d/2}} \quad (9.5.2)$$

where C_d is chosen so that $\int d\theta h_\theta(0, 1) = 1$.

Lemma 9.5.11. $\Delta h_\theta = 0$ in H

Proof. (a) Ignoring C_d and differentiating gives

$$\begin{aligned} D_{x_i} h_\theta &= -\frac{d}{2} \cdot \frac{2(x_i - \theta_i)y}{(|x - \theta|^2 + y^2)^{(d+2)/2}} \\ D_{x_i x_i} h_\theta &= -d \frac{y}{(|x - \theta|^2 + y^2)^{(d+2)/2}} + \frac{d(d+2)(x_i - \theta_i)^2 y}{(|x - \theta|^2 + y^2)^{(d+4)/2}} \\ D_y h_\theta &= \frac{1}{(|x - \theta|^2 + y^2)^{d/2}} - d \cdot \frac{y^2}{(|x - \theta|^2 + y^2)^{(d+2)/2}} \\ D_{yy} h_\theta &= -d \cdot \frac{y + 2y}{(|x - \theta|^2 + y^2)^{(d+2)/2}} + \frac{d(d+2)y^3}{(|x - \theta|^2 + y^2)^{(d+4)/2}} \end{aligned}$$

Adding up we see that

$$\begin{aligned} \sum_{i=1}^{d-1} D_{x_i x_i} h_\theta + D_{yy} h_\theta &= \frac{\{(-d)(d-1) + (-d) \cdot 3\}y}{(|x - \theta|^2 + y^2)^{(d+2)/2}} \\ &\quad + \frac{d(d+2)y(|x - \theta|^2 + y^2)}{(|x - \theta|^2 + y^2)^{(d+4)/2}} = 0 \end{aligned}$$

which completes the proof. $\square$

Theorem 9.5.12. Let f be bounded and continuous and

$$u(x, y) = \int d\theta h_\theta(x, y) f(\theta, 0)$$

u satisfies the Dirichlet problem in H with boundary value f . From this it follows that $h_\theta(x, y)$ gives the probability density $P_{x,y}(B(\tau_H) = (\theta, 0))$.

Proof. Interchanging the derivative and integral we see that $\Delta u = 0$. The next step is to show that it satisfies the boundary condition.

Clearly $\int d\theta h_\theta(x, y)$ is independent of x . To show that it is independent of y , let $x = 0$ and change variables $\theta_i = y\phi_i$ for $1 \leq i \leq d-1$ to get

$$\int d\theta h_\theta(0, y) = \int \frac{C_d y}{(y^2|\phi|^2 + y^2)^{d/2}} \cdot y^{d-1} d\phi = \int d\phi h_\phi(0, 1) = 1$$

Changing variables $\theta_i = x_i + r_i y$ and using dominated convergence

$$\int_{D(x, \epsilon)^c} d\theta h_\theta(x, y) = \int_{D(0, \epsilon/y)^c} dr h_r(0, 1) \rightarrow 0$$

Since $P_{x,y}(\tau < \infty) = 1$ for all $x \in H$, the last conclusion follows from Theorem 9.5.2. $\square$

When $d = 2$

$$h_\theta(0, y) = \frac{y}{\pi(|\theta|^2 + y^2)}$$

is Cauchy density. To see why this should be true start the Brownian motion at $(0, 0)$ and let $T_y = \inf\{t : B_t^2 = y\}$. The strong Markov property implies that $B^1(T_y)$ has independent increments. Brownian scaling implies $B^1(T_y) =_d y B^1(T_1)$. Since $B^1(T_y) =_d -B^1(T_y)$ the distribution must be the symmetric stable law with $\alpha = 1$.

Exit distributions for the ball. Let $D = \{x : |x| < 1\}$, let $\tau_D = \inf\{t : B_t \notin D\}$, and define the Poisson kernel by

$$k_y(x) = \frac{1 - |x|^2}{|x - y|^d} \quad (9.5.3)$$

Theorem 9.5.13. *If f is bounded and continuous, then*

$$E_x f(B_\tau) = u(x) = \int_{\partial D} k_y(x) f(y) \pi(dy) \quad (9.5.4)$$

where π is surface measure on ∂D normalized to be a probability measure.

The proof technique is the same as for the result in the half-space. We show that u satisfies the Dirichlet problem with boundary condition f . We leave it to the reader to show that for fixed y , $\Delta k_y(x) = 0$ and then complete the proof as above.

EXERCISES

9.5.1. Suppose h is bounded and harmonic on $\mathbf{R}^d$. Prove that h is constant.

9.5.2. Suppose $f \in C^2$ is superharmonic, takes values in $[0, \infty]$ and is continuous. Show that in dimensions 1 and 2, f must be constant. Give an example of the form $f(x) = g(|x|)$ in $d \geq 3$.

9.5.3. Let B_t be a d -dimensional Brownian motion and let $\tau = \inf\{t : B_t \notin G\}$. Suppose G is a connected open set G , all boundary points are regular and $P_x(\tau < \infty) < 1$ in G . Show that all bounded solutions of the Dirichlet problem with boundary value 0 on ∂G have the form $u(x) = c P_x(\tau = \infty)$. Hint: Define $\bar{u} = u + \|u\|_\infty P_x(\tau = \infty)$ in G , $\bar{u} = 0$ on G^c , and show that $u(B_y)$ is a submartingale.

9.5.4. Let $\tau = \inf\{t : B_t \notin G\}$ and consider now the Dirichlet problem in a connected open set G in $d \geq 3$ where all boundary points are regular and which has $P_x(\tau < \infty) < 1$ in G . Suppose that the boundary condition $f \equiv 0$. Generalize the previous result to conclude that all bounded solutions have the form $u(x) = cP_x(\tau = \infty)$.

9.5.5. Define a flat cone $\bar{V}(y, v, a)$ to be the intersection of $V(y, v, a)$ with a $d - 1$ dimensional hyperplane that contains the line $\{y + \theta v : \theta \in \mathbf{R}\}$. Show that Theorem 9.5.10 remains true if “cone” is replaced by “flat cone.”

9.6 Green's Functions and Potential Kernels

Let $D = B(0, r) = \{y : |y| < r\}$ the ball of radius r centered at 0. B_t will return to D i.o. in $d \leq 2$ but not in $d \geq 3$. In this section we will investigate the occupation time $\int_0^\infty 1_D(B_t) dt$. The first step is to show that for any x

$$P_x \left(\int_0^\infty 1_D(B_t) dt = \infty \right) = 1 \quad \text{in } d \leq 2, \quad (9.6.1)$$

$$E_x \int_0^\infty 1_D(B_t) dt < \infty \quad \text{in } d \geq 3. \quad (9.6.2)$$

Proof. Let $T_0 = 0$ and $G = B(0, 2r)$. For $k \geq 1$, let

$$\begin{aligned} S_k &= \inf\{t > T_{k-1} : B_t \in D\} \\ T_k &= \inf\{t > S_k : B_t \in G\} \end{aligned}$$

Writing τ for T_1 and using the strong Markov property, we get for $k \geq 1$

$$P_x \left(\int_{S_k}^{T_k} 1_D(B_t) dt \geq s \middle| \mathcal{F}_{S_k} \right) = P_{B(S_k)} \left(\int_0^\tau 1_D(B_t) dt \geq s \right) = H(s)$$

From the strong Markov property it follows that

$$\int_{S_k}^{T_k} 1_D(B_t) dt \quad \text{are i.i.d.}$$

Since these random variables have positive mean it follows from the strong law of large numbers that

$$\int_0^\infty 1_D(B_t) dt \geq \lim_{n \rightarrow \infty} \sum_{k=1}^n \int_{S_k}^{T_k} 1_D(B_t) dt = \infty \quad \text{a.s.}$$

proving (9.6.1).

If f is a nonnegative function, then Fubini's theorem implies

$$\begin{aligned} E_x \int_0^\infty f(B_t) dt &= \int_0^\infty E_x f(B_t) dt = \int_0^\infty \int p_t(x, y) f(y) dy dt \\ &= \int \int_0^\infty p_t(x, y) dt f(y) dy \end{aligned}$$

where $p_t(x, y) = (2\pi t)^{-d/2} e^{-|x-y|^2/2t}$ is the transition density for Brownian motion. As $t \rightarrow \infty$, $p_t(x, y) \sim (2\pi t)^{-d/2}$, so if $d \leq 2$ then $\int p_t(x, y) dt = \infty$. When $d \geq 3$, changing variables $t = |x - y|^2/2s$ gives

$$\begin{aligned} \int_0^\infty p_t(x, y) dt &= \int_0^\infty \frac{1}{(2\pi t)^{d/2}} e^{-|y-x|^2/2t} dt \\ &= \int_\infty^0 \left(\frac{s}{\pi|x-y|^2} \right)^{d/2} e^{-s} \left(-\frac{|x-y|^2}{2s^2} \right) ds \\ &= \frac{|x-y|^{2-d}}{2\pi^{d/2}} \int_0^\infty s^{(d/2)-2} e^{-s} ds \\ &= \frac{\Gamma(\frac{d}{2}-1)}{2\pi^{d/2}} |x-y|^{2-d} \end{aligned} \tag{9.6.3}$$

where $\Gamma(\alpha) = \int_0^\infty s^{\alpha-1} e^{-s} ds$ is the usual gamma function. If we define

$$G(x, y) = \int_0^\infty p_t(x, y) dt$$

then in $d \geq 3$, $G(x, y) < \infty$ for $x \neq y$, and

$$(2.4) \quad E_x \int_0^\infty f(B_t) dt = \int G(x, y) f(y) dy$$

To complete the proof now we observe that taking $f = 1_D$ with $D = B(0, r)$ and changing to polar coordinates

$$\int_D G(0, y) dy = \int_0^r s^{d-1} C_d s^{2-d} ds = \frac{C_d}{2} r^2 < \infty$$

To extend the last conclusion to $x \neq 0$ observe that applying the strong Markov property at the exit time τ from $B(0, |x|)$ for a Brownian motion starting at 0 and using the rotational symmetry we have

$$E_x \int_0^\infty 1_D(B_s) ds = E_0 \int_\tau^\infty 1_D(B_s) ds \leq E_0 \int_0^\infty 1_D(B_s) ds$$

which completes the proof. $\square$

We call $G(x, y)$ the **potential kernel**, because $G(\cdot, y)$ is the electrostatic potential of a unit charge at y . See Chapter 3 of Port and Stone

(1978) for more on this. In $d \leq 2$, $\int_0^\infty p_t(x, y) dt \equiv \infty$ so we have to take another approach to define a useful G :

$$G(x, y) = \int_0^\infty (p_t(x, y) - a_t) dt \quad (9.6.4)$$

where the a_t are constants we will choose to make the integral converge (at least when $x \neq y$). To see why this modified definition might be useful note that if $\int f(y) dy = 0$ then

$$\int G(x, y) f(y) dy = \lim_{t \rightarrow \infty} \int_0^t E_x f(B_s) ds$$

When $d = 1$ we let $a_t = p_t(0, 0)$. With this choice,

$$G(x, y) = \frac{1}{\sqrt{2\pi}} \int_0^\infty (e^{-(y-x)^2/2t} - 1) t^{-1/2} dt$$

and the integral converges, since the integrand is ≤ 0 and $\sim -(y-x)^2/2t^{3/2}$ as $t \rightarrow \infty$. Changing variables $t = (y-x)^2/2u$ gives

$$\begin{aligned} G(x, y) &= \frac{1}{\sqrt{2\pi}} \int_\infty^0 (e^{-u} - 1) \left(\frac{2u}{(y-x)^2} \right)^{1/2} \frac{-(y-x)^2}{2u^2} du \\ &= -\frac{|y-x|}{2\sqrt{\pi}} \int_0^\infty \left(\int_0^u e^{-s} ds \right) u^{-3/2} du \\ &= -\frac{|y-x|}{\sqrt{\pi}} \int_0^\infty ds e^{-s} \int_s^\infty \frac{u^{-3/2}}{2} du \\ &= -\frac{|y-x|}{\sqrt{\pi}} \int_0^\infty ds e^{-s} s^{-1/2} = -|y-x| \end{aligned} \quad (9.6.5)$$

since

$$\int_0^\infty ds e^{-s} s^{-1/2} = \int_0^\infty dr e^{-r^2/2} \sqrt{2} = \frac{1}{2} \sqrt{2} \cdot \sqrt{2\pi} = \sqrt{\pi}$$

The computation is almost the same for $d = 2$. The only thing that changes is the choice of a_t . If we try $a_t = p_t(0, 0)$ again, then for $x \neq y$ the integrand $\sim -t^{-1}$ as $t \rightarrow 0$ and the integral diverges, so we let

$a_t = p_t(0, e_1)$ where $e_1 = (1, 0)$. With this choice of a_t , we get

$$\begin{aligned}
 G(x, y) &= \frac{1}{2\pi} \int_0^\infty (e^{-|x-y|^2/2t} - e^{-1/2t}) t^{-1} dt \\
 &= \frac{1}{2\pi} \int_0^\infty \left(\int_{|x-y|^2/2t}^{1/2t} e^{-s} ds \right) t^{-1} dt \\
 &= \frac{1}{2\pi} \int_0^\infty ds e^{-s} \int_{|x-y|^2/2s}^{1/2s} t^{-1} dt \\
 &= \frac{1}{2\pi} \left(\int_0^\infty ds e^{-s} \right) (-\log(|x-y|^2)) \\
 &= \frac{-1}{\pi} \log(|x-y|)
 \end{aligned} \tag{9.6.6}$$

To sum up, the potential kernels are given by

$$G(x, y) = \begin{cases} (\Gamma(d/2 - 1)/2\pi^{d/2}) \cdot |x-y|^{2-d} & d \geq 3 \\ (-1/\pi) \cdot \log(|x-y|) & d = 2 \\ -1 \cdot |x-y| & d = 1 \end{cases} \tag{9.6.7}$$

The reader should note that in each case $G(x, y) = C\varphi(|x-y|)$ where φ is the harmonic function we used in Section 9.1. This is, of course, no accident. $x \rightarrow G(x, 0)$ is obviously spherically symmetric and, as we will see, satisfies $\Delta G(x, 0) = 0$ for $x \neq 0$, so the results above imply $G(x, 0) = A + B\varphi(|x|)$.

The formulas above correspond to $A = 0$, which is nice and simple. But what about the weird looking B 's? What is special about them? The answer is simple: They are chosen to make $\frac{1}{2}\Delta G(x, 0) = -\delta_0$ (a point mass at 0) in the distributional sense. It is easy to see that this happens in $d = 1$. In that case $\phi(x) = |x|$ then

$$\varphi'(x) = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}$$

so if $B = -1$ then $B\varphi''(x) = -2\delta_0$. More sophisticated readers can check that this is also true in $d \geq 2$. (See F. John (1982), pages 96-97.)

9.7 Poisson's Equation

In this section, we study the following equation in a bounded open set G :

- (a) $u \in C^2$ and $\frac{1}{2}\Delta u = -g$ in G , where g is bounded and continuous.
- (b) At each point of ∂G , u is continuous and $u = 0$.

We can add a solution of the Dirichlet problem to replace $u = 0$ in (b) by $u = f$. It follows from Theorem 9.1.1 that

Theorem 9.7.1. *Let $\tau = \inf\{t > 0 : B_t \notin G\}$. If u satisfies (a), then*

$$M_t = u(B_t) + \int_0^t g(B_s) ds$$

is a local martingale on $[0, \tau)$.

The next step is to prove a uniqueness result.

Theorem 9.7.2. *If there is a solution of (a) that is bounded, it must be*

$$v(x) \equiv E_x \left(\int_0^\tau g(B_t) dt \right)$$

Note that if $g \equiv 1$ the solution is $v(x) = E_x \tau$.

Proof. If u satisfies (a) then M_t defined is a local martingale on $[0, \tau)$. If G is bounded, then $E_x \tau < \infty$ for all $x \in G$. If u and g are bounded then for $t < \tau$

$$|M_t| \leq \|u\|_\infty + \tau \|g\|_\infty$$

Since M_t is dominated by an integrable random variable it follows that

$$M_\tau \equiv \lim_{t \uparrow \tau} M_t = \int_0^\tau g(B_t) dt$$

and $u(x) = E_x(M_\tau)$. □

As in our treatment of the Dirichlet problem:

Theorem 9.7.3. *If $v \in C^2$, then it satisfies (a).*

Proof. Let $x \in D$ and $B(x, r) \subset D$. The strong Markov property implies

$$v(x) = E_x \int_0^{\tau_B} g(B_s) ds + E_x v(B(\tau_B))$$

Using Taylor's theorem

$$\begin{aligned} E_x v(B(\tau_B)) - v(x) &= E_x \sum_{i=1}^d \frac{\partial v}{\partial x_i}(x) [B_i(\tau_B) - x_i] \\ &\quad + \frac{1}{2} E_x \sum_{1 \leq i, j \leq d} \frac{\partial^2 v}{\partial x_i \partial x_j}(x) [B_i(\tau_B) - x_i] [B_j(\tau_B) - x_j] + o(r^2) \end{aligned}$$

Since for $i \neq j$, $B_i(t)$, $B_i^2(t) - t$ and $B_i(t)B_j(t)$ are martingales, we have

$$E_x v(B(\tau_B)) - v(x) = \frac{1}{2} \Delta v(x) E_x \tau_B + o(r^2)$$

On the other hand $E_x \int_0^{\tau_B} g(B_s) ds = [g(x) + o(1)] E_x \tau_B$ so letting $r \rightarrow 0$ we have $(1/2) \Delta v = -g$ □

After the extensive discussion in the last section on the Dirichlet problem, the conditions needed to guarantee that the boundary conditions hold should come as no surprise.

Theorem 9.7.4. *Suppose that G and g are bounded. Let y be a regular point of ∂G . If $x_n \in G$ and $x_n \rightarrow y$, then $v(x_n) \rightarrow 0$.*

Proof. We begin by observing:

- (i) It follows from Lemma 9.5.9 that if $\epsilon > 0$, then $P_{x_n}(\tau > \epsilon) \rightarrow 0$.
- (ii) If G is bounded, then $C = \sup_x E_x \tau < \infty$ and, hence, $\|v\|_\infty \leq C\|g\|_\infty < \infty$.

Let $\epsilon > 0$. Beginning with some elementary inequalities then using the Markov property we have

$$\begin{aligned} |v(x_n)| &\leq E_{x_n} \left(\int_0^{\tau \wedge \epsilon} |g(B_s)| ds \right) + E_{x_n} \left(\left| \int_\epsilon^\tau g(B_s) ds \right| ; \tau > \epsilon \right) \\ &\leq \epsilon \|g\|_\infty + E_{x_n}(|v(B_\epsilon)|; \tau > \epsilon) \leq \epsilon \|g\|_\infty + \|v\|_\infty P_{x_n}(\tau > \epsilon) \end{aligned}$$

Letting $n \rightarrow \infty$, and using (i) and (ii) proves the desired result since ϵ is arbitrary. $\square$

Last, but not least, we come to the question of proving $v \in C^2$. For these developments we will assume that g is defined on $\mathbf{R}^d$ not just in G and has compact support. Recall we are supposing G is bounded and notice that the values of g on G^c are irrelevant, so there will be no loss of generality if we later want to suppose that $\int g(x) dx = 0$. We begin with the case $d \geq 3$, because in this case implies

$$\bar{w}(x) = E_x \int_0^\infty |g(B_t)| dt < \infty$$

and, moreover, is a bounded function of x . This means we can define

$$w(x) = E_x \int_0^\infty g(B_t) dt,$$

use the strong Markov property to conclude

$$w(x) = E_x \int_0^\tau g(B_t) dt + E_x w(B_\tau),$$

and change notation to get

$$v(x) = w(x) - E_x w(B_\tau). \quad (9.7.1)$$

Theorem 9.5.4 tells us that the second term is C^∞ in G , so to verify that $v \in C^2$ we need only prove that w is, a task in $d \geq 3$ that is made simpler by the fact that gives the following explicit formula

$$w(x) = C_d \int |x - y|^{2-d} g(y) dy$$

The details are tedious, so we will content ourselves to state the result.

Theorem 9.7.5. *In $d \geq 2$ if g is Hölder continuous, then w is C^2 . In $d = 1$ it is sufficient to suppose g is continuous*

The last result settles the question of smoothness in $d \geq 3$. To extend the result to $d \leq 2$, we use the potential kernels defined in (9.6.4). Using this interpretation of w , it is easy to see that (9.7.1) holds, so again our problem is reduced to proving that w is C^2 , which is a problem in calculus. Once all the computations are done, we find that Theorem 9.7.5 holds in $d = 2$ and that in $d = 1$, it is sufficient to assume that g is continuous. The reader can find details in The reader can find a proof either in Port and Stone (1978), pages 116-117, or in Gilbarg and Trudinger (1977), pages 53-55.

9.7.1 Occupation times

Half space. To explain why we want to define the potential kernels, we return to Brownian motion in the half space, $H = \{y : y_d > 0\}$. Let $\tau = \inf\{t : B_t \notin H\}$, and let

$$\bar{y} = (y_1, \dots, y_{d-1}, -y_d)$$

be the reflection of y through the plane $\{y \in \mathbf{R}^d : y_d = 0\}$

Theorem 9.7.6. *If $x \in H$, $f \geq 0$ has compact support, and $\{x : f(x) > 0\} \subset H$ then*

$$E_x \left(\int_0^\tau f(B_t) dt \right) = \int G(x, y) f(y) dy - \int G(x, \bar{y}) f(y) dy$$

Proof. Using Fubini's theorem which is justified since $f \geq 0$, then using the reflection principle and the fact that $\{x : f(x) > 0\} \subset H$ we have

$$\begin{aligned} E_x \int_0^\tau f(B_t) dt &= \int_0^\infty E_x(f(B_t); \tau > t) dt \\ &= \int_0^\infty \int_H (p_t(x, y) - p_t(x, \bar{y})) f(y) dy dt \\ &= \int_0^\infty \int_H (p_t(x, y) - a_t) f(y) dy dt \\ &\quad - \int_0^\infty \int_H (p_t(x, \bar{y}) - a_t) f(y) dy dt \\ &= \int G(x, y) f(y) dy - \int G(x, \bar{y}) f(y) dy \end{aligned}$$

The compact support of f and the formulas for G imply that $\int |G(x, y) f(y)| dy$ and $\int |G(x, \bar{y}) f(y)| dy$ are finite, so the last two equalities are valid. $\square$

The proof given above simplifies considerably in the case $d \geq 3$; however, part of the point of the proof above is that, with the definition we have chosen for G in the recurrent case, the formulas and proofs can be the same for all d .

Let $G_H(x, y) = G(x, y) - G(x, \bar{y})$. We think of $G_H(x, y)$ as the “expected occupation time (density) at y for a Brownian motion starting at x and killed when it leaves H .” The rationale for this interpretation is that (for suitable f)

$$E_x \left(\int_0^\tau f(B_t) dt \right) = \int G_H(x, y) f(y) dy$$

The answer is very simple in $d = 1$.

$$G_H(x, y) = 2(x \wedge y) \quad \text{for all } x, y > 0 \quad (9.7.2)$$

It is somewhat surprising that $y \rightarrow G_H(x, y)$ is constant $= 2x$ for $y \geq x$, that is, all points $y > x$ have the same expected occupation time!

Proof. $G(x, y) = -|x - y|$, so $G_H(x, y) = -|x - y| + |x + y|$. Separating things into cases, we see that

$$G_H(x, y) = \begin{cases} -(x - y) + (x + y) = 2y & \text{when } 0 < y < x \\ -(y - x) + (x + y) = 2x & \text{when } x < y \end{cases}$$

which gives the desired formula. □

Ball. Let $G(x, y)$ be the Green's functions defined in (9.6.7)

Theorem 9.7.7. *If g is bounded and measurable then*

$$E_x \int_0^\tau g(B_t) dt = \int G_D(x, y) g(y) dy$$

where

$$G_D(x, y) = G(x, y) - \int \frac{1 - |x|^2}{|x - z|^d} G(z, y) \pi(dz) \quad (9.7.3)$$

This follows from (9.7.1) and (9.5.4). This formula can be simplified considerably. Turning to page 123 in Folland (1976) we find

Theorem 9.7.8. *In $d \geq 3$, if $0 < |y| < 1$ then*

$$G_D(x, y) = G(x, y) - |y|^{2-d} G(x, y/|y|^2)$$

Proof. By (9.7.1) it is enough to show that (i) the second term is harmonic in D and (ii) $= G(x, y)$ on the boundary. $y \rightarrow y/|y|^2$ maps points $y \in D$

to D^c , so (i) holds. To prove (ii), we note

$$\begin{aligned} G(x, y) - |y|^{2-d} G(x, y/|y|^2) &= \frac{C_d}{|x - y|^{d-2}} - \frac{C_d}{|y|^{d-2}} \left| x - \frac{y}{|y|^2} \right|^{-(d-2)} \\ &= \frac{C_d}{|x - y|^{d-2}} - \frac{C_d}{|x|y| - y|y|^{-1}|^{d-2}} = 0 \end{aligned}$$

The last equality follows from a fact useful for the next proof:

Lemma 9.7.9. *If $|x| = 1$ then $|x|y| - y|y|^{-1}| = |x - y|$.*

Using $|z|^2 = z \cdot z$ and then $|x|^2 = 1$ we have

$$\begin{aligned} |x|y| - y|y|^{-1}|^2 &= |x|^2|y|^2 - 2x \cdot y + 1 \\ &= |y|^2 - 2x \cdot y + |x|^2 = |x - y|^2 \end{aligned}$$

which completes the proof. $\square$

Again turning to page 123 of Folland (1976) we “guess”

Theorem 9.7.10. *In $d = 2$ if $0 < |y| < 1$ then*

$$G_D(x, y) = \frac{-1}{\pi} (\ln |x - y| - \ln |x|y| - y|y|^{-1}|)$$

We leave the proof to the reader.

In $d = 1$, where $D = (-1, 1)$, (9.7.3) tells us that

$$\begin{aligned} G_D(x, y) &= G(x, y) - \frac{x+1}{2} G(1, y) - \frac{1-x}{2} G(-1, y) \\ &= -|x - y| + \frac{x+1}{2} (1 - y) + \frac{1-x}{2} (y + 1) \\ &= -|x - y| + 1 - xy \end{aligned}$$

Considering the two cases $x \geq y$ and $x \leq y$ leads to

$$G_D(x, y) = \begin{cases} (1-x)(1+y) & -1 \leq y \leq x \leq 1 \\ (1-y)(1+x) & -1 \leq x \leq y \leq 1 \end{cases} \quad (9.7.4)$$

Geometrically, if we fix y then $x \rightarrow G_D(x, y)$ is determined by the conditions that it is linear on $[0, y]$ and on $[y, 1]$ with $G_D(0, y) = G_D(1, y) = 0$, and $(1/2)G_D''(y, y) = -\delta_y$, a point mass at y . That is, G_D' is discontinuous at y and $G_D'(y-, y) - G_D'(y+, y) = 2$.

EXERCISES

Exercise 9.7.1. Let G be a bounded open set and let $\tau = \inf\{t : B_t \notin G\}$. What equation should we solve to find $w(x) = E_x \tau^2$?

9.8 Schrödinger Equation

The developments here follow Chung and Zhao (1995). In this section, we will consider what happens when we add cu where c is a bounded continuous function to the left-hand side of the Dirichlet problem. That is, given a bounded continuous function f on ∂G , we will study

- (a) $u \in C^2$ and $\frac{1}{2}\Delta u + cu = 0$ in G .
- (b) At each point of ∂G , u is continuous and $u = f$.

As always, the first step in solving the PDE is to find a local martingale.

Theorem 9.8.1. *Let $\tau = \inf\{t > 0 : B_t \notin G\}$. If u satisfies (a) then*

$$M_t = u(B_t) \exp\left(\int_0^t c(B_s) ds\right)$$

is a local martingale on $[0, \tau)$.

Proof. Let $c_t = \int_0^t c(B_s) ds$. Applying the version of Itô's from (9.4.1) gives

$$\begin{aligned} u(B_t) \exp(c_t) - u(B_0) &= \int_0^t \exp(c_s) \nabla u(B_s) \cdot dB_s + \int_0^t u(B_s) \exp(c_s) c(B_s) ds \\ &\quad + \frac{1}{2} \int_0^t \Delta u(B_s) \exp(c_s) ds \end{aligned}$$

for $t < \tau$. This proves result, since $\frac{1}{2}\Delta u + cu = 0$, and using theorem 7.6.4 the first term on the right-hand side is a local martingale on $[0, \tau)$. $\square$

At this point, the reader might expect that the next step, as it has been five times before, is to assume that everything is bounded and conclude that if there is a solution that is bounded, it must be

$$v(x) \equiv E_x(f(B_\tau) \exp(c_\tau))$$

We will not do this, however, because the following simple example shows that this result is false.

Example 9.8.2. Let $d = 1$, $G = (-a, a)$, $c \equiv \gamma$, and $f \equiv 1$. The equation we are considering is

$$\frac{1}{2}u'' + \gamma u = 0 \quad u(a) = u(-a) = 1$$

The general solution is $A \cos bx + B \sin bx$, where $b = \sqrt{2\gamma}$. So if we want the boundary condition to be satisfied we must have

$$\begin{aligned} 1 &= A \cos ba + B \sin ba \\ 1 &= A \cos(-ba) + B \sin(-ba) = A \cos ba - B \sin ba \end{aligned}$$

Adding the two equations and then subtracting them it follows that

$$2 = 2A \cos ba \quad 0 = 2B \sin ba$$

From this we see that $B = 0$ always works and we may or may not be able to solve for A .

If $\cos ba = 0$ then there is no solution. If $\cos ba \neq 0$ then $u(x) = \cos bx / \cos ba$ is a solution. We will see later that if $ab < \pi/2$ then

$$v(x) = \cos bx / \cos ba$$

However this cannot possibly hold for $ab > \pi/2$ since $v(x) \geq 0$ while the right-hand side is < 0 for some values of x .

We will see below that the trouble with the last example is that if $ab > \pi/2$ then $c \equiv \gamma$ is too large, or to be precise, if we let

$$w(x) = E_x \exp \left(\int_0^\tau c(B_s) ds \right)$$

then $w \equiv \infty$ in $(-a, a)$. The rest of this section is devoted to showing that if $w \not\equiv \infty$, then “everything is fine.” The development will require several stages. The first step is to show

Lemma 9.8.3. *Let $\theta > 0$. There is a $\mu > 0$ so that if H is an open set with Lebesgue measure $|H| \leq \mu$ and $\tau_H = \inf\{t > 0 : B_t \notin H\}$ then*

$$\sup_x E_x (\exp(\theta \tau_H)) \leq 2$$

Proof. Pick $\gamma > 0$ so that $e^{\theta\gamma} \leq 4/3$. Clearly,

$$P_x(\tau_H > \gamma) \leq \int_H \frac{1}{(2\pi\gamma)^{d/2}} e^{-|x-y|^2/2\gamma} dy \leq \frac{|H|}{(2\pi\gamma)^{d/2}} \leq \frac{1}{4}$$

if we pick μ small enough. Using the Markov property we can conclude that

$$\begin{aligned} P_x(\tau_H > k\gamma) &= E_x(P_{B_\gamma}(\tau_H > (k-1)\gamma); \tau_H > \gamma) \\ &\leq \frac{1}{4} \sup_y P_y(\tau_H > (k-1)\gamma) \end{aligned}$$

So it follows by induction that for all integers $k \geq 0$ we have

$$\sup_x P_x(\tau_H > k\gamma) \leq \frac{1}{4^k}$$

Since e^x is increasing, and $e^{\theta\gamma} \leq 4/3$ by assumption, we have

$$\begin{aligned} E_x \exp(\theta \tau_H) &\leq \sum_{k=1}^{\infty} \exp(\theta \gamma k) P_x((k-1)\gamma < \tau_H \leq k\gamma) \\ &\leq \sum_{k=1}^{\infty} \left(\frac{4}{3}\right)^k \frac{1}{4^{k-1}} = \frac{4}{3} \sum_{k=1}^{\infty} \frac{1}{3^{k-1}} = \frac{4}{3} \cdot \frac{1}{1 - \frac{1}{3}} = 2 \end{aligned}$$

Careful readers may have noticed that we left $\tau_H = 0$ out of the expected value. However, by the Blumenthal 0-1 law either $P_x(\tau_H = 0) = 0$ in which case our computation is valid, or $P_x(\tau_H = 0) = 1$ in which case $E_x \exp(\theta \tau_H) = 1$. $\square$

Let $c^* = \sup_x |c(x)|$. By Lemma 9.8.3, we can pick r_0 so small that if $T_r = \inf\{t : |B_t - B_0| > r\}$ and $r \leq r_0$, then $E_x \exp(c^* T_r) \leq 2$ for all x .

Lemma 9.8.4. *Let $2\delta \leq r_0$. If $D(x, 2\delta) \subset G$ and $y \in D(x, \delta)$, then*

$$w(y) \leq 2^{d+2} w(x)$$

The reason for our interest in this is that it shows $w(x) < \infty$ implies $w(y) < \infty$ for $y \in D(x, \delta)$.

Proof. If $D(y, r) \subset G$, and $r \leq r_0$ then the strong Markov property implies

$$\begin{aligned} w(y) &= E_y[\exp(c_{T_r}) w(B(T_r))] \leq E_y[\exp(c^* T_r) w(B(T_r))] \\ &= E_y[\exp(c^* T_r)] \int_{\partial D(y, r)} w(z) \pi(dz) \leq 2 \int_{\partial D(y, r)} w(z) \pi(dz) \end{aligned} \quad (9.8.1)$$

where π is surface measure on $\partial D(y, r)$ normalized to be a probability measure, since the exit time, T_r , and exit location, $B(T_r)$, are independent.

If $\delta \leq r_0$ and $D(y, \delta) \subset G$, multiplying the last inequality by r^{d-1} and integrating from 0 to δ gives

$$\frac{\delta^d}{d} w(y) \leq 2 \cdot \frac{1}{\sigma_d} \int_{D(y, \delta)} w(z) dz$$

where σ_d is the surface area of $\{x : |x| = 1\}$. Rearranging we have

$$\int_{D(y, \delta)} w(z) dz \geq 2^{-1} \frac{\delta^d}{C_o} w(y) \quad (9.8.2)$$

where $C_o = d/\sigma_d$ is a constant that depends only on d .

Repeating the argument in (9.8.1) with $y = x$ and using the fact that $c_{T_r} \geq -c^* T_r$ gives

$$\begin{aligned} w(x) &= E_x[\exp(c_{T_r}) w(B(T_r))] \geq E_x[\exp(-c^* T_r) w(B(T_r))] \\ &= E_x[\exp(-c^* T_r)] \int_{\partial D(x, r)} w(z) \pi(dz) \end{aligned}$$

Since $1/x$ is convex, Jensen's inequality implies

$$E_x[\exp(-c^* T_r)] \geq 1/E_x[\exp(c^* T_r)] \geq 1/2$$

Combining the last two displays, multiplying by r^{d-1} , and integrating from 0 to 2δ we get the lower bound

$$\frac{(2\delta)^d}{d} w(x) \geq 2^{-1} \cdot \frac{1}{\sigma_d} \int_{D(x, 2\delta)} w(z) dz$$

Rearranging and using $D(x, 2\delta) \supset D(y, \delta)$, $w \geq 0$ we have

$$w(x) \geq 2^{-1} \frac{C_o}{(2\delta)^d} \int_{D(y, \delta)} w(z) dz \quad (9.8.3)$$

where again $C_o = d/\sigma_d$. Combining (9.8.3) and (9.8.2) it follows that

$$\begin{aligned} &\geq 2^{-1} \frac{C_o}{(2\delta)^d} \int_{D(y, \delta)} w(z) dz \\ &\geq 2^{-1} \frac{C_o}{(2\delta)^d} \cdot 2^{-1} \frac{\delta^d}{C_o} w(y) = 2^{-(d+2)} w(y) \end{aligned}$$

which gives the desired result. $\square$

Lemma 9.8.4 and a simple covering argument lead to

Theorem 9.8.5. *Let G be a connected open set. If $w \not\equiv \infty$ then*

$$w(x) < \infty \quad \text{for all } x \in G$$

Proof. From Lemma 9.8.4, we see that if $w(x) < \infty$, $2\delta \leq r_0$, and $D(x, 2\delta) \subset G$, then $w < \infty$ on $D(x, \delta)$. From this result, it follows that $G_0 = \{x : w(x) < \infty\}$ is an open subset of G . To argue now that G_0 is also closed (when considered as a subset of G) we observe that if $2\delta < r_0$, $D(y, 3\delta) \subset G$, and we have $x_n \in G_0$ with $x_n \rightarrow y \in G$ then for n sufficiently large, $y \in D(x_n, \delta)$ and $D(x_n, 2\delta) \subset G$, so $w(y) < \infty$. $\square$

Before we proceed to the uniqueness result, we want to strengthen the last conclusion.

Theorem 9.8.6. *Let G be a connected open set with finite Lebesgue measure, $|G| < \infty$. If $w \not\equiv \infty$ then*

$$\sup_x w(x) < \infty$$

Proof. Let $K \subset G$ be compact so that $|G - K| < \mu$ the constant in Lemma 9.8.3 for $\theta = c^*$. For each $x \in K$ we can pick a δ_x so that $2\delta_x \leq r_0$ and $D(x, 2\delta_x) \subset G$. The open sets $D(x, \delta_x)$ cover K so there is a finite subcover $D(x_i, \delta_{x_i})$, $1 \leq i \leq I$. Clearly,

$$\sup_{1 \leq i \leq I} w(x_i) < \infty$$

Lemma 9.8.4 implies $w(y) \leq 2^{d+2}w(x_i)$ for $y \in D(x_i, \delta_{x_i})$, so

$$M = \sup_{y \in K} w(y) < \infty$$

If $y \in H = G - K$, then $E_y(\exp(c^* \tau_H)) \leq 2$ by (6.3a) so using the strong Markov property

$$\begin{aligned} w(y) &= E_y(\exp(c_{\tau_H}); \tau_H = \tau) + E_y(\exp(c_{\tau_H})w(B_{\tau_H}); B_{\tau_H} \in K) \\ &\leq 2 + ME_y(\exp(c_{\tau_H}); B_{\tau_H} \in K) \leq 2 + 2M \end{aligned}$$

we have the desired result. $\square$

With Theorem 9.8.6 established, we are now more than ready to prove our uniqueness result. To simplify the statements of the results that follow we will now list the assumptions that we will make for the rest of the section.

- (A1) G is a bounded connected open set.
- (A2) f and c are bounded and continuous.
- (A3) $w \not\equiv \infty$.

Theorem 9.8.7. *If there is a solution that is bounded, it must be*

$$v(x) \equiv E_x(f(B_\tau) \exp(c_\tau))$$

Proof. Theorem 9.8.1 implies that $M_s = u(B_s) \exp(c_s)$ is a local martingale on $[0, \tau)$. Since f , c , and u are bounded, letting $s \uparrow \tau \wedge t$ and using the bounded convergence theorem gives

$$u(x) = E_x(f(B_\tau) \exp(c_\tau); \tau \leq t) + E_x(u(B_t) \exp(c_t); \tau > t)$$

Since f is bounded and $w(x) = E_x \exp(c_\tau) < \infty$, the dominated convergence theorem implies that as $t \rightarrow \infty$, the first term converges to $E_x(f(B_\tau) \exp(c_\tau))$. To show that the second term $\rightarrow 0$, we begin with the observation that since $\{\tau > t\} \in \mathcal{F}_t$, the definition of conditional expectation and the Markov property imply

$$\begin{aligned} E_x(u(B_t) \exp(c_t); \tau > t) &= E_x(E_x(u(B_t) \exp(c_t) | \mathcal{F}_t); \tau > t) \\ &= E_x(u(B_t) \exp(c_t) w(B_t); \tau > t) \end{aligned}$$

Now we claim that for all $y \in G$

$$w(y) \geq \exp(-c^*)P_y(\tau \leq 1) \geq \epsilon > 0$$

The first inequality is trivial. The last two follow easily from (A1).

Replacing $w(B_t)$ by ϵ ,

$$\begin{aligned} E_x(|u(B_t)| \exp(c_t); \tau > t) &\leq \epsilon^{-1} E_x(|u(B_t)| \exp(c_t); \tau > t) \\ &\leq \epsilon^{-1} \|u\|_\infty E_x(\exp(c_t); \tau > t) \rightarrow 0 \end{aligned}$$

as $t \rightarrow \infty$, by the dominated convergence theorem since $w(x) = E_x \exp(c_\tau) < \infty$ and $P_x(\tau < \infty) = 1$. Going back to the first equation in the proof, we have shown $u(x) = v(x)$ and the proof is complete. $\square$

This completes our consideration of uniqueness. The next stage in our program, fortunately, is as easy as it always has been. Recall that here and in what follows we are assuming (A1)–(A3).

Theorem 9.8.8. *If $v \in C^2$, then it satisfies (a) in G .*

Proof. Let $x \in D$ and $B(x, r) \subset D$. Let σ be the exit time from $B(x, r)$. The strong Markov property implies

$$v(x) = E_x v(B_\sigma \exp(c_\sigma))$$

Using Taylor's theorem

$$\begin{aligned} E_x v(B(\sigma) \exp(c_\sigma)) - v(x) &= E_x \sum_{i=1}^d \frac{\partial v}{\partial x_i}(x) [B_i(\tau_D) - x_i] \\ &\quad + \frac{1}{2} E_x \sum_{1 \leq i, j \leq d} \frac{\partial^2 v}{\partial x_i \partial x_j}(x) [B_i(\tau_D) - x_i] [B_j(\tau_D) - x_j] + o(r^2) \end{aligned}$$

Since for $i \neq j$, $B_i(t)$, $B_i^2(t) - t$ and $B_i(t)B_j(t)$ are martingales, we have

$$E_x v(B(\tau_D)) - v(x) = \frac{1}{2} \Delta v(x) E_x \tau_B + o(r^2)$$

On the other hand $E_x \int_0^{\tau_B} g(B_s) ds = [g(x) + o(1)] E_x \tau_B$ so letting $r \rightarrow 0$ we have $(1/2) \Delta v = -g$ $\square$

Having proved Theorem 9.8.8, the next step is to consider the boundary condition. As in the last two sections, we need the boundary to be regular.

Theorem 9.8.9. *v satisfies (b) at each regular point of ∂G .*

Proof. Let y be a regular point of ∂G . We showed in (4.5a) and (4.5b) that if $x_n \rightarrow y$, then $P_{x_n}(\tau \leq \delta) \rightarrow 1$ and $P_{x_n}(B_\tau \in D(y, \delta)) \rightarrow 1$ for all $\delta > 0$. Since c is bounded and f is bounded and continuous, the bounded convergence theorem implies that

$$E_{x_n}(\exp(c_\tau) f(B_\tau); \tau \leq 1) \rightarrow f(y)$$

To control the contribution from the rest of the space, we observe that if $|c| \leq M$ then using the Markov property and the boundedness of w established in Theorem 9.8.6 we have

$$\begin{aligned} E_{x_n}(\exp(c_\tau) f(B_\tau); \tau > 1) &\leq e^M \|f\|_\infty E_{x_n}(w(B_1); \tau > 1) \\ &\leq e^M \|f\|_\infty \|w\|_\infty P_{x_n}(\tau > 1) \rightarrow 0 \end{aligned}$$

which completes the proof. $\square$

This brings us finally to the problem of determining when v is smooth enough to be a solution.

Theorem 9.8.10. *If in addition to (A1)–(A3), c is Hölder continuous, then $v \in C^2$ and, hence, satisfies (a).*

Return to Example 9.8.2. Recall $d = 1$, $G = (-a, a)$, $c(x) \equiv \gamma > 0$, and $f \equiv 1$.

Theorem 9.8.11. *Let $\tau = \inf\{t : B_t \notin (-a, a)\}$. If $0 < \gamma < \pi^2/8a^2$, then*

$$E_x e^{\gamma\tau} = \frac{\cos(x\sqrt{2\gamma})}{\cos(a\sqrt{2\gamma})}$$

If $\gamma \geq \pi^2/8a^2$ then $E_x e^{\gamma\tau} \equiv \infty$.

Proof. By results in Example 9.8.2

$$u(x) = \cos(x\sqrt{2\gamma}) / \cos(a\sqrt{2\gamma})$$

is a nonnegative solution of

$$\frac{1}{2}u'' + \gamma u = 0 \quad u(-a) = u(a) = 1$$

To check that $w \not\equiv \infty$, we observe that Theorem 9.8.1 implies that $M_t = u(B_t)e^{\gamma t}$ is a local martingale on $[0, \tau)$. If we let T_n be a sequence of stopping times that reduces M_t , then

$$u(x) = E_x(u(B_{T_n \wedge n})e^{\gamma(T_n \wedge n)})$$

Letting $n \rightarrow \infty$ and noting $T_n \wedge n \uparrow \tau$, it follows from Fatou's lemma that

$$u(x) \geq E_x e^{\gamma\tau}$$

The last equation implies $w \not\equiv \infty$, so Theorem 9.8.8 implies the result for $\gamma < \pi^2/8a^2$.

To see that $w(x) \equiv \infty$ when $\gamma \geq \pi^2/8a^2$, suppose not. In this case (6.3) implies $v(x) = E_x(f(B_\tau) \exp(c_\tau))$ is the unique solution of our equation but computations in Example 9.8.2 imply that in this case there is no nonnegative solution. $\square$

Appendix A

Measure Theory Details

This Appendix proves the results from measure theory that were stated but not proved in the text.

A.1 Caratheódory's Extension Theorem

This section is devoted to the proof of:

Theorem A.1.1. *Let $\mathcal{S}$ be a semialgebra and let μ defined on $\mathcal{S}$ have $\mu(\emptyset) = 0$. Suppose (i) if $S \in \mathcal{S}$ is a finite disjoint union of sets $S_i \in \mathcal{S}$ then $\mu(S) = \sum_i \mu(S_i)$, and (ii) if $S_i, S \in \mathcal{S}$ with $S = +_{i \geq 1} S_i$ then $\mu(S) \leq \sum_i \mu(S_i)$. Then μ has a unique extension $\bar{\mu}$ that is a measure on $\bar{\mathcal{S}}$ the algebra generated by $\mathcal{S}$. If the extension is σ -finite then there is a unique extension ν that is a measure on $\sigma(\mathcal{S})$.*

Proof. Lemma 1.1.7 shows that $\bar{\mathcal{S}}$ is the collection of finite disjoint unions of sets in $\mathcal{S}$. We define $\bar{\mu}$ on $\bar{\mathcal{S}}$ by $\bar{\mu}(A) = \sum_i \mu(S_i)$ whenever $A = +_i S_i$. To check that $\bar{\mu}$ is well defined, suppose that $A = +_j T_j$ and observe $S_i = +_j (S_i \cap T_j)$ and $T_j = +_i (S_i \cap T_j)$, so (i) implies

$$\sum_i \mu(S_i) = \sum_{i,j} \mu(S_i \cap T_j) = \sum_j \mu(T_j)$$

In Section 1.1 we proved:

Lemma A.1.2. *Suppose only that (i) holds.*

(a) *If $A, B_i \in \bar{\mathcal{S}}$ with $A = +_{i=1}^n B_i$ then $\bar{\mu}(A) = \sum_i \bar{\mu}(B_i)$.*

(b) *If $A, B_i \in \bar{\mathcal{S}}$ with $A \subset \cup_{i=1}^n B_i$ then $\bar{\mu}(A) \leq \sum_i \bar{\mu}(B_i)$.*

To extend the additivity property to $A \in \bar{\mathcal{S}}$ that are countable disjoint unions $A = +_{i \geq 1} B_i$, where $B_i \in \bar{\mathcal{S}}$, we observe that each $B_i = +_j S_{i,j}$ with $S_{i,j} \in \mathcal{S}$ and $\sum_{i \geq 1} \bar{\mu}(B_i) = \sum_{i \geq 1, j} \mu(S_{i,j})$, so replacing the B_i 's by $S_{i,j}$'s we can without loss of generality suppose that the $B_i \in \mathcal{S}$. Now $A \in \bar{\mathcal{S}}$

implies $A = +_j T_j$ (a finite disjoint union) and $T_j = +_{i \geq 1} T_j \cap B_i$, so (ii) implies

$$\mu(T_j) \leq \sum_{i \geq 1} \mu(T_j \cap B_i)$$

Summing over j and observing that nonnegative numbers can be summed in any order,

$$\bar{\mu}(A) = \sum_j \mu(T_j) \leq \sum_{i \geq 1} \sum_j \mu(T_j \cap B_i) = \sum_{i \geq 1} \mu(B_i)$$

the last equality following from (i). To prove the opposite inequality, let $A_n = B_1 + \cdots + B_n$, and $C_n = A \cap A_n^c$. $C_n \in \bar{\mathcal{S}}$, since $\bar{\mathcal{S}}$ is an algebra, so finite additivity of $\bar{\mu}$ implies

$$\bar{\mu}(A) = \bar{\mu}(B_1) + \cdots + \bar{\mu}(B_n) + \bar{\mu}(C_n) \geq \bar{\mu}(B_1) + \cdots + \bar{\mu}(B_n)$$

and letting $n \rightarrow \infty$, $\bar{\mu}(A) \geq \sum_{i \geq 1} \bar{\mu}(B_i)$.

Having defined a measure on the algebra $\bar{\mathcal{S}}$ we now complete the proof by establishing

Theorem A.1.3. Carathéodory's Extension Theorem. *Let μ be a σ -finite measure on an algebra $\mathcal{A}$. Then μ has a unique extension to $\sigma(\mathcal{A}) =$ the smallest σ -algebra containing $\mathcal{A}$.*

Uniqueness. We will prove that the extension is unique before tackling the more difficult problem of proving its existence. The key to our uniqueness proof is Dynkin's $\pi - \lambda$ theorem, a result that we will use many times in the book. As usual, we need a few definitions before we can state the result. $\mathcal{P}$ is said to be a **π -system** if it is closed under intersection, i.e., if $A, B \in \mathcal{P}$ then $A \cap B \in \mathcal{P}$. For example, the collection of rectangles $(a_1, b_1] \times \cdots \times (a_d, b_d]$ is a π -system. $\mathcal{L}$ is said to be a **λ -system** if it satisfies: (i) $\Omega \in \mathcal{L}$. (ii) If $A, B \in \mathcal{L}$ and $A \subset B$ then $B - A \in \mathcal{L}$. (iii) If $A_n \in \mathcal{L}$ and $A_n \uparrow A$ then $A \in \mathcal{L}$. The reader will see in a moment that the next result is just what we need to prove uniqueness of the extension.

Theorem A.1.4. $\pi - \lambda$ Theorem. *If $\mathcal{P}$ is a π -system and $\mathcal{L}$ is a λ -system that contains $\mathcal{P}$ then $\sigma(\mathcal{P}) \subset \mathcal{L}$.*

Proof. We will show that

(a) if $\ell(\mathcal{P})$ is the smallest λ -system containing $\mathcal{P}$ then $\ell(\mathcal{P})$ is a σ -field.

The desired result follows from (a). To see this, note that since $\sigma(\mathcal{P})$ is the smallest σ -field and $\ell(\mathcal{P})$ is the smallest λ -system containing $\mathcal{P}$ we have

$$\sigma(\mathcal{P}) \subset \ell(\mathcal{P}) \subset \mathcal{L}$$

To prove (a) we begin by noting that a λ -system that is closed under intersection is a σ -field since

$$\begin{aligned} \text{if } A \in \mathcal{L} \text{ then } A^c &= \Omega - A \in \mathcal{L} \\ A \cup B &= (A^c \cap B^c)^c \\ \bigcup_{i=1}^n A_i \uparrow \bigcup_{i=1}^\infty A_i &\text{ as } n \uparrow \infty \end{aligned}$$

Thus, it is enough to show

(b) $\ell(\mathcal{P})$ is closed under intersection.

To prove (b), we let $\mathcal{G}_A = \{B : A \cap B \in \ell(\mathcal{P})\}$ and prove

(c) if $A \in \ell(\mathcal{P})$ then $\mathcal{G}_A$ is a λ -system.

To check this, we note: (i) $\Omega \in \mathcal{G}_A$ since $A \in \ell(\mathcal{P})$.

(ii) if $B, C \in \mathcal{G}_A$ and $B \supset C$ then $A \cap (B - C) = (A \cap B) - (A \cap C) \in \ell(\mathcal{P})$ since $A \cap B, A \cap C \in \ell(\mathcal{P})$ and $\ell(\mathcal{P})$ is a λ -system.

(iii) if $B_n \in \mathcal{G}_A$ and $B_n \uparrow B$ then $A \cap B_n \uparrow A \cap B \in \ell(\mathcal{P})$ since $A \cap B_n \in \ell(\mathcal{P})$ and $\ell(\mathcal{P})$ is a λ -system.

To get from (c) to (b), we note that since $\mathcal{P}$ is a π -system,

if $A \in \mathcal{P}$ then $\mathcal{G}_A \supset \mathcal{P}$ and so (c) implies $\mathcal{G}_A \supset \ell(\mathcal{P})$

i.e., if $A \in \mathcal{P}$ and $B \in \ell(\mathcal{P})$ then $A \cap B \in \ell(\mathcal{P})$. Interchanging A and B in the last sentence: if $A \in \ell(\mathcal{P})$ and $B \in \mathcal{P}$ then $A \cap B \in \ell(\mathcal{P})$ but this implies

if $A \in \ell(\mathcal{P})$ then $\mathcal{G}_A \supset \mathcal{P}$ and so (c) implies $\mathcal{G}_A \supset \ell(\mathcal{P})$.

This conclusion implies that if $A, B \in \ell(\mathcal{P})$ then $A \cap B \in \ell(\mathcal{P})$, which proves (b) and completes the proof. $\square$

To prove that the extension in Theorem A.1.3 is unique, we will show:

Theorem A.1.5. *Let $\mathcal{P}$ be a π -system. If ν_1 and ν_2 are measures (on σ -fields $\mathcal{F}_1$ and $\mathcal{F}_2$) that agree on $\mathcal{P}$ and there is a sequence $A_n \in \mathcal{P}$ with $A_n \uparrow \Omega$ and $\nu_i(A_n) < \infty$, then ν_1 and ν_2 agree on $\sigma(\mathcal{P})$.*

Proof. Let $A \in \mathcal{P}$ have $\nu_1(A) = \nu_2(A) < \infty$. Let

$$\mathcal{L} = \{B \in \sigma(\mathcal{P}) : \nu_1(A \cap B) = \nu_2(A \cap B)\}$$

We will now show that $\mathcal{L}$ is a λ -system. Since $A \in \mathcal{P}$, $\nu_1(A) = \nu_2(A)$ and $\Omega \in \mathcal{L}$. If $B, C \in \mathcal{L}$ with $C \subset B$ then

$$\begin{aligned} \nu_1(A \cap (B - C)) &= \nu_1(A \cap B) - \nu_1(A \cap C) \\ &= \nu_2(A \cap B) - \nu_2(A \cap C) = \nu_2(A \cap (B - C)) \end{aligned}$$

Here we use the fact that $\nu_i(A) < \infty$ to justify the subtraction. Finally, if $B_n \in \mathcal{L}$ and $B_n \uparrow B$, then part (iii) of Theorem 1.1.1 implies

$$\nu_1(A \cap B) = \lim_{n \rightarrow \infty} \nu_1(A \cap B_n) = \lim_{n \rightarrow \infty} \nu_2(A \cap B_n) = \nu_2(A \cap B)$$

Since $\mathcal{P}$ is closed under intersection by assumption, the $\pi - \lambda$ theorem implies $\mathcal{L} \supset \sigma(\mathcal{P})$, i.e., if $A \in \mathcal{P}$ with $\nu_1(A) = \nu_2(A) < \infty$ and $B \in \sigma(\mathcal{P})$ then $\nu_1(A \cap B) = \nu_2(A \cap B)$. Letting $A_n \in \mathcal{P}$ with $A_n \uparrow \Omega$, $\nu_1(A_n) = \nu_2(A_n) < \infty$, and using the last result and part (iii) of Theorem 1.1.1, we have the desired conclusion. $\square$

Exercise A.1.1. Give an example of two probability measures $\mu \neq \nu$ on $\mathcal{F}$ = all subsets of $\{1, 2, 3, 4\}$ that agree on a collection of sets $\mathcal{C}$ with $\sigma(\mathcal{C}) = \mathcal{F}$, i.e., the smallest σ -algebra containing $\mathcal{C}$ is $\mathcal{F}$.

Existence. Our next step is to show that a measure (not necessarily σ -finite) defined on an algebra $\mathcal{A}$ has an extension to the σ -algebra generated by $\mathcal{A}$. If $E \subset \Omega$, we let $\mu^*(E) = \inf \sum_i \mu(A_i)$ where the infimum is taken over all sequences from $\mathcal{A}$ so that $E \subset \cup_i A_i$. Intuitively, if ν is a measure that agrees with μ on $\mathcal{A}$, then it follows from part (ii) of Theorem 1.1.1 that

$$\nu(E) \leq \nu(\cup_i A_i) \leq \sum_i \nu(A_i) = \sum_i \mu(A_i)$$

so $\mu^*(E)$ is an upper bound on the measure of E . Intuitively, the measurable sets are the ones for which the upper bound is tight. Formally, we say that E is **measurable** if

$$\mu^*(F) = \mu^*(F \cap E) + \mu^*(F \cap E^c) \quad \text{for all sets } F \subset \Omega \quad (\text{A.1.1})$$

The last definition is not very intuitive, but we will see in the proofs below that it works very well.

It is immediate from the definition that μ^* has the following properties:

- (i) **monotonicity.** If $E \subset F$ then $\mu^*(E) \leq \mu^*(F)$.
- (ii) **subadditivity.** If $F \subset \cup_i F_i$, a countable union, then $\mu^*(F) \leq \sum_i \mu^*(F_i)$.

Any set function with $\mu^*(\emptyset) = 0$ that satisfies (i) and (ii) is called an **outer measure**. Using (ii) with $F_1 = F \cap E$ and $F_2 = F \cap E^c$ (and $F_i = \emptyset$ otherwise), we see that to prove a set is measurable, it is enough to show

$$\mu^*(F) \geq \mu^*(F \cap E) + \mu^*(F \cap E^c) \quad (\text{A.1.2})$$

We begin by showing that our new definition extends the old one.

Lemma A.1.6. *If $A \in \mathcal{A}$ then $\mu^*(A) = \mu(A)$ and A is measurable.*

Proof. Part (ii) of Theorem 1.1.1 implies that if $A \subset \cup_i A_i$ then

$$\mu(A) \leq \sum_i \mu(A_i)$$

so $\mu(A) \leq \mu^*(A)$. Of course, we can always take $A_1 = A$ and the other $A_i = \emptyset$ so $\mu^*(A) \leq \mu(A)$.

To prove that any $A \in \mathcal{A}$ is measurable, we begin by noting that the inequality is (A.1.2) trivial when $\mu^*(F) = \infty$, so we can without loss of generality assume $\mu^*(F) < \infty$. To prove that (A.1.2) holds when $E = A$, we observe that since $\mu^*(F) < \infty$ there is a sequence $B_i \in \mathcal{A}$ so that $\cup_i B_i \supset F$ and

$$\sum_i \mu(B_i) \leq \mu^*(F) + \epsilon$$

Since μ is additive on $\mathcal{A}$, and $\mu = \mu^*$ on $\mathcal{A}$ we have

$$\mu(B_i) = \mu^*(B_i \cap A) + \mu^*(B_i \cap A^c)$$

Summing over i and using the subadditivity of μ^* gives

$$\mu^*(F) + \epsilon \geq \sum_i \mu^*(B_i \cap A) + \sum_i \mu^*(B_i \cap A^c) \geq \mu^*(F \cap A) + \mu^*(F^c \cap A)$$

which proves the desired result since ϵ is arbitrary. $\square$

Lemma A.1.7. *The class $\mathcal{A}^*$ of measurable sets is a σ -field, and the restriction of μ^* to $\mathcal{A}^*$ is a measure.*

Remark. This result is true for any outer measure.

Proof. It is clear from the definition that:

(a) If E is measurable then E^c is.

Our first nontrivial task is to prove:

(b) If E_1 and E_2 are measurable then $E_1 \cup E_2$ and $E_1 \cap E_2$ are.

Proof of (b). To prove the first conclusion, let G be any subset of Ω . Using subadditivity, the measurability of E_2 (let $F = G \cap E_1^c$ in (A.1.1)), and the measurability of E_1 , we get

$$\begin{aligned} \mu^*(G \cap (E_1 \cup E_2)) + \mu^*(G \cap (E_1^c \cap E_2^c)) \\ \leq \mu^*(G \cap E_1) + \mu^*(G \cap E_1^c \cap E_2) + \mu^*(G \cap E_1^c \cap E_2^c) \\ = \mu^*(G \cap E_1) + \mu^*(G \cap E_1^c) = \mu^*(G) \end{aligned}$$

To prove that $E_1 \cap E_2$ is measurable, we observe $E_1 \cap E_2 = (E_1^c \cup E_2^c)^c$ and use (a). $\square$

(c) Let $G \subset \Omega$ and $E_1, \dots, E_n$ be disjoint measurable sets. Then

$$\mu^*(G \cap \cup_{i=1}^n E_i) = \sum_{i=1}^n \mu^*(G \cap E_i)$$

Proof of (c). Let $F_m = \cup_{i \leq m} E_i$. E_n is measurable, $F_n \supset E_n$, and $F_{n-1} \cap E_n = \emptyset$, so

$$\begin{aligned} \mu^*(G \cap F_n) &= \mu^*(G \cap F_n \cap E_n) + \mu^*(G \cap F_n \cap E_n^c) \\ &= \mu^*(G \cap E_n) + \mu^*(G \cap F_{n-1}) \end{aligned}$$

The desired result follows from this by induction. $\square$

(d) If the sets E_i are measurable then $E = \cup_{i=1}^\infty E_i$ is measurable.

Proof of (d). Let $E'_i = E_i \cap (\cap_{j < i} E_j^c)$. (a) and (b) imply E'_i is measurable, so we can suppose without loss of generality that the E_i are pairwise disjoint. Let $F_n = E_1 \cup \dots \cup E_n$. F_n is measurable by (b), so using monotonicity and (c) we have

$$\begin{aligned} \mu^*(G) &= \mu^*(G \cap F_n) + \mu^*(G \cap F_n^c) \geq \mu^*(G \cap F_n) + \mu^*(G \cap E^c) \\ &= \sum_{i=1}^n \mu^*(G \cap E_i) + \mu^*(G \cap E^c) \end{aligned}$$

Letting $n \rightarrow \infty$ and using subadditivity

$$\mu^*(G) \geq \sum_{i=1}^\infty \mu^*(G \cap E_i) + \mu^*(G \cap E^c) \geq \mu^*(G \cap E) + \mu^*(G \cap E^c)$$

which is (A.1.2). $\square$

The last step in the proof of Theorem A.1.7 is

(e) If $E = \cup_i E_i$ where $E_1, E_2, \dots$ are disjoint and measurable, then

$$\mu^*(E) = \sum_{i=1}^\infty \mu^*(E_i)$$

Proof of (e). Let $F_n = E_1 \cup \dots \cup E_n$. By monotonicity and (c)

$$\mu^*(E) \geq \mu^*(F_n) = \sum_{i=1}^n \mu^*(E_i)$$

Letting $n \rightarrow \infty$ now and using subadditivity gives the desired conclusion.

$\square$

A.2 Which Sets Are Measurable?

The proof of Theorem A.1.3 given in the last section defines an extension to $\mathcal{A}^* \supset \sigma(\mathcal{A})$. Our next goal is to describe the relationship between these two σ -algebras. Let $\mathcal{A}_\sigma$ denote the collection of countable unions of sets in $\mathcal{A}$, and let $\mathcal{A}_{\sigma\delta}$ denote the collection of countable intersections of sets in $\mathcal{A}_\sigma$. Our first goal is to show that every measurable set is almost a set in $\mathcal{A}_{\sigma\delta}$.

Define the symmetric difference by $A \Delta B = (A - B) \cup (B - A)$.

Lemma A.2.1. *Let E be any set with $\mu^*(E) < \infty$.*

- (i) *For any $\epsilon > 0$, there is an $A \in \mathcal{A}_\sigma$ with $A \supset E$ and $\mu^*(A) \leq \mu^*(E) + \epsilon$.*
- (ii) *For any $\epsilon > 0$, there is a $B \in \mathcal{A}$ with $\mu(B \Delta E) \leq 2\epsilon$, where*
- (iii) *There is a $C \in \mathcal{A}_{\sigma\delta}$ with $C \supset E$ and $\mu^*(C) = \mu^*(E)$.*

Proof. By the definition of μ^* , there is a sequence A_i so that $A \equiv \cup_i A_i \supset E$ and $\sum_i \mu(A_i) \leq \mu^*(E) + \epsilon$. The definition of μ^* implies $\mu^*(A) \leq \sum_i \mu(A_i)$, establishing (i).

For (ii) we note that there is a finite union $B = \cup_{i=1}^n A_i$ so that $\mu(A - B) \leq \epsilon$, and hence $\mu(E - B) \leq \epsilon$. Since $\mu(B - E) \leq \mu(A - E) \leq \epsilon$ the desired result follows.

For (iii), let $A_n \in \mathcal{A}_\sigma$ with $A_n \supset E$ and $\mu^*(A_n) \leq \mu^*(E) + 1/n$, and let $C = \cap_n A_n$. Clearly, $C \in \mathcal{A}_{\sigma\delta}$, $B \supset E$, and hence by monotonicity, $\mu^*(C) \geq \mu^*(E)$. To prove the other inequality, notice that $B \subset A_n$ and hence $\mu^*(C) \leq \mu^*(A_n) \leq \mu^*(E) + 1/n$ for any n . $\square$

Theorem A.2.2. *Suppose μ is σ -finite on $\mathcal{A}$. $B \in \mathcal{A}^*$ if and only if there is an $A \in \mathcal{A}_{\sigma\delta}$ and a set N with $\mu^*(N) = 0$ so that $B = A - N (= A \cap N^c)$.*

Proof. It follows from Lemma A.1.6 and A.1.7 if $A \in \mathcal{A}_{\sigma\delta}$ then $A \in \mathcal{A}^*$. A.1.2 in Section A.1 and monotonicity imply sets with $\mu^*(N) = 0$ are measurable, so using Lemma A.1.7 again it follows that $A \cap N^c \in \mathcal{A}^*$. To prove the other direction, let Ω_i be a disjoint collection of sets with $\mu(\Omega_i) < \infty$ and $\Omega = \cup_i \Omega_i$. Let $B_i = B \cap \Omega_i$ and use Lemma A.2.1 to find $A_i^n \in \mathcal{A}_\sigma$ so that $A_i^n \supset B_i$ and $\mu(A_i^n) \leq \mu^*(B_i) + 1/n2^i$. Let $A_n = \cup_i A_i^n$. $B \subset A_n$ and

$$A_n - B \subset \sum_{i=1}^{\infty} (A_i^n - B_i)$$

so, by subadditivity,

$$\mu^*(A_n - B) \leq \sum_{i=1}^{\infty} \mu^*(A_i^n - B_i) \leq 1/n$$

Since $A_n \in \mathcal{A}_\sigma$, the set $A = \cap_n A_n \in \mathcal{A}_{\sigma\delta}$. Clearly, $A \supset B$. Since $N \equiv A - B \subset A_n - B$ for all n , monotonicity implies $\mu^*(N) = 0$, and the proof is complete. $\square$

A measure space $(\Omega, \mathcal{F}, \mu)$ is said to be **complete** if $\mathcal{F}$ contains all subsets of sets of measure 0. In the proof of Theorem A.2.2, we showed that $(\Omega, \mathcal{A}^*, \mu^*)$ is complete. Our next result shows that $(\Omega, \mathcal{A}^*, \mu^*)$ is the completion of $(\Omega, \sigma(\mathcal{A}), \mu)$.

Theorem A.2.3. *If $(\Omega, \mathcal{F}, \mu)$ is a measure space, then there is a complete measure space $(\Omega, \bar{\mathcal{F}}, \bar{\mu})$, called the **completion** of $(\Omega, \mathcal{F}, \mu)$, so that: (i) $E \in \bar{\mathcal{F}}$ if and only if $E = A \cup B$, where $A \in \mathcal{F}$ and $B \subset N \in \mathcal{F}$ with $\mu(N) = 0$, (ii) $\bar{\mu}$ agrees with μ on $\mathcal{F}$.*

Proof. The first step is to check that $\bar{\mathcal{F}}$ is a σ -algebra. If $E_i = A_i \cup B_i$ where $A_i \in \mathcal{F}$ and $B_i \subset N_i$ where $\mu(N_i) = 0$, then $\cup_i A_i \in \mathcal{F}$ and subadditivity implies $\mu(\cup_i N_i) \leq \sum_i \mu(N_i) = 0$, so $\cup_i E_i \in \bar{\mathcal{F}}$. As for complements, if $E = A \cup B$ and $B \subset N$, then $B^c \supset N^c$ so

$$E^c = A^c \cap B^c = (A^c \cap N^c) \cup (A^c \cap B^c \cap N)$$

$A^c \cap N^c$ is in $\mathcal{F}$ and $A^c \cap B^c \cap N \subset N$, so $E^c \in \bar{\mathcal{F}}$.

We define $\bar{\mu}$ in the obvious way: If $E = A \cup B$ where $A \in \mathcal{F}$ and $B \subset N$ where $\mu(N) = 0$, then we let $\bar{\mu}(E) = \mu(A)$. The first thing to show is that $\bar{\mu}$ is well defined, i.e., if $E = A_i \cup B_i$, $i = 1, 2$, are two decompositions, then $\mu(A_1) = \mu(A_2)$. Let $A_0 = A_1 \cap A_2$ and $B_0 = B_1 \cup B_2$. $E = A_0 \cup B_0$ is a third decomposition with $A_0 \in \mathcal{F}$ and $B_0 \subset N_1 \cup N_2$, and has the pleasant property that if $i = 1$ or 2

$$\mu(A_0) \leq \mu(A_i) \leq \mu(A_0) + \mu(N_1 \cup N_2) = \mu(A_0)$$

The last detail is to check that $\bar{\mu}$ is measure, but that is easy. If $E_i = A_i \cup B_i$ are disjoint, then $\cup_i E_i$ can be decomposed as $\cup_i A_i \cup (\cup_i B_i)$, and the $A_i \subset E_i$ are disjoint, so

$$\bar{\mu}(\cup_i E_i) = \mu(\cup_i A_i) = \sum_i \mu(A_i) = \sum_i \bar{\mu}(E_i) \quad \square$$

Theorem 1.1.11 allows us to construct Lebesgue measure λ on $(\mathbf{R}^d, \mathcal{R}^d)$. Using Theorem A.2.3, we can extend λ to be a measure on $(\mathbf{R}, \bar{\mathcal{R}}^d)$ where $\bar{\mathcal{R}}^d$ is the completion of $\mathcal{R}^d$. Having done this, it is natural (if somewhat optimistic) to ask: Are there any sets that are not in $\bar{\mathcal{R}}^d$? The answer is “Yes” and we will now give an example of a nonmeasurable B in $\mathbf{R}$.

A nonmeasurable subset of $[0,1)$

The key to our construction is the observation that λ is translation invariant: i.e., if $A \in \bar{\mathcal{R}}$ and $x + A = \{x + y : y \in A\}$, then $x + A \in \bar{\mathcal{R}}$ and $\lambda(A) = \lambda(x + A)$. We say that $x, y \in [0, 1)$ are equivalent and write $x \sim y$ if $x - y$ is a rational number. By the axiom of choice, there is a set B that contains exactly one element from each equivalence class. B is our nonmeasurable set, that is,

Theorem A.2.4. $B \notin \bar{\mathcal{R}}$.

Proof. The key is the following:

Lemma A.2.5. If $E \subset [0, 1)$ is in $\bar{\mathcal{R}}$, $x \in (0, 1)$, and $x +' E = \{(x + y) \bmod 1 : y \in E\}$, then $\lambda(E) = \lambda(x +' E)$.

Proof. Let $A = E \cap [0, 1 - x)$ and $B = E \cap [1 - x, 1)$. Let $A' = x + A = \{x + y : y \in A\}$ and $B' = x - 1 + B$. $A, B \in \bar{\mathcal{R}}$, so by translation invariance $A', B' \in \bar{\mathcal{R}}$ and $\lambda(A) = \lambda(A')$, $\lambda(B) = \lambda(B')$. Since $A' \subset [x, 1)$ and $B' \subset [0, x)$ are disjoint,

$$\lambda(E) = \lambda(A) + \lambda(B) = \lambda(A') + \lambda(B') = \lambda(x +' E) \quad \square$$

From Lemma A.2.5, it follows easily that B is not measurable; if it were, then $q +' B$, $q \in \mathbf{Q} \cap [0, 1)$ would be a countable disjoint collection of measurable subsets of $[0, 1)$, all with the same measure α and having

$$\bigcup_{q \in \mathbf{Q} \cap [0, 1)} (q +' B) = [0, 1)$$

If $\alpha > 0$ then $\lambda([0, 1)) = \infty$, and if $\alpha = 0$ then $\lambda([0, 1)) = 0$. Neither conclusion is compatible with the fact that $\lambda([0, 1)) = 1$ so $B \notin \bar{\mathcal{R}}$. $\square$

Exercise A.2.1. Let B be the nonmeasurable set constructed in Theorem A.2.4. (i) Let $B_q = q +' B$ and show that if $D_q \subset B_q$ is measurable, then $\lambda(D_q) = 0$. (ii) Use (i) to conclude that if $A \subset \mathbf{R}$ has $\lambda(A) > 0$, there is a nonmeasurable $S \subset A$.

Letting $B' = B \times [0, 1]^{d-1}$ where B is our nonmeasurable subset of $(0, 1)$, we get a nonmeasurable set in $d > 1$. In $d = 3$, there is a much more interesting example, but we need the reader to do some preliminary work. In Euclidean geometry, two subsets of $\mathbf{R}^d$ are said to be **congruent** if one set can be mapped onto the other by translations and rotations.

Claim. Two congruent measurable sets must have the same Lebesgue measure.

Exercise A.2.2. Prove the claim in $d = 2$ by showing (i) if B is a rotation of a rectangle A then $\lambda^*(B) = \lambda(A)$. (ii) If C is congruent to D then $\lambda^*(C) = \lambda^*(D)$.

Banach-Tarski Theorem

Banach and Tarski (1924) used the axiom of choice to show that it is possible to partition the sphere $\{x : |x| \leq 1\}$ in $\mathbf{R}^3$ into a finite number of sets $A_1, \dots, A_n$ and find congruent sets $B_1, \dots, B_n$ whose union is two disjoint spheres of radius 1! Since congruent sets have the same Lebesgue measure, at least one of the sets A_i must be nonmeasurable. The construction relies on the fact that the group generated by rotations

in $\mathbf{R}^3$ is not Abelian. Lindenbaum (1926) showed that this cannot be done with any bounded set in $\mathbf{R}^2$. For a popular account of the Banach-Tarski theorem, see French (1988).

Solovay's Theorem

The axiom of choice played an important role in the last two constructions of nonmeasurable sets. Solovay (1970) proved that its use is unavoidable. In his own words, "We show that the existence of a non-Lebesgue measurable set cannot be proved in Zermelo-Frankel set theory if the use of the axiom of choice is disallowed." This should convince the reader that all subsets of $\mathbf{R}^d$ that arise "in practice" are in $\bar{\mathcal{R}}^d$.

A.3 Kolmogorov's Extension Theorem

To construct some of the basic objects of study in probability theory, we will need an existence theorem for measures on infinite product spaces. Let $\mathbf{N} = \{1, 2, \dots\}$ and

$$\mathbf{R}^{\mathbf{N}} = \{(\omega_1, \omega_2, \dots) : \omega_i \in \mathbf{R}\}$$

We equip $\mathbf{R}^{\mathbf{N}}$ with the product σ -algebra $\mathcal{R}^{\mathbf{N}}$, which is generated by the **finite dimensional rectangles** = sets of the form $\{\omega : \omega_i \in (a_i, b_i]\}$ for $i = 1, \dots, n\}$, where $-\infty \leq a_i < b_i \leq \infty$.

Theorem A.3.1. Kolmogorov's extension theorem. *Suppose we are given probability measures μ_n on $(\mathbf{R}^n, \mathcal{R}^n)$ that are consistent, that is,*

$$\mu_{n+1}((a_1, b_1] \times \dots \times (a_n, b_n] \times \mathbf{R}) = \mu_n((a_1, b_1] \times \dots \times (a_n, b_n])$$

Then there is a unique probability measure P on $(\mathbf{R}^{\mathbf{N}}, \mathcal{R}^{\mathbf{N}})$ with

$$(*) \quad P(\omega : \omega_i \in (a_i, b_i], 1 \leq i \leq n) = \mu_n((a_1, b_1] \times \dots \times (a_n, b_n])$$

An important example of a consistent sequence of measures is

Example A.3.2. Let $F_1, F_2, \dots$ be distribution functions and let μ_n be the measure on $\mathbf{R}^n$ with

$$\mu_n((a_1, b_1] \times \dots \times (a_n, b_n]) = \prod_{m=1}^n (F_m(b_m) - F_m(a_m))$$

In this case, if we let $X_n(\omega) = \omega_n$, then the X_n are independent and X_n has distribution F_n .

Proof of Theorem A.3.1. Let $\mathcal{S}$ be the sets of the form $\{\omega : \omega_i \in (a_i, b_i], 1 \leq i \leq n\}$, and use (*) to define P on $\mathcal{S}$. $\mathcal{S}$ is a semialgebra, so by Theorem A.1.1 it is enough to show that if $A \in \mathcal{S}$ is a disjoint

union of $A_i \in \mathcal{S}$, then $P(A) \leq \sum_i P(A_i)$. If the union is finite, then all the A_i are determined by the values of a finite number of coordinates and the conclusion follows from the proof of Theorem 1.1.11.

Suppose now that the union is infinite. Let $\mathcal{A} = \{\text{finite disjoint unions of sets in } \mathcal{S}\}$ be the algebra generated by $\mathcal{S}$. Since $\mathcal{A}$ is an algebra (by Lemma 1.1.7)

$$B_n \equiv A - \cup_{i=1}^n A_i$$

is a finite disjoint union of rectangles, and by the result for finite unions,

$$P(A) = \sum_{i=1}^n P(A_i) + P(B_n)$$

It suffices then to show

Lemma A.3.3. *If $B_n \in \mathcal{A}$ and $B_n \downarrow \emptyset$ then $P(B_n) \downarrow 0$.*

Proof. Suppose $P(B_n) \downarrow \delta > 0$. By repeating sets in the sequence, we can suppose

$$B_n = \cup_{k=1}^{K_n} \{\omega : \omega_i \in (a_i^k, b_i^k], 1 \leq i \leq n\} \quad \text{where } -\infty \leq a_i^k < b_i^k \leq \infty$$

The strategy of the proof is to approximate the B_n from within by compact rectangles with almost the same probability and then use a diagonal argument to show that $\cap_n B_n \neq \emptyset$. There is a set $C_n \subset B_n$ of the form

$$C_n = \cup_{k=1}^{K_n} \{\omega : \omega_i \in [\bar{a}_i^k, \bar{b}_i^k], 1 \leq i \leq n\} \quad \text{with } -\infty < \bar{a}_i^k < \bar{b}_i^k < \infty$$

that has $P(B_n - C_n) \leq \delta/2^{n+1}$. Let $D_n = \cap_{m=1}^n C_m$.

$$P(B_n - D_n) \leq \sum_{m=1}^n P(B_m - C_m) \leq \delta/2$$

so $P(D_n) \downarrow$ a limit $\geq \delta/2$. Now there are sets $C_n^*, D_n^* \subset \mathbf{R}^n$ so that

$$C_n = \{\omega : (\omega_1, \dots, \omega_n) \in C_n^*\} \quad \text{and} \quad D_n = \{\omega : (\omega_1, \dots, \omega_n) \in D_n^*\}$$

Note that

$$C_n = C_n^* \times \mathbf{R} \times \mathbf{R} \times \dots \quad \text{and} \quad D_n = D_n^* \times \mathbf{R} \times \mathbf{R} \times \dots$$

so C_n and C_n^* (and D_n and D_n^*) are closely related but $C_n \subset \Omega$ and $C_n^* \subset \mathbf{R}^n$.

C_n^* is a finite union of closed rectangles, so

$$D_n^* = C_n^* \cap_{m=1}^{n-1} (C_m^* \times \mathbf{R}^{n-m})$$

is a compact set. For each m , let $\omega_m \in D_m$. $D_m \subset D_1$ so $\omega_{m,1}$ (i.e., the first coordinate of ω_m) is in D_1^* . Since D_1^* is compact, we can pick a subsequence $m(1, j) \geq j$ so that as $j \rightarrow \infty$,

$$\omega_{m(1,j),1} \rightarrow \text{a limit } \theta_1$$

For $m \geq 2$, $D_m \subset D_2$ and hence $(\omega_{m,1}, \omega_{m,2}) \in D_2^*$. Since D_2^* is compact, we can pick a subsequence of the previous subsequence (i.e., $m(2, j) = m(1, i_j)$ with $i_j \geq j$) so that as $j \rightarrow \infty$

$$\omega_{m(2,j),2} \rightarrow \text{a limit } \theta_2$$

Continuing in this way, we define $m(k, j)$ a subsequence of $m(k-1, j)$ so that as $j \rightarrow \infty$,

$$\omega_{m(k,j),k} \rightarrow \text{a limit } \theta_k$$

Let $\omega'_i = \omega_{m(i,i)}$. ω'_i is a subsequence of all the subsequences so $\omega'_{i,k} \rightarrow \theta_k$ for all k . Now $\omega'_{i,1} \in D_1^*$ for all $i \geq 1$ and D_1^* is closed so $\theta_1 \in D_1^*$. Turning to the second set, $(\omega'_{i,1}, \omega'_{i,2}) \in D_2^*$ for $i \geq 2$ and D_2^* is closed, so $(\theta_1, \theta_2) \in D_2^*$. Repeating the last argument, we conclude that $(\theta_1, \dots, \theta_k) \in D_k^*$ for all k , so $\omega = (\theta_1, \theta_2, \dots) \in D_k$ (no star here since we are now talking about subsets of Ω) for all k and

$$\emptyset \neq \cap_k D_k \subset \cap_k B_k$$

a contradiction that proves the desired result. $\square$

A.4 Radon-Nikodym Theorem

In this section, we prove the Radon-Nikodym theorem. To develop that result, we begin with a topic that at first may appear to be unrelated. Let $(\Omega, \mathcal{F})$ be a measurable space. α is said to be a **signed measure** on $(\Omega, \mathcal{F})$ if (i) α takes values in $(-\infty, \infty]$, (ii) $\alpha(\emptyset) = 0$, and (iii) if $E = \bigcup_i E_i$ is a disjoint union then $\alpha(E) = \sum_i \alpha(E_i)$, in the following sense:

If $\alpha(E) < \infty$, the sum converges absolutely and $\alpha(E) = \sum_i \alpha(E_i)$.

If $\alpha(E) = \infty$, then $\sum_i \alpha(E_i)^- < \infty$ and $\sum_i \alpha(E_i)^+ = \infty$.

Clearly, a signed measure cannot be allowed to take both the values ∞ and $-\infty$, since $\alpha(A) + \alpha(B)$ might not make sense. In most formulations, a signed measure is allowed to take values in either $(-\infty, \infty]$ or $[-\infty, \infty)$. We will ignore the second possibility to simplify statements later. As usual, we turn to examples to help explain the definition.

Example A.4.1. Let μ be a measure, f be a function with $\int f^- d\mu < \infty$, and let $\alpha(A) = \int_A f d\mu$. Exercise 5.8 implies that α is a signed measure.

Example A.4.2. Let μ_1 and μ_2 be measures with $\mu_2(\Omega) < \infty$, and let $\alpha(A) = \mu_1(A) - \mu_2(A)$.

The Jordan decomposition, Theorem A.4.6 below, will show that Example A.4.2 is the general case. To derive that result, we begin with two definitions. A set A is **positive** if every measurable $B \subset A$ has $\alpha(B) \geq 0$. A set A is **negative** if every measurable $B \subset A$ has $\alpha(B) \leq 0$.

Exercise A.4.1. In Example A.4.1, A is positive if and only if $\mu(A \cap \{x : f(x) < 0\}) = 0$.

Lemma A.4.3. (i) Every measurable subset of a positive set is positive.
(ii) If the sets A_n are positive then $A = \cup_n A_n$ is also positive.

Proof. (i) is trivial. To prove (ii), observe that

$$B_n = A_n \cap \left(\bigcap_{m=1}^{n-1} A_m^c \right) \subset A_n$$

are positive, disjoint, and $\cup_n B_n = \cup_n A_n$. Let $E \subset A$ be measurable, and let $E_n = E \cap B_n$. $\alpha(E_n) \geq 0$ since B_n is positive, so $\alpha(E) = \sum_n \alpha(E_n) \geq 0$. $\square$

The conclusions in Lemma A.4.3 remain valid if positive is replaced by negative. The next result is the key to the proof of Theorem A.4.5.

Lemma A.4.4. Let E be a measurable set with $\alpha(E) < 0$. Then there is a negative set $F \subset E$ with $\alpha(F) < 0$.

Proof. If E is negative, this is true. If not, let n_1 be the smallest positive integer so that there is an $E_1 \subset E$ with $\alpha(E_1) \geq 1/n_1$. Let $k \geq 2$. If $F_k = E - (E_1 \cup \dots \cup E_{k-1})$ is negative, we are done. If not, we continue the construction letting n_k be the smallest positive integer so that there is an $E_k \subset F_k$ with $\alpha(E_k) \geq 1/n_k$. If the construction does not stop for any $k < \infty$, let

$$F = \cap_k F_k = E - (\cup_k E_k)$$

Since $0 > \alpha(E) > -\infty$ and $\alpha(E_k) \geq 0$, it follows from the definition of signed measure that

$$\alpha(E) = \alpha(F) + \sum_{k=1}^{\infty} \alpha(E_k)$$

$\alpha(F) \leq \alpha(E) < 0$, and the sum is finite. From the last observation and the construction, it follows that F can have no subset G with $\alpha(G) > 0$, for then $\alpha(G) \geq 1/N$ for some N and we would have a contradiction. $\square$

Theorem A.4.5. Hahn decomposition. Let α be a signed measure. Then there is a positive set A and a negative set B so that $\Omega = A \cup B$ and $A \cap B = \emptyset$.

Proof. Let $c = \inf\{\alpha(B) : B \text{ is negative}\} \leq 0$. Let B_i be negative sets with $\alpha(B_i) \downarrow c$. Let $B = \cup_i B_i$. By Lemma A.4.3, B is negative, so by the definition of c , $\alpha(B) \geq c$. To prove $\alpha(B) \leq c$, we observe that $\alpha(B) = \alpha(B_i) + \alpha(B - B_i) \leq \alpha(B_i)$, since B is negative, and let $i \rightarrow \infty$. The last two inequalities show that $\alpha(B) = c$, and it follows from our definition of a signed measure that $c > -\infty$. Let $A = B^c$. To show

A is positive, observe that if A contains a set with $\alpha(E) < 0$, then by Lemma A.4.4, it contains a negative set F with $\alpha(F) < 0$, but then $B \cup F$ would be a negative set that has $\alpha(B \cup F) = \alpha(B) + \alpha(F) < c$, a contradiction. $\square$

The Hahn decomposition is not unique. In Example A.4.1, A can be any set with

$$\{x : f(x) > 0\} \subset A \subset \{x : f(x) \geq 0\} \quad \text{a.e.}$$

where $B \subset C$ a.e. means $\mu(B \cap C^c) = 0$. The last example is typical of the general situation. Suppose $\Omega = A_1 \cup B_1 = A_2 \cup B_2$ are two Hahn decompositions. $A_2 \cap B_1$ is positive and negative, so it is a **null set**: All its subsets have measure 0. Similarly, $A_1 \cap B_2$ is a null set.

Two measures μ_1 and μ_2 are said to be **mutually singular** if there is a set A with $\mu_1(A) = 0$ and $\mu_2(A^c) = 0$. In this case, we also say μ_1 is **singular with respect to μ_2** and write $\mu_1 \perp \mu_2$.

Exercise A.4.2. Show that the uniform distribution on the Cantor set (Example 1.2.7) is singular with respect to Lebesgue measure.

Theorem A.4.6. Jordan decomposition. *Let α be a signed measure. There are mutually singular measures α_+ and α_- so that $\alpha = \alpha_+ - \alpha_-$. Moreover, there is only one such pair.*

Proof. Let $\Omega = A \cup B$ be a Hahn decomposition. Let

$$\alpha_+(E) = \alpha(E \cap A) \quad \text{and} \quad \alpha_-(E) = -\alpha(E \cap B)$$

Since A is positive and B is negative, α_+ and α_- are measures. $\alpha_+(A^c) = 0$ and $\alpha_-(A) = 0$, so they are mutually singular. To prove uniqueness, suppose $\alpha = \nu_1 - \nu_2$ and D is a set with $\nu_1(D) = 0$ and $\nu_2(D^c) = 0$. If we set $C = D^c$, then $\Omega = C \cup D$ is a Hahn decomposition, and it follows from the choice of D that

$$\nu_1(E) = \alpha(C \cap E) \quad \text{and} \quad \nu_2(E) = -\alpha(D \cap E)$$

Our uniqueness result for the Hahn decomposition shows that $A \cap D = A \cap C^c$ and $B \cap C = A^c \cap C$ are null sets, so $\alpha(E \cap C) = \alpha(E \cap (A \cup C)) = \alpha(E \cap A)$ and $\nu_1 = \alpha_+$. $\square$

Exercise A.4.3. Show that $\alpha_+(E) = \sup\{\alpha(F) : F \subset E\}$.

Remark. Let α be a **finite signed measure** (i.e., one that does not take the value ∞ or $-\infty$) on $(\mathbf{R}, \mathcal{R})$. Let $\alpha = \alpha_+ - \alpha_-$ be its Jordan decomposition. Let $A(x) = \alpha((-\infty, x])$, $F(x) = \alpha_+((-\infty, x])$, and $G(x) = \alpha_-((-\infty, x])$. $A(x) = F(x) - G(x)$ so the distribution function for a finite signed measure can be written as a difference of two bounded increasing functions. It follows from Example A.4.2 that the converse is

also true. Let $|\alpha| = \alpha^+ + \alpha^-$. $|\alpha|$ is called the **total variation** of α , since in this example $|\alpha|((a, b])$ is the total variation of A over $(a, b]$ as defined in analysis textbooks. See, for example, Royden (1988), p. 103. We exclude the left endpoint of the interval since a jump there makes no contribution to the total variation on $[a, b]$, but it does appear in $|\alpha|$.

Our third and final decomposition is:

Theorem A.4.7. Lebesgue decomposition. *Let μ, ν be σ -finite measures. ν can be written as $\nu_r + \nu_s$, where ν_s is singular with respect to μ and*

$$\nu_r(E) = \int_E g \, d\mu$$

Proof. By decomposing $\Omega = \bigcup_i \Omega_i$, we can suppose without loss of generality that μ and ν are finite measures. Let $\mathcal{G}$ be the set of $g \geq 0$ so that $\int_E g \, d\mu \leq \nu(E)$ for all E .

(a) If $g, h \in \mathcal{G}$ then $g \vee h \in \mathcal{G}$.

Proof of (a). Let $A = \{g > h\}$, $B = \{g \leq h\}$.

$$\int_E g \vee h \, d\mu = \int_{E \cap A} g \, d\mu + \int_{E \cap B} h \, d\mu \leq \nu(E \cap A) + \nu(E \cap B) = \nu(E)$$

Let $\kappa = \sup\{\int g \, d\mu : g \in \mathcal{G}\} \leq \nu(\Omega) < \infty$. Pick g_n so that $\int g_n \, d\mu > \kappa - 1/n$ and let $h_n = g_1 \vee \dots \vee g_n$. By (a), $h_n \in \mathcal{G}$. As $n \uparrow \infty$, $h_n \uparrow h$. The definition of κ , the monotone convergence theorem, and the choice of g_n imply that

$$\kappa \geq \int h \, d\mu = \lim_{n \rightarrow \infty} \int h_n \, d\mu \geq \lim_{n \rightarrow \infty} \int g_n \, d\mu = \kappa$$

Let $\nu_r(E) = \int_E h \, d\mu$ and $\nu_s(E) = \nu(E) - \nu_r(E)$. The last detail is to show:

(b) ν_s is singular with respect to μ .

Proof of (b). Let $\epsilon > 0$ and let $\Omega = A_\epsilon \cup B_\epsilon$ be a Hahn decomposition for $\nu_s - \epsilon\mu$. Using the definition of ν_r and then the fact that A_ϵ is positive for $\nu_s - \epsilon\mu$ (so $\epsilon\mu(A_\epsilon \cap E) \leq \nu_s(A_\epsilon \cap E)$),

$$\int_E (h + \epsilon 1_{A_\epsilon}) \, d\mu = \nu_r(E) + \epsilon\mu(A_\epsilon \cap E) \leq \nu(E)$$

This holds for all E , so $k = h + \epsilon 1_{A_\epsilon} \in \mathcal{G}$. It follows that $\mu(A_\epsilon) = 0$, for if not, then $\int k \, d\mu > \kappa$ a contradiction. Letting $A = \bigcup_n A_{1/n}$, we have $\mu(A) = 0$. To see that $\nu_s(A^c) = 0$, observe that if $\nu_s(A^c) > 0$, then $(\nu_s - \epsilon\mu)(A^c) > 0$ for small ϵ , a contradiction since $A^c \subset B_\epsilon$, a negative set. $\square$

Exercise A.4.4. Prove that the Lebesgue decomposition is unique. Note that you can suppose without loss of generality that μ and ν are finite.

We are finally ready for the main business of the section. We say a measure ν is **absolutely continuous with respect to μ** (and write $\nu \ll \mu$) if $\mu(A) = 0$ implies that $\nu(A) = 0$.

Exercise A.4.5. If $\mu_1 \ll \mu_2$ and $\mu_2 \perp \nu$ then $\mu_1 \perp \nu$.

Theorem A.4.8. Radon-Nikodym theorem. *If μ, ν are σ -finite measures and ν is absolutely continuous with respect to μ , then there is a $g \geq 0$ so that $\nu(E) = \int_E g d\mu$. If h is another such function then $g = h$ μ a.e.*

Proof. Let $\nu = \nu_r + \nu_s$ be any Lebesgue decomposition. Let A be chosen so that $\nu_s(A^c) = 0$ and $\mu(A) = 0$. Since $\nu \ll \mu$, $0 = \nu(A) \geq \nu_s(A)$ and $\nu_s \equiv 0$. To prove uniqueness, observe that if $\int_E g d\mu = \int_E h d\mu$ for all E , then letting $E \subset \{g > h, g \leq n\}$ be any subset of finite measure, we conclude $\mu(g > h, g \leq n) = 0$ for all n , so $\mu(g > h) = 0$, and, similarly, $\mu(g < h) = 0$. $\square$

Example A.4.9. Theorem A.4.8 may fail if μ is not σ -finite. Let $(\Omega, \mathcal{F}) = (\mathbf{R}, \mathcal{R})$, μ = counting measure and ν = Lebesgue measure.

The function g whose existence is proved in Theorem A.4.8 is often denoted $d\nu/d\mu$. This notation suggests the following properties, whose proofs are left to the reader.

Exercise A.4.6. If $\nu_1, \nu_2 \ll \mu$ then $\nu_1 + \nu_2 \ll \mu$

$$d(\nu_1 + \nu_2)/d\mu = d\nu_1/d\mu + d\nu_2/d\mu$$

Exercise A.4.7. If $\nu \ll \mu$ and $f \geq 0$ then $\int f d\nu = \int f \frac{d\nu}{d\mu} d\mu$.

Exercise A.4.8. If $\pi \ll \nu \ll \mu$ then $d\pi/d\mu = (d\pi/d\nu) \cdot (d\nu/d\mu)$.

Exercise A.4.9. If $\nu \ll \mu$ and $\mu \ll \nu$ then $d\mu/d\nu = (d\nu/d\mu)^{-1}$.

A.5 Differentiating under the Integral

At several places in the text, we need to interchange differentiate inside a sum or an integral. This section is devoted to results that can be used to justify those computations.

Theorem A.5.1. *Let $(S, \mathcal{S}, \mu)$ be a measure space. Let f be a complex valued function defined on $\mathbf{R} \times S$. Let $\delta > 0$, and suppose that for $x \in (y - \delta, y + \delta)$ we have*

$$(i) \ u(x) = \int_S f(x, s) \mu(ds) \text{ with } \int_S |f(x, s)| \mu(ds) < \infty$$

(ii) for fixed s , $\partial f/\partial x(x, s)$ exists and is a continuous function of x ,

(iii) $v(x) = \int_S \frac{\partial f}{\partial x}(x, s) \mu(ds)$ is continuous at $x = y$,

and (iv) $\int_S \int_{-\delta}^{\delta} \left| \frac{\partial f}{\partial x}(y + \theta, s) \right| d\theta \mu(ds) < \infty$

then $u'(y) = v(y)$.

Proof. Letting $|h| \leq \delta$ and using (i), (ii), (iv), and Fubini's theorem in the form given in Exercise 1.7.4, we have

$$\begin{aligned} u(y+h) - u(y) &= \int_S f(y+h, s) - f(y, s) \mu(ds) \\ &= \int_S \int_0^h \frac{\partial f}{\partial x}(y+\theta, s) d\theta \mu(ds) \\ &= \int_0^h \int_S \frac{\partial f}{\partial x}(y+\theta, s) \mu(ds) d\theta \end{aligned}$$

The last equation implies

$$\frac{u(y+h) - u(y)}{h} = \frac{1}{h} \int_0^h v(y+\theta) d\theta$$

Since v is continuous at y by (iii), letting $h \rightarrow 0$ gives the desired result. $\square$

Example A.5.2. For a result in Section 3.3, we need to know that we can differentiate under the integral sign in

$$u(x) = \int \cos(xs) e^{-s^2/2} ds$$

For convenience, we have dropped a factor $(2\pi)^{-1/2}$ and changed variables to match Theorem A.5.1. Clearly, (i) and (ii) hold. The dominated convergence theorem implies (iii)

$$x \rightarrow \int -s \sin(sx) e^{-s^2/2} ds$$

is a continuous. For (iv), we note

$$\int \left| \frac{\partial f}{\partial x}(x, s) \right| ds = \int |s| e^{-s^2/2} ds < \infty$$

and the value does not depend on x , so (iv) holds.

For some examples the following form is more convenient:

Theorem A.5.3. Let $(S, \mathcal{S}, \mu)$ be a measure space. Let f be a complex valued function defined on $\mathbf{R} \times S$. Let $\delta > 0$, and suppose that for $x \in (y - \delta, y + \delta)$ we have

(i) $u(x) = \int_S f(x, s) \mu(ds)$ with $\int_S |f(x, s)| \mu(ds) < \infty$

(ii) for fixed s , $\partial f / \partial x(x, s)$ exists and is a continuous function of x ,

(iii')
$$\int_S \sup_{\theta \in [-\delta, \delta]} \left| \frac{\partial f}{\partial x}(y + \theta, s) \right| \mu(ds) < \infty$$

then $u'(y) = v(y)$.

Proof. In view of Theorem A.5.1 it is enough to show that (iii) and (iv) of that result hold. Since

$$\int_{-\delta}^{\delta} \left| \frac{\partial f}{\partial x}(y + \theta, s) \right| d\theta \leq 2\delta \sup_{\theta \in [-\delta, \delta]} \left| \frac{\partial f}{\partial x}(y + \theta, s) \right|$$

it is clear that (iv) holds. To check (iii), we note that

$$|v(x) - v(y)| \leq \int_S \left| \frac{\partial f}{\partial x}(x, s) - \frac{\partial f}{\partial x}(y, s) \right| \mu(ds)$$

(ii) implies that the integrand $\rightarrow 0$ as $x \rightarrow y$. The desired result follows from (iii') and the dominated convergence theorem. $\square$

To indicate the usefulness of the new result, we prove:

Example A.5.4. If $\phi(\theta) = Ee^{\theta Z} < \infty$ for $\theta \in [-\epsilon, \epsilon]$ then $\phi'(0) = EZ$.

Proof. Here θ plays the role of x , and we take μ to be the distribution of Z . Let $\delta = \epsilon/2$. $f(x, s) = e^{xs} \geq 0$, so (i) holds by assumption. $\partial f / \partial x = se^{xs}$ is clearly a continuous function, so (ii) holds. To check (iii'), we note that there is a constant C so that if $x \in (-\delta, \delta)$, then $|s|e^{xs} \leq C(e^{-\epsilon s} + e^{\epsilon s})$. $\square$

Taking $S = \mathbf{Z}$ with $\mathcal{S} =$ all subsets of S and $\mu =$ counting measure in Theorem A.5.3 gives the following:

Theorem A.5.5. Let $\delta > 0$. Suppose that for $x \in (y - \delta, y + \delta)$ we have

(i) $u(x) = \sum_{n=1}^{\infty} f_n(x)$ with $\sum_{n=1}^{\infty} |f_n(x)| < \infty$

(ii) for each n , $f'_n(x)$ exists and is a continuous function of x ,

and (iii) $\sum_{n=1}^{\infty} \sup_{\theta \in (-\delta, \delta)} |f'_n(y + \theta)| < \infty$

then $u'(x) = v(x)$.

Example A.5.6. In Section 2.6 we want to show that if $p \in (0, 1)$ then

$$\left(\sum_{n=1}^{\infty} (1-p)^n \right)' = - \sum_{n=1}^{\infty} n(1-p)^{n-1}$$

Proof. Let $f_n(x) = (1-x)^n$, $y = p$, and pick δ so that $[y-\delta, y+\delta] \subset (0, 1)$. Clearly (i) $\sum_{n=1}^{\infty} |(1-x)^n| < \infty$ and (ii) $f'_n(x) = n(1-x)^{n-1}$ is continuous for x in $[y-\delta, y+\delta]$. To check (iii), we note that if we let $2\eta = y - \delta$ then there is a constant C so that if $x \in [y-\delta, y+\delta]$ and $n \geq 1$, then

$$n(1-x)^{n-1} = \frac{n(1-x)^{n-1}}{(1-\eta)^{n-1}} \cdot (1-\eta)^{n-1} \leq C(1-\eta)^{n-1} \quad \square$$

Table of the Normal Distribution

$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy$$

To illustrate the use of the table: $\Phi(0.36) = 0.6406$, $\Phi(1.34) = 0.9099$

	0	1	2	3	4	5	6	7	8	9
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7703	0.7734	0.7764	0.7793	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8943	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9924	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9986	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

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